

Claims Encounters and Financial Services Administrator

Installation Guide

Medicaid Management Solutions

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Document Revision History

The *HPA 5.0 Installation Guide* was published in January 2012 and was renamed to *Administrator Installation Guide* in April 2021. The following table lists subsequent changes that have been made to the document.

Date	Location	Description
2/17/12	All chapters	Changed installation names with specific version number to <version>.
2/17/12	Introduction	Replaced server graphics.
3/16/12	Introduction	Removed references to HPA View database servers.
4/6/12	All chapters	Added level 2 headings to chapter TOC.
4/6/12	Introduction, page 8	Added “4.0” and error message graphic to .NET Framework requirements.
4/6/12	Operating System and Web Browser Requirements, page 33	Defined differences between Windows Server 2012 R2 and Windows Server User Account Control settings.
4/6/12	Operating System and Web Browser Requirements, page 34	Added note to configure all workstations.
4/6/12	QFramework Requirement page 138, HPA Requirements page 160, HPA Connect	Added note to refer to Compatibility Matrices.
2/4/16	Requirements page 174 Configuring Windows Server, page 40	Added IIS configuration changes and additional information on silent install.
12/22//20	Requirements page, Configuring Windows Server, Installing SQL Server, SQL Reporting Services, Installing Biztalk Server Configuring Biztalk Server	Updated the Operating system Windows Server 16, SQL Server 16, MS Office 16, Biztalk 16 and .Net framework 4.8 upgrade changes.
04/09/2021	Installing HPA, Installing McKesson and Micro-Dyn product, Updating the config version, Trademark, HPA UI on Application server	Updated the installation screenshot with claimsXten feature checked and disabled, removed ClaimCheck installation and configuration steps, renamed <i>HPA Installation Guide</i> to <i>Administrator Installation Guide</i> , updated header of all pages.
07/23/2021	Installing HPA Connect HIPAA API, Verifying HPA Connect Installation	Added steps to follow for installing HPA HIPAA API and verifying it.
10/08/2021	Installing HPA Connect HIPAA API	Updated step in Transaction Processing Servers window for specifying the API Server name.
10/08/2021	Configuring GPCS Certificate	Added GPCS Certificate configuration.
08/12/2022	Core Installation Requirements, Performing a new Installation	Removed i.e. configuration information and updated Browser compatibility details for Chrome and Edge Chromium.

Date	Location	Description
04/01/2022	Installing QFramework, HPA and Connect, Updating the HPA Service Account database permissions	Updated steps for HTTPS protocol and steps to update service account database permissions.
05/12/2023	Installing HPA Connect API Server, Database Restoration and Update, Installing SQL Server, Installing SQL Reporting Services, Configuring SQL Reporting Services	Updated EDIFabricKey registry entry details, SQL Encryption updates, SQL Server 2022 installation changes.
09/18/2023	SQL Encryption, HPA Connect API Server, QEnterprise Configuration	Updated details about SQL Encryption and configuring UploadReadAheadSize for Connect API and Distribution Hub.
01/09/2025	Cover and headers	In Release 31.05 (version 5.0.31.00), implemented NuGet packages so installation will deploy DLLs to respective bin folder instead of GAC. Product version has changed from 5.0.31.00 (in R30) to 5.31.1.0 (in R31.05).
10/01/2025	Guidelines to Implement SSO All Chapters	Added a Note as part of R34. Renamed 3M GPCS to GPCS.
01/09/2026	XV Environment setup for Pharmacy Claims API Integration Configuring the Web configuration files for DH and EU	R35 Updates

I Introduction

Overview

This guide provides information and procedures to install the HPA™ suite. Each component includes step-by-step installation instructions. Before starting the installation, make sure you have all prerequisites in place and that you have correctly configured all applicable Microsoft® software. For more information, see [Requirements](#).

For more information about HPA components or the installation process, contact your account representative.

NOTE: Information in this document is based on minimum supported configuration requirements for HPA 5.0. If you plan to deviate from the recommended configuration, contact Product Support for assistance.

HPA Upgrades

Upgrades to version 5.0 from all previous HPA versions are supported. However, each HPA software version requires manual preparation procedures for upgrades. Required information to upgrade from the previous major software version is documented in the installation guide for each version of HPA.

System Components

The following table provides brief descriptions of the applications that work with the HPA enterprise application system.

Application	Description
HPA	Designed to encompass an "n-tier" environment and service-oriented architecture, HPA offers the flexibility and functionality that payers need to achieve optimal growth and efficiency with the lowest total cost of ownership.
Microsoft BizTalk Server®	Translates X12 standard files to Extensible Markup Language (XML) data for HPA Connect, which maps the data between the XML format and the proprietary HPA database schema. XML schemas are defined by Microsoft.
QFramework	Allows you to track and configure HPA system message and transaction processing.

Application	Description
HPA Connect	A set of tools and web services that enables electronic data interchange (EDI) transactions of Health Insurance Portability and Accountability Act of 1996 (HIPAA) standard transactions between a health plan and its trading partners. HPA Connect supports: • Batch and real-time processing of eligibility benefit inquiries and responses • Claim status inquiries and responses • Referral certification and authorization requests and responses • Benefit enrollment and maintenance transactions • Claims submissions for professional, institutional, dental, and pharmacy • Claims payment and remittance advice • Payroll-deducted and other group premium payment for insurance HIPAA EDI transactions implemented in HPA by the HPA Connect product are made up of several transaction sets. Each set is externally defined and contains generic abstractions of tables and fields from the HPA PlanData database.
HPA Reports	Provides reports and analytics to help monitor and plan system functions. You can also configure system reports for restricted access.
HPA Help	Includes detailed feature descriptions, parameters, and procedures to configure and maintain HPA system modules.
HPA Rules	Installs industry and system-related rules used by the HPA system.

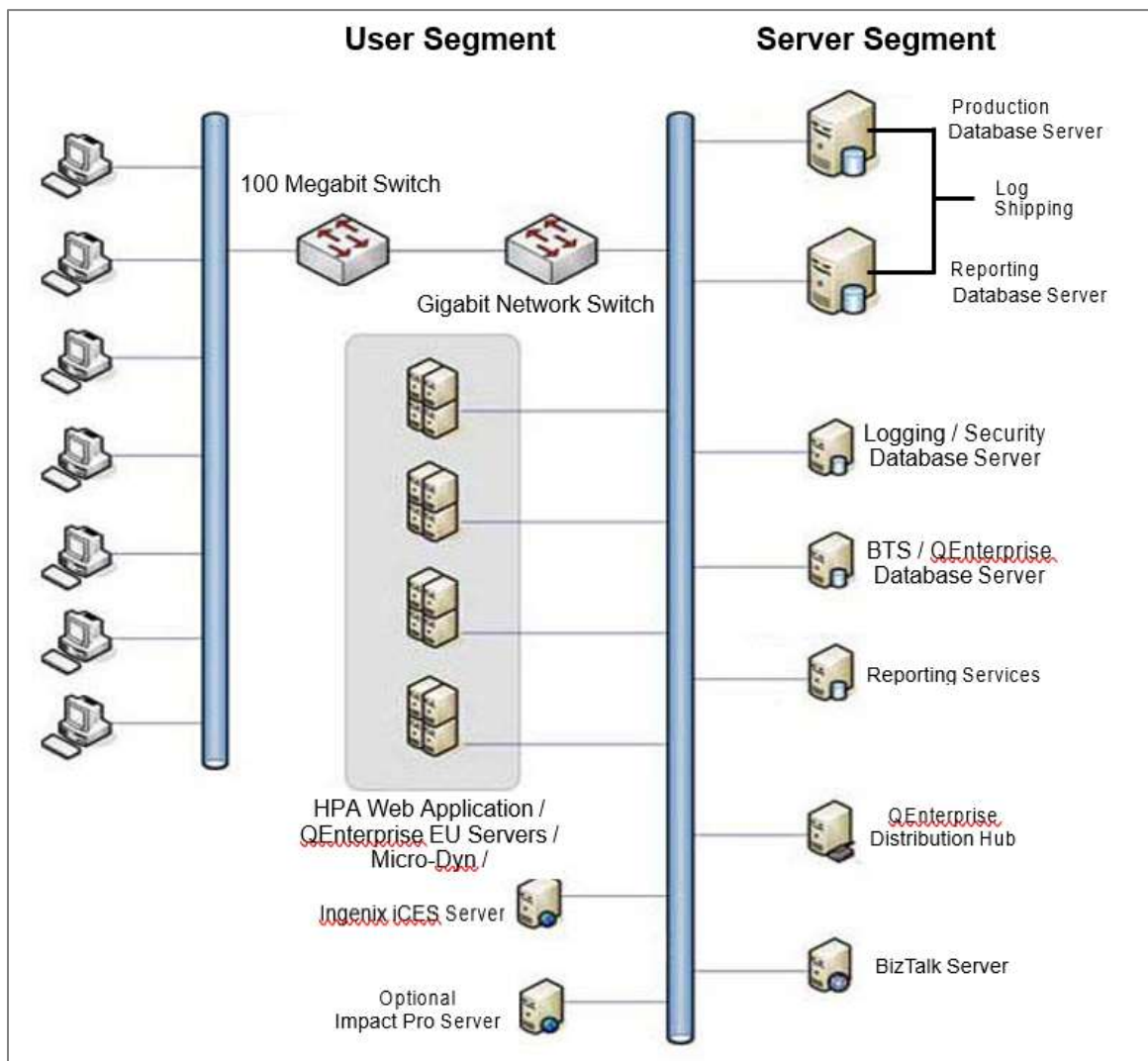
System Architecture

The following sections describe system architecture and server configurations.

Minimum Supported Configuration

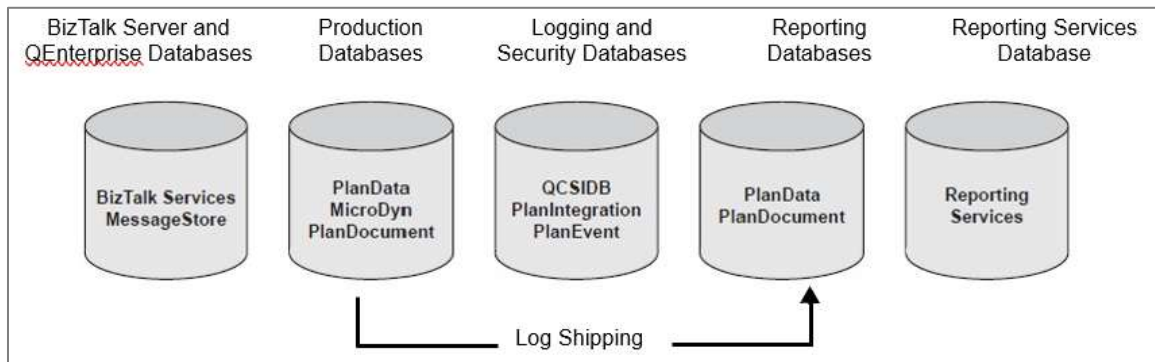
While there are multiple ways to set up and configure an HPA system, thorough testing has determined the minimum supported configuration, as shown in graphic below. This configuration provides flexibility and helps ensure a stable environment. HPA recommends that you follow the configuration and deployment order.

NOTE: HPA recommends that the QEnterprise Execution Unit (EU) and the user interface (UI) feature be installed on separate servers. Performance issues may occur if the QEnterprise EU and UI feature are installed on the same server.



SQL Server Configuration

The following graphic shows a minimum supported Microsoft SQL Server® configuration with the HPA Core database servers.



The PlanIntegration database cannot reside on the same physical SQL Server as the BizTalk Server database or MessageStore database.

Production databases provide log shipping to Reporting databases. According to Microsoft, log shipping "feeds transaction logs from one database to another on a constant basis. Continually backing up the transaction logs from a source database and then copying and restoring the logs to a destination database keeps the destination database synchronized with the source database. This allows you to have a backup server and also provides a way to offload query processing from the main computer (the source server) to read-only destination servers."

Requirements

The following sections describe installation requirements necessary to successfully install HPA.

Administrator Requirements

You must understand and be able to successfully install the following software:

- **Windows Configuration:** Microsoft Windows Server operating system administration skills
- **Server Database Configuration and Installation:**
 - Microsoft SQL Server database administration skills (and access rights)
 - Microsoft BizTalk Server administration skills (and access rights)
- **HPA Installation:** Microsoft Windows Server® system administration skills

Core Installation Requirements

You must install and configure the software listed below before installing any other software.

Operating system installation requirements: Administrator must be installed on servers where the operating system has been installed manually from a valid Microsoft operating system CD. **Never install the operating system from an image.** A server built from an image is not a stable platform and therefore not a supported product installation.

After the operating system has been installed, do **not** change the server name—this causes errors with Internet Information Services (IIS) and .NET Framework.

- **Microsoft Windows Server®:** Install this operating system (OS) on all servers except where BizTalk Server is installed. See [Configuring Microsoft Windows Server](#) for more information.
- **Microsoft Critical Updates:** Install all critical updates for Windows Server that exist prior to the latest release date for this product. 5.0
- **Browsers Supported:** HPA is certified on Google Chrome (Version: 104.0.5112.81) and Microsoft Edge Chromium (Version 104.0.1293.47) browsers.

NOTE: Ensure that **Always open pop-ups** is selected when a pop-up blocker is encountered on accessing the HPA application

- **Microsoft .NET Framework 4.8:** To install, right-click the setup file. Then, click **Run as administrator**. This gives you limited administrative access. If you are a member of the local administrator's group, you have the ability to do this without entering administrator credentials.
- **Microsoft XML Core Services (MSXML):** Required for applications to build Windows-native XML-based applications.
- **Microsoft Excel®:** Before installing HPA, you must install Microsoft Excel either as a standalone version or through one of the Microsoft Office® 2016 installations. The HPA UI feature requires that Microsoft Excel .NET Programmability Support and Microsoft Query be installed prior to installing this feature.

NOTE: HPA supports only Microsoft Office 32-bit applications. Microsoft Office 64-bit applications are not supported.

- Install **.NET Programmability Support** to run from the local disk.
- Install **Microsoft Query** under the **Office Tools** feature to run from the local disk.

- **Microsoft Access Database Engine 2016 Redistributable (32 bit):** After the Microsoft Office installation is complete, you must install Microsoft Access Database Engine 2016 Redistributable (32 bit) to transfer data between Microsoft Office Access (*.mdb and *.accdb) files and Microsoft Office Excel (*.xls, *.xlsx, and *.xlsb) files to other data sources such as Microsoft SQL Server.

To download the Microsoft Access Database Engine 2016 Redistributable, go to <https://www.microsoft.com/en-us/download/details.aspx>.

Important Prerequisite Notes

The following is important installation information. Read these notes carefully before beginning any installation.

- Close all programs prior to installing HPA.
- Make sure you have administrator permissions. The person performing the installation must be the administrator of that machine.
- Add a local administrator account. Before beginning the SQL Server database installation, a local administrator account must be added to Microsoft SQL Server. The local administrator account is used to setup both the SQL Server and SQL Server Agent services.
- Perform installations locally. Copy all installation files to a local drive.
- Performing all installations from a local drive helps eliminate the possibility that network connectivity issues might impact a successful installation.
- Accept default settings unless instructed otherwise. When installing third-party software, accept the default selections unless otherwise noted.
- Rebuild the servers. Servers that previously ran Microsoft Windows 2012 R2 and Microsoft .NET Framework 4.0 require a complete rebuild of the machine including a fresh installation of the new operating system before proceeding with HPA installation. **HPA requires Microsoft Windows Server 2019 and Microsoft .NET Framework 4.8.**
- Follow BizTalk Server installation requirements.

NOTE: HPA 5.0 supports BizTalk Server Enterprise or BizTalk Server Standard 64-bit versions only. BizTalk Server 32-bit versions are not supported.

For more information about BizTalk Server installation requirements, refer to the following Microsoft articles:

- *BizTalk Server may not perform as expected on a Terminal Server that is running in Application mode* at <http://support.microsoft.com/?kbid=832027>.
- *Configuring Enterprise SSO Using the Configuration Manager* at <http://msdn.microsoft.com/en-us/library/aa561969.aspx>.

NOTE: Depending on your version of BizTalk, you may need to install a Microsoft hotfix before installing HPA Connect. For Microsoft BizTalk service pack and update information, refer to the *Service Pack and Cumulative Update list for BizTalk Server* at <http://support.microsoft.com/kb/2555976>.

- **Map server names.** Take time before beginning this installation to clearly map out the names for each server. Having those naming conventions in place and clearly mapped out simplifies the installation process.
- **Disable third-party pop-up blocker applications.** Make sure that there are no third-party pop-up blocker applications installed on the client machine. Failure to remove pop-up blocker applications results in errors and unpredictable HPA processing results.

Follow server name requirements. All server names must conform to Microsoft's naming convention for NetBIOS and DNS names. The convention states that Unicode characters, numbers, or white space cannot be used. Symbols such as—! @ # \$ % ^ & ') (. - _ { } ~ and period (.) have special reserved meaning. Failure to comply to Microsoft's standard could yield unwanted results or behavior

Software Considerations

The following are HPA and third-party software considerations:

- **HPA environment names:** HPA environment names cannot include the < > & “ or ‘ characters.
- **Adobe® Reader®:** Download the most current version of Adobe Acrobat Reader to view release notes, installation instructions, and other documentation in PDF format. To download Adobe Acrobat Reader, go to www.adobe.com.
- **SQL Server Database:** You must install Microsoft SQL Server. You must set the dbplatformtype column in the dbalias table to **128** for the applicable HPA environment Id (envId).
- **HPA Reports:** HPA Reports only supports SQL Reporting Services on a single instance of SQL Server. See [Installing HPA Reports](#) for more information.
- **Microsoft Hotfix support in HPA:** Gainwell Technologies validates its products to a specific Microsoft Windows operating system and service pack but does not support Microsoft product hotfixes, unless otherwise noted. While clients are not dissuaded from installing Microsoft hotfixes, TriZetto cannot confirm that the TriZetto product will operate properly once the operating system hotfix has been installed. To minimize the need for hotfixes in a production environment, TriZetto recommends that all application servers be isolated from the Internet.
- **Network Load Balancing (NLB):** NLB must be configured in Single Affinity mode for clusters that use NLB. Refer to Microsoft Help for information on configuring NLB.
- **Web services:** Some Gainwell Technologies products create web services. In creating these web services, the local system accounts, IUSR, IWAM, and Network Service are given specific rights to Gainwell Technologies product directories and virtual directories. Changing directory security once the installation process is complete may result in Gainwell Technologies products not functioning or uninstalling properly.
- **Hyper-Threading:** Gainwell Technologies supports hyper-threading functionality on Intel® single-core processors running on the application, web, Distribution Hub (DH), or EU servers. Intel multi-core processors running on the application, web, DH, or EU servers is also supported.

Workstation Hardware

The HPA application requires that all workstations have a monitor and video card that supports 1024 x 768 resolution.

Installation Package

This section lists the files and components included with the HPA Suite. Copy all of the installation files to a local drive. Performing the installations from a local drive prevents the possibility of network connectivity issues that might interfere with the installation process.

Installation Files

The following files start the HPA component installation wizards:

- HPA_<version>_x64.exe (HealthPAS Administrator)
- HPA_Connect_<version>_x64.exe (HPA Connect)
- HPA_Help_<version>_x64.exe (HPA Help)
- HPA_QFramework_<version>_x64.exe (HealthPAS Framework)
- HPA_Rules_<version>_x64.exe (HPA Rules)

Database Files

The HPA system uses the following databases:

- **Model:** SQL Server system database used as the template for all databases created on an instance of SQL Server. Because Tempdb is created every time SQL Server is started, the Model database must always exist on a SQL Server system.
- **Tempdb:** SQL Server system database that is used as a global resource available to all users connected to an instance of SQL Server. The Tempdb database is used to store user objects, internal objects, and version stores.
- **MessageStore:** Stores messages exchanged between the HPA application server and HPA DH and EU servers.
- **PlanData:** Primary database for the HPA application. Stores data for Carriers, Programs, Members, Providers, Benefit Plans, Claims, etc.
- **PlanDocument:** Contains document images captured or imported in Case Management.
- **PlanEvent:** Stores application tracing log data, HPA data access log data, and HPA Service Transaction information.
- **PlanIntegration:** Holds process logging information related to HPA background processing (for example, HPA data import and export and mass claims adjudication).
- **QCSIDB:** HPA security and configuration database. Stores information for HPA Environments, Environment Configuration, Users, User Roles and User Activities.
- **HPA_System_Manager:** Contains detailed upgrade steps necessary to upgrade HPA application databases. This database must be present on each of the database servers housing a HPA application database.

Reference Database Backup Files

Use the following backup (.bak) files to restore HPA databases. For more information, see [Initiating a New Client Database](#).

- HPA<version>_MessageStore.bak
- HPA<version>_PlanData.bak
- HPA<version>_PlanDocument.bak
- HPA<version>_PlanEvent.bak

- HPA<version>_PlanIntegration.bak
- HPA<version>_QCSIDB.bak

Update Tempdb Scripts

Use the following scripts to update the Tempdb and Model databases. For more information, see [Updating the Tempdb, Model, and QCSIDB Databases](#).

- HPA Create UDT.sql
- Linked NT Login.sql

Update Report View Scripts

Run the following scripts to update report views. For more information, see [Updating Report Views](#) (New Client Database) or [Updating Report Views](#) (Upgrading an Existing Client).

- ViewLinkedServer.sql
- ReportViews.sql

Picklist Table Script

Run the following script to install the Picklist table. For more information, see [Installing the Picklist Table](#).

- 01 tlReportPickList Reference Data.sql

Create SPQ Exec Upgrade

Run the following script against all databases that is upgraded using the Automated Database Upgrade tool:

- 07 HPA <version> Create_SPQ_ExecUpgrade.sql

Update PlanEvent Script

Run the following script to upgrade to HPA <version> Update PlanEvent:

- HPA <version> Update PlanEvent.sql

Update aCOM3 Status Script

Run the following script to update the aCOM3 status:

- acom3status_update.sql

License Files

Run the following Registry Data File (.reg) license files prior to launching the HPA installation wizards:

- QNXTAR.reg
- QNXTCM.reg
- QNXTIMPACTPRO.reg

- QNXTINGENIX.reg
- QNXTMICRODYN.reg

Documentation Files (5.0)

For more information about HPA configuration options and instructions, refer to the HPA Help application.

For instructions on installing HPA Help, see [Installing HPA Help](#).

- HPA <version> Database Schema Changes.pdf
- HPA <version> EDI Report Conversion Guide.pdf
- HPA <version> InstallGuide.pdf
- QXI <version> Reference.zip
- HPA <version> RelNotes.pdf
- HPA Connect <version> 4010 User Guide.pdf
- HPA Connect <version> 5010 User Guide.pdf
- HPA EDI 4010 <version> Eligibility Workbook.pdf
- HPA EDI 4010 <version> Institutional Claim Workbook.pdf
- HPA EDI 4010 <version> Professional Claim Workbook.pdf
- HPA EDI 4010 <version> Provider Workbook.pdf
- HPA EDI 4010 <version> Referral Workbook.pdf

Installation Workflow

This section provides steps that must be completed to successfully install HPA components.

Performing a New Installation

NOTE: Previous versions of HPA component installation programs included programming logic that notified users if a HPA component was being installed without the correct prerequisites in place. Currently, HPA 5.0 component installation wizards **do not** automatically check for installation prerequisites. This is a known issue and will be resolved in a future change package. Make sure that all prerequisites are in place before installing each HPA component. For prerequisite information, refer to the “Requirements” section at the beginning of the chapter for the applicable HPA component.

The following is an overview of the steps required to install and configure HPA components:

1. Configure the Servers. See [Operating System and Web Browser Configuration](#).
2. Install SQL Databases. See [SQL Server Installation and Configuration](#) and [Database Restoration and Update](#).
3. Install SQL Server Reporting Services. See [SQL Reporting Services](#).
4. Install BizTalk Server. See [BizTalk Server Installation and Configuration](#).
5. Install QFramework. See [QFramework Installation](#).
6. Install HPA. See [HPA Installation](#).
7. Install HPA Connect. See [HPA Connect Installation](#).
8. Install HPA Reports. See [HPA Reports Installation](#).
9. Install HPA ITS (optional). For more information, refer to the *HPA ITS Installation Guide Release ITS 11.3.5.0*.
10. Install HPA Help. See [HPA Help Installation](#).
11. Install HPA Rules. See [HPA Rules Installation](#).
12. Install optional third-party applications. See [Micro-Dyne Preparation and Setup](#).

After Installation, shortcut menu icons will be created for HPA application, Distribution Hub console, the Execution Unit console, the Master Service, and the Orphan Retry services. These icons can be accessed with the default browser that is configured in the server.

NOTE: Google Chrome and Microsoft Edge Chromium are the supported browsers for HPA components.

Upgrading to HPA Version 5.0

If you are upgrading to HPA 5.0 from a previous version, you must uninstall the existing HPA software first. For version-specific instructions, refer to the applicable installation guide for the version of HPA that you are installing.

When upgrading to HPA 5.0, you must convert the Excel Production Reports from .xls format to .xlsx format. This procedure is described in the *HPA 5.0 EDI Report Conversion Guide*.

If you are upgrading and have existing databases, apply the applicable upgrade scripts. See [Upgrading an Existing Client Database](#) for the names of the database scripts that must be applied.

Uninstall software using the **Control Panel | Programs | Uninstall a program** tool.

If you are an existing client you must perform the procedures outlined in [Manually Upgrading the PlanEvent Database for HPA 5.0](#) before you uninstall the previous version of HPA.

Uninstall HPA software components in the following order (as applicable):

1. HPA Rules
2. HPA Help
3. HPA Reports
4. HPA Connect
5. HPA
6. QFramework

After uninstalling all components, you can reinstall HPA components as described in [Performing a New Installation](#).

II Operating System and Web Browser Configuration

Operating system and web browser configuration for the HPA™ suite involves the following activity:

- [Configuring Security Permissions](#)
- [Configuring the Required Service Account](#)
- [Configuring the Firewall Settings for the Domain Profile](#)
- [Configuring Microsoft Windows Server](#)
- [Creating a Desktop Folder](#)

Configuring Security Permissions

After installing the core requirements (see [Installation Package](#)), make sure that the following security permission settings are configured:

- **Registry Security:** No policies or permission changes are implemented on the systems registry.
- **IIS Security:** No policies or permission changes are applied to the Internet Information Services.
- **Directory Security:** No policies are applied to the directories.

HPA installs and tests using the Microsoft® Default Directory Permissions.

Windows Server

The HPA 5.0 suite of products has been validated on the default New Technology File System (NTFS) permissions of Windows® Server®. If you are deploying this product on a secondary drive (that is, not the C: drive), the permissions for the drive must be the same as the permissions of the C: drive.

Configuring the Required Service Account

A domain level account, referred to as the HPA Service Account, is required for IIS configuration of HPA components. The IIS configuration is used to run HPA Windows Services and for HPA application code to authenticate to the HPA SQL Server databases. This account should be configured on each of the servers where a HPA application component is installed.

This account must have local administrator rights on all servers where the HPA suite of products is installed. The account is configured for SQL access when the HPA databases are restored and configured. For more information, see [Updating the HPA Service Account Database Permissions](#).

The HPA service account is used during various HPA product installations to run the following:

- QCSI Services
- COM+ Identity
- IIS Application Pools

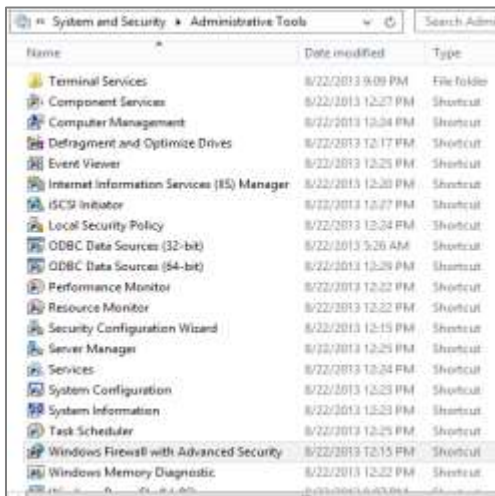
Configuring Firewall Settings for the Domain Profile

To complete the installation procedures, you must disable the firewall for the Domain profile. This disables the firewall for all domain applications.

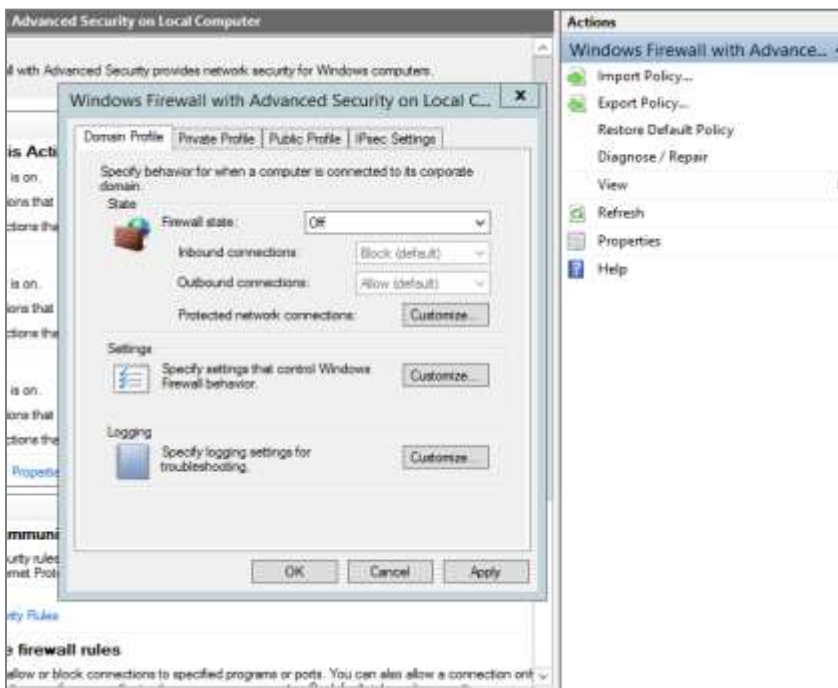
NOTE: The instructions in this guide use interface elements and instructions based on the **Windows Server 2019** operating systems. If you are using another Windows operating system, the interface elements and instructions may or may not be the same as described in the guide.

To disable the firewall for the Domain profile:

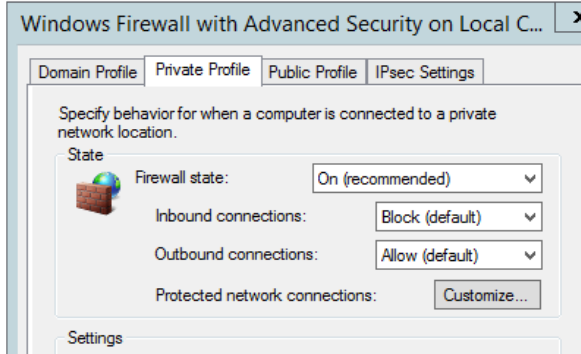
1. Open Windows Firewall by selecting **Start | Control Panel | System and security | Administrative Tools | Windows Firewall with Advanced Security**, as shown below.



2. In the right (Actions) pane of the Windows Firewall with Advanced Security on Local Computer screen, Click **Properties**. The Windows Firewall with Advanced Security on Local Computer Profile dialog box appears.



3. Complete the following actions on the Domain Profile tab.
 - 3.1 Select **Off** as the Firewall state In the State area.
 - 3.2 Click **OK** to save the change.
4. Make sure the Firewall state is **On (recommended)** on the Private Profile and Public Profile Firewall tabs, as shown below.



Configuring Microsoft Windows Server

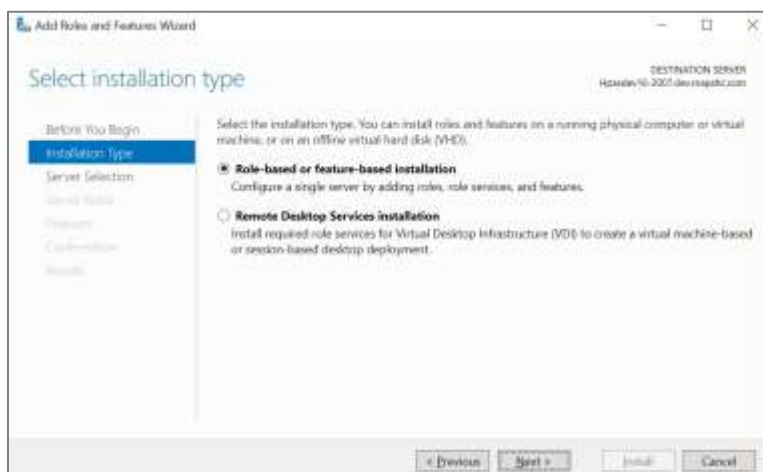
This section describes the configuration modifications required after the Microsoft Windows Server® operating system is installed. Perform these modifications on all servers.

NOTE: The instructions in this guide use interface elements and instructions based on the Windows Server 2019 operating systems. If you are using another Windows operating system, the interface elements and instructions may or may not be the same as described in the guide.

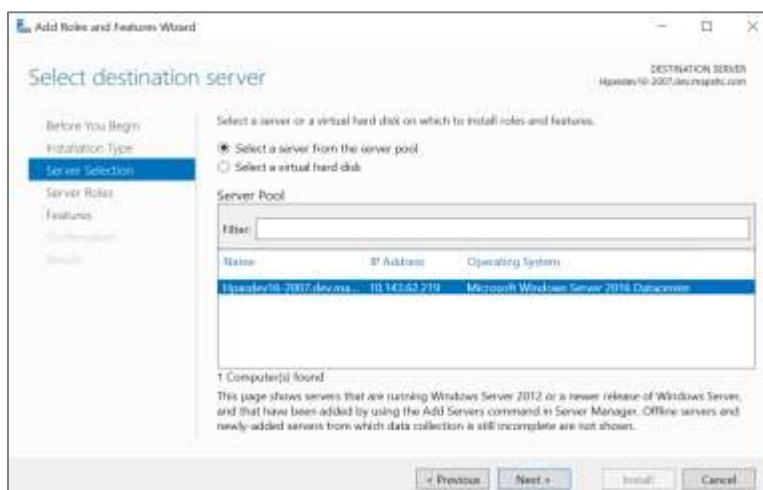
Installing Required Roles and Features

To install required roles and services:

1. Select **Start | Administrative Tools | Server Manager**. The Server Manager dialog box appears.
2. Select **Add roles and features** from Server Manager Dashboard section. The Add Roles and Features Wizard appears.
3. If the Before You Begin dialog box appears, Click **Next**. The Installation type dialog box appears.
4. Select **Role-based or feature-based installation** and click **Next**.

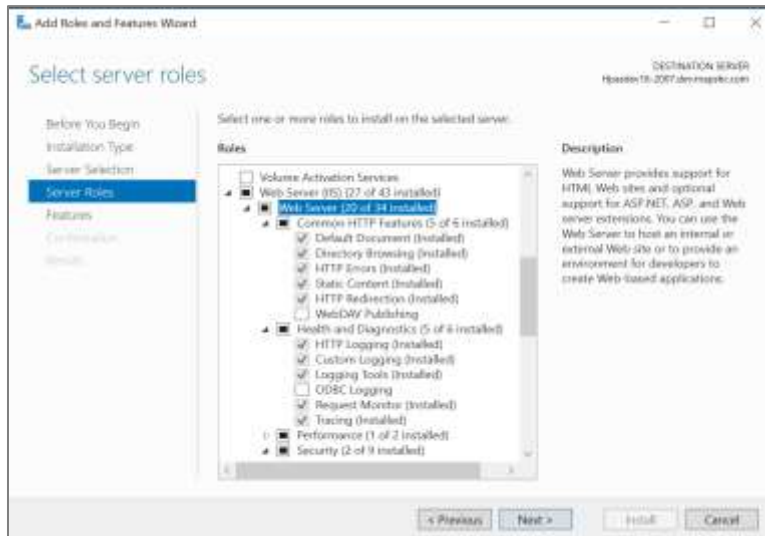


4. Choose **Select a server from the server pool** and click **Next** on the Select destination server dialog box.



5. Select **Web Server (IIS)** and other options listed below in the Roles area and click **Next** on the Selected server roles dialog box.

NOTE: Some options are selected (enabled) by default. There is no need to disable them.



From the...	Select...
Common HTTP Features list	• Static Content • Default Document • Directory Browsing • HTTP Errors • HTTP Redirection
Application Development list	• ASP.NET • NET Extensibility • ISAPI Extensions • ISAPI Filters • Server Side Includes
Health and Diagnostics list	• HTTP Logging • Logging Tools • Request Monitor • Tracing • Custom Logging
Security list	• Windows Authentication • Request Filtering
Performance list	• Static Content Compression
Management Tools list	• IIS Management Console • IIS 6 Management Compatibility
NOTE: Selecting IIS 6 Management Compatibility automatically selects IIS 6 Metabase Compatibility .	

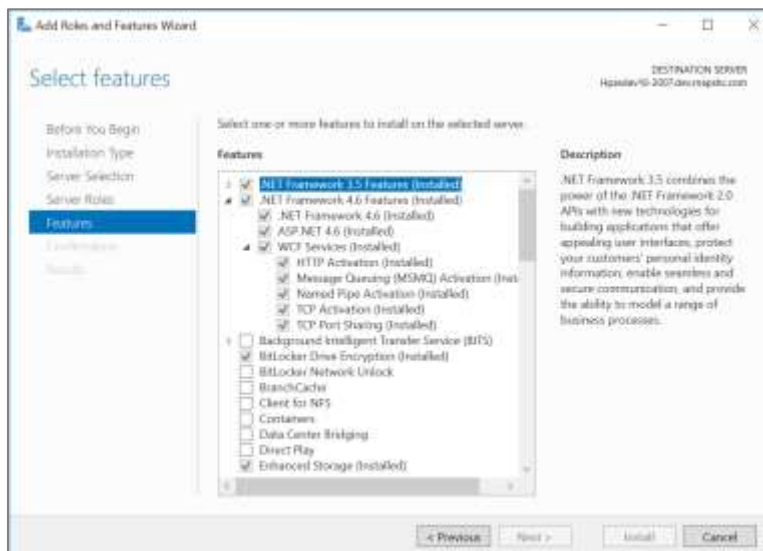
6. Proceed to [Installing Required Features](#) when the Select features dialog box appears.

Installing Required Features

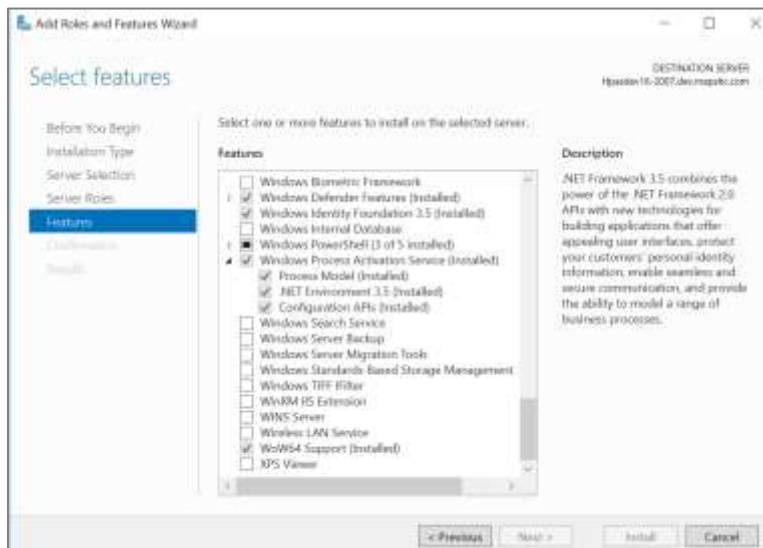
The following features must be configured:

- .NET Framework 4.6 Features
 - .NET Framework 4.6
 - ASP.NET 4.6
 - WCF Services
 - Http Activation
 - Message Queuing (MSMQ)
 - Named Pipe Activation
 - TCP Activation
 - TCP Port Sharing
- .NET Framework 3.5 Features

NOTE: Features in the dialog box are selected (enabled) by default in Windows Server 2019.



NOTE: .NET Framework 4.6 feature will be installed by default with Windows Server 16.



1. Ensure that **.Net Framework 4.6** and **Web Server(IIS) Support** are selected and click **Next**.

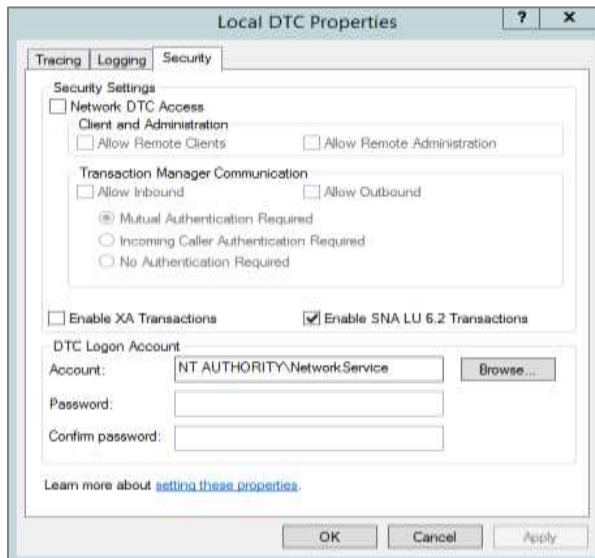
NOTE: If a system message appears, click **Add Required Role Services**.

2. When the Confirm Installation Selections dialog box appears, confirm your selections and click **Install**. After the installation is complete, the Installation Results dialog box appears.
3. Click **Close**.

Enabling Network DTC Access

To enable Network DTC Access:

1. Select **Start | All Programs | Administrative Tools | Component Services** to display the Component Services dialog box.
2. In the left pane, select **Component Services | Computers | My Computer | Distributed Transaction Coordinator**.
3. Right-click **Local DTC**, then click **Properties** to open the Local DTC Properties dialog box.



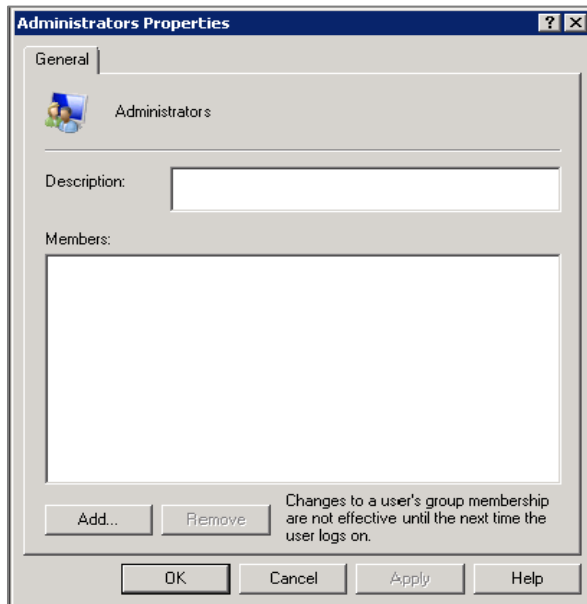
4. Complete the following actions on the Security tab.
 - 4.1 Select **Network DTC Access** in the Security Settings area.
 - 4.2 Select **Allow Inbound** and **Allow Outbound** in the Transaction Manager Communication area.
 - 4.3. Click **Apply**, then click **OK**.
5. Click **Yes** to system message that appears, if applicable.

Adding the Services Account to Administrators

To add the Services Account to the Administrators group:

1. Select **Start | All Programs | Administrative Tools | Computer Management** to display the Computer Management dialog box.
2. Expand **Local Users and Groups** in the left pane.

- Click **Groups**, then double-click **Administrators** to open the Administrators Properties dialog box.

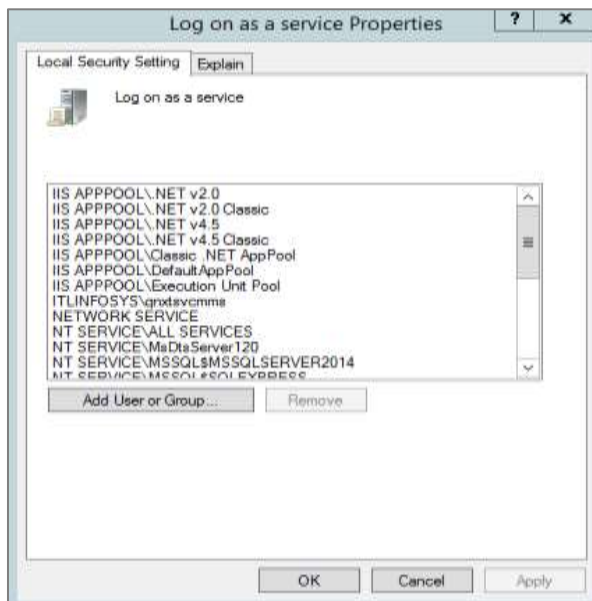


- Add or verify the service account and click **OK**.

Adding User Right Assignments

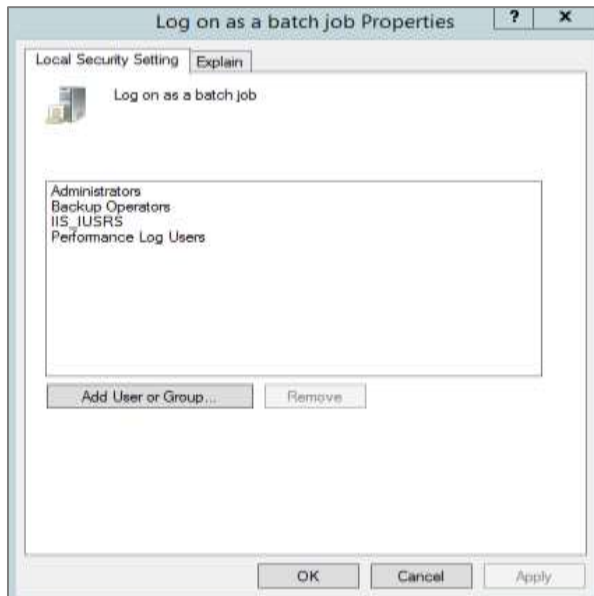
To add User Right Assignments:

- Select **Start | All Programs | Administrative Tools | Local Security Policy** to open the Local Security Policy dialog box.
- Expand **Local Policies** in the left pane and click **User Rights Assignment**.
- Double-click **Log on as service** in the Policy area to display the Log on as a service Properties dialog box.



- Add the name, then click **OK**.

5. Double-click **Log on as a batch job** in the Policy area to open the Log on as a batch job Properties dialog box.



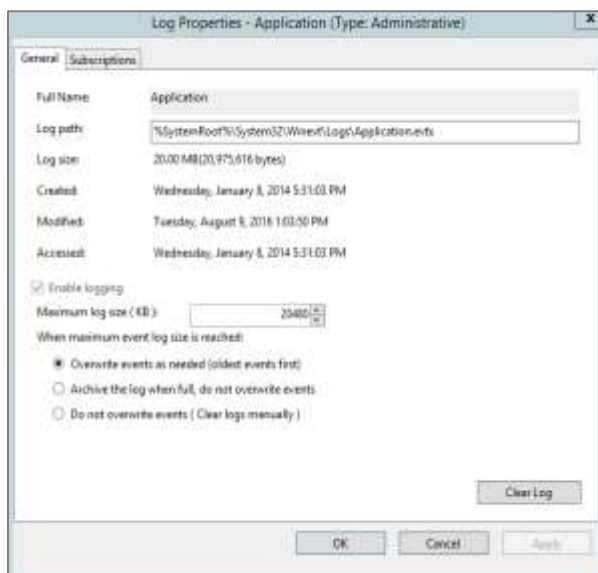
6. Add or verify the name as applicable, then click **OK**.

Configuring the Event Viewer

Event Viewer settings are required for all event logs (including Administrator - generated event logs) to support Administrator software processing requirements.

To configure the Event Viewer:

1. Select **Start | All Programs | Administrative Tools | Event Viewer** to open the Event Viewer dialog box.
2. Select **Windows Logs** in the left pane. Event logs appear in the center Windows Logs area.
3. Right-click the application event log, then select **Properties** to display the Log Properties - Application dialog box.



4. Select Overwrite events as needed (oldest events first), then click Apply and/or OK.
5. Repeat steps 3 and 4 for remaining event logs in the Windows Logs area:
 - Security
 - Setup
 - System

Setting User Account Controls

To allow HPA to write to the Program Files directory and parts of the registry, set the User Account Control settings as follows:

- **Windows Server 2019:** Select the lowest setting for the User Account Control.
- **Windows Server 2019:** Turn off the User Account Controls.

Both procedures are described below.

Adjusting Windows Server 2019 User Account Control Settings

To adjust the User Account Control settings for Windows Server 2019:

1. Select **Start | Control Panel**.
2. Select **User Accounts**.
3. Click **User Accounts** in the User Accounts area.
4. Click **Change User Account Control Settings** in the Make changes to your account area.
5. Click and drag the control to lowest setting in the Choose when to be notified about changes to your computer area, then click **OK**.

Turning Off Windows Server 2019 User Account Control Settings

To turn off User Account Control settings for Windows Server 2019:

1. Select **Start | Control Panel**.
2. Select **User Accounts**.
3. Click **User Accounts** in the User Accounts area.
4. Click **Change User Account Control Settings** in the Make changes to your account area.
5. Move the slider bar down to **Never notify selection** and click **OK**.
6. Reboot the machine for changes to take effect.

Creating a Desktop Folder

To avoid issues that could be encountered between the Windows Server® operating system and Microsoft Excel®, you must create a Desktop folder in the systemprofile directory.

To create a desktop folder in the systemprofile directory:

1. Right-click **Start** and select **Explore**.
2. In the Folders area select **Desktop | My Computer | Local Disk (C:) | WINDOWS | SysWOW64 | config | systemprofile** to navigate to the systemprofile directory.
3. Add a new folder and name it **Desktop**.

III SQL Server Installation and Configuration

This chapter includes the following sections:

- [SQL Server Requirements](#)
- [Installing SQL Server](#)
- [Installing Third-Party Servers](#)

SQL Server Requirements

You must add a local administrator account before installing Microsoft® SQL Server®. The local administrator can set up both the SQL Server and SQL Server Agent services.

If you are installing the Reporting Services component of SQL Server, see [SQL Reporting Services Installation and Configuration](#) before you begin the installation process.

Highest supported Microsoft® SQL Server® must be SQL Server 2022. SQL Server includes the following configuration considerations:

- SQL Server data and log files **should not be** on the local C:\ drive.
- SQL Database log files should be on a separate device from the data file.
- The Microsoft Windows swap file should be on a separate device.
- The physical location of database file groups should be determined by the hardware configuration of the machine.
- Data files should be resized according to the hardware specifications of the machine.
- The database `Tempdb` should be configured with multiple files:
 - One file for each CPU is a sufficient starting point.
 - All files must be the identical size.
 - Autogrow should be disabled.
- Use Log Shipping to replicate data for reporting purposes.
- Gainwell Technologies recommends a minimum supported SQL Server configuration. For more information, see [SQL Server Configuration](#).

Installing SQL Server

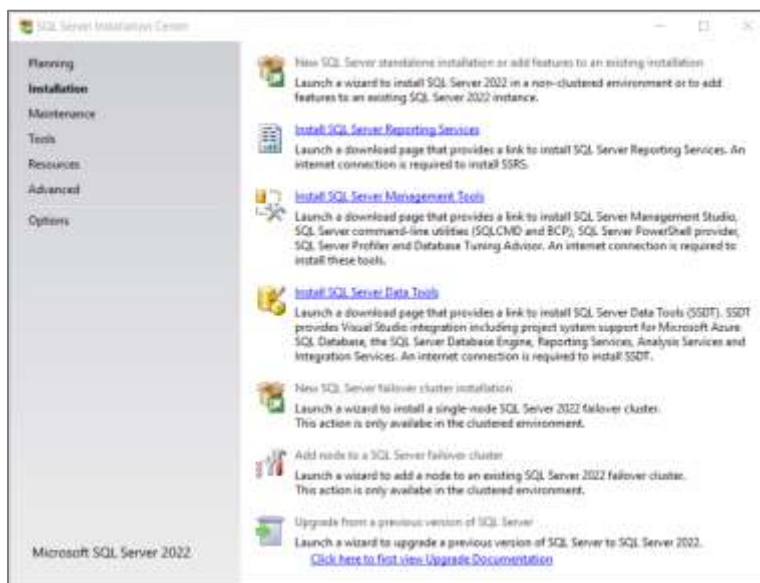
Information in this section describes how to install Microsoft® SQL Server®. Before beginning the installation, review [Important Prerequisite Notes](#).

To install SQL Server:

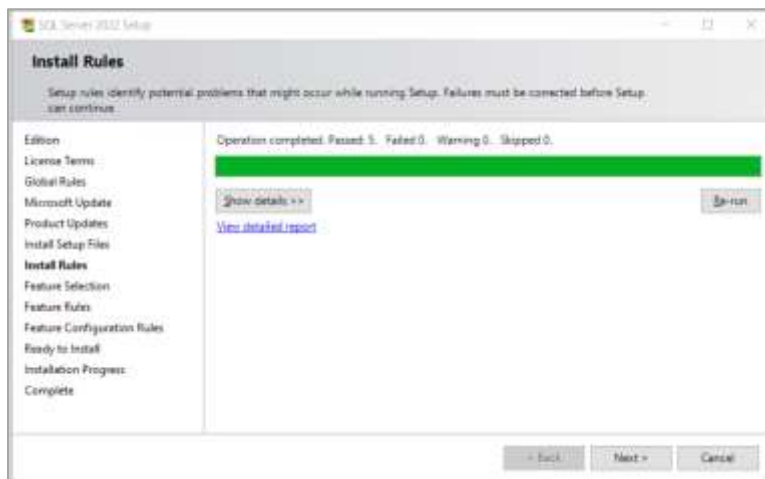
1. Access the SQL Server installation program.
2. Right-click the setup file, then click **Run as administrator**.

NOTE: If your Windows Server installation does not have the prerequisite version of the Microsoft .NET Framework installed, Microsoft Windows Installation version, or Windows Updates installed, the installer initiates an upgrade of these components automatically. Allow the updates, restart your server, and re-launch the SQL Server installation program.

3. Click **Installation** on the SQL Server Installation Center dialog box.

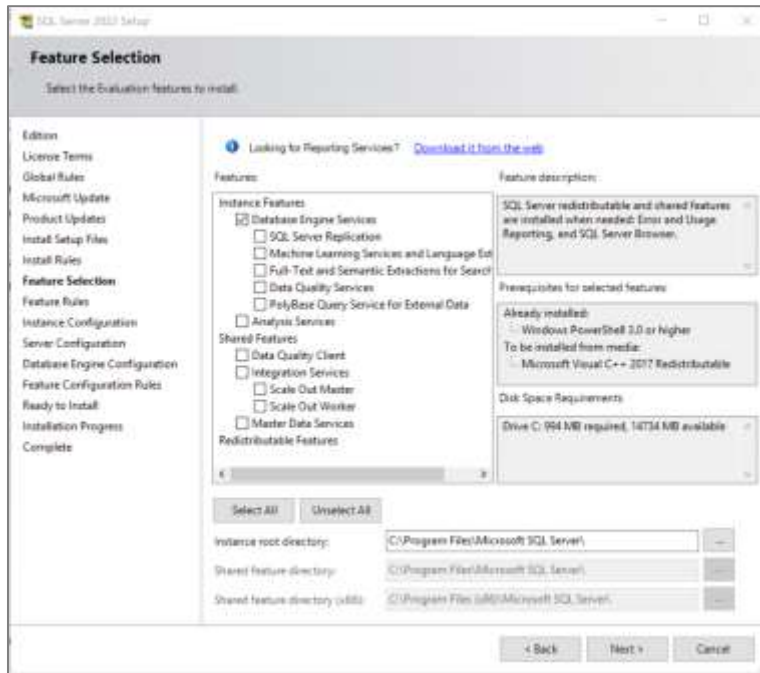


4. Select **New installation or add features to an existing installation** on the SQL Server Setup wizard dialog box. The Setup Support Install Rules dialog box appears and display progress.



5. Click **Show details** to verify that no support rule checks failed, then click **OK**.
6. Click **Install** on the Setup Support Files dialog box.

7. Click **Show details** to verify that no support rule checks failed and click **Next** to display the Install Type dialog box.
8. Select **New Install or add shared features** and click **Next** to display the Product Key dialog box.
9. Enter your product key and click **Next** to display the License Terms dialog box.
10. Select **I accept the license terms** and click **Next** to display the Setup Support Files dialog box.
11. Click **Next** to open the Setup Role dialog box.
12. Select **SQL Server Feature Installation** and click **Next** to open the Feature Selection dialog box.

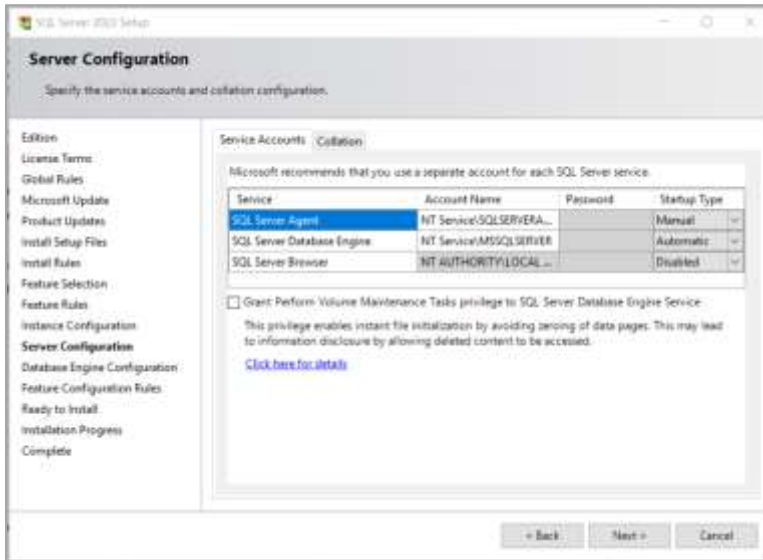


13. Complete the following action on the Feature Selection dialog box
 - 13.1 Select the following options from the Features area:
 - Instance Features: **Database Engine Services**
 - 13.2 If necessary, change the value in the Shared feature directory box. Otherwise, leave the default value.
 - 13.3 click **Next** to display the Installation Rules dialog box.
14. Click **Show details** to verify that no support rule checks failed and click **Next** to display the Instance Configuration dialog box.

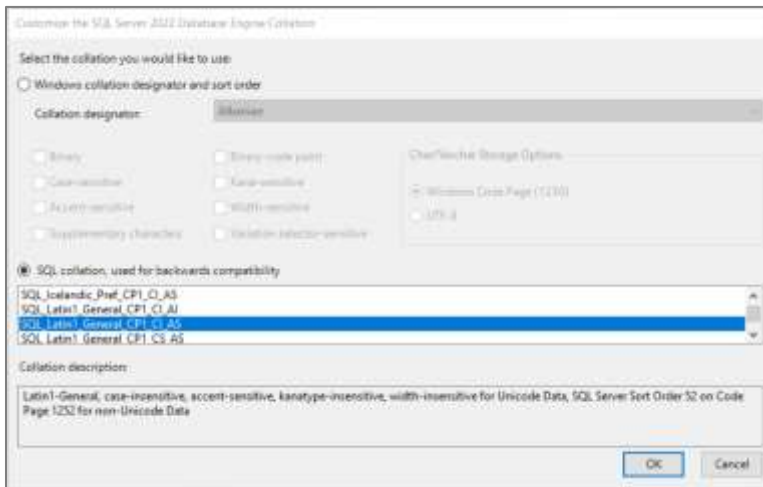
NOTE: You may see a number of rules with a status of **Not applicable**. This is expected.

15. Change the values in the Named instance, Instance ID, and Instance root directory fields if needed and click **Next** to display the Disk Space Requirements dialog box.

16. Verify information in the Disk Usage Summary area to ensure enough disk space is available on target drives and click **Next** to display the Server Configuration dialog box.



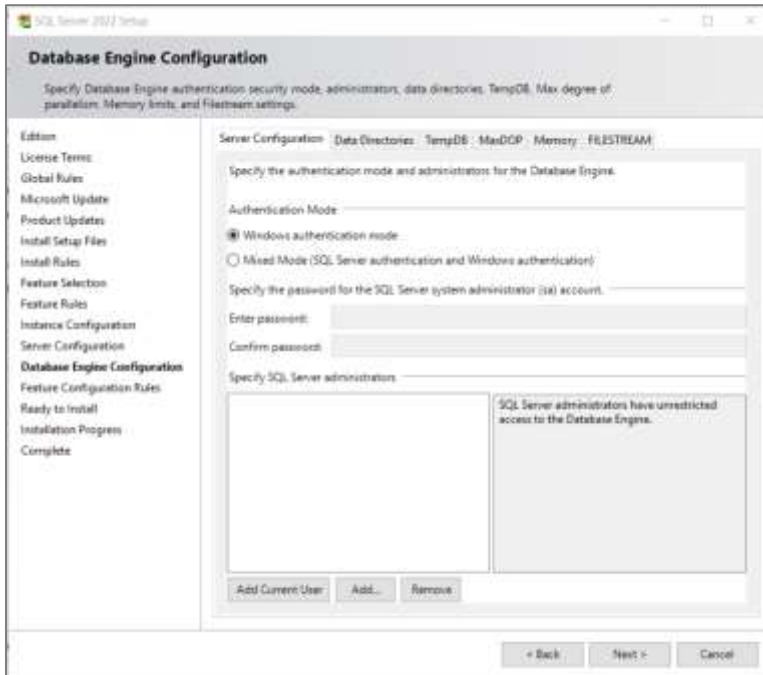
17. Enter the Account Name and Password for the user under which each SQL Server service should run on the Service Accounts tab.
18. Click **Customize** on the Collation tab to display the Customize the SQL Server Database Engine Collation dialog box and complete the following actions:



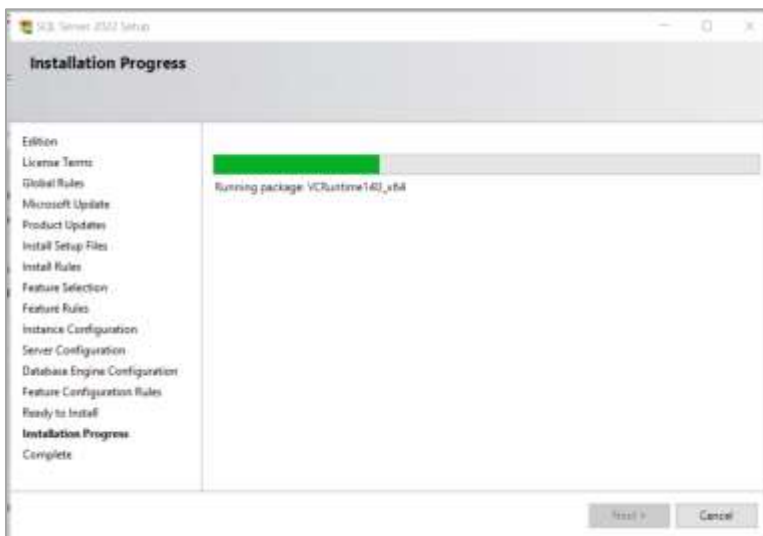
NOTE: HPA applications require the SQL Server to manage data in a certain order.

- 18.1 Select **SQL collation, used for backwards compatibility**.
- 18.2 Choose the **SQL_Latin1_General_Pref_CP1_CI_AS** collation value.
- 18.3 Click **OK** to close the dialog box.

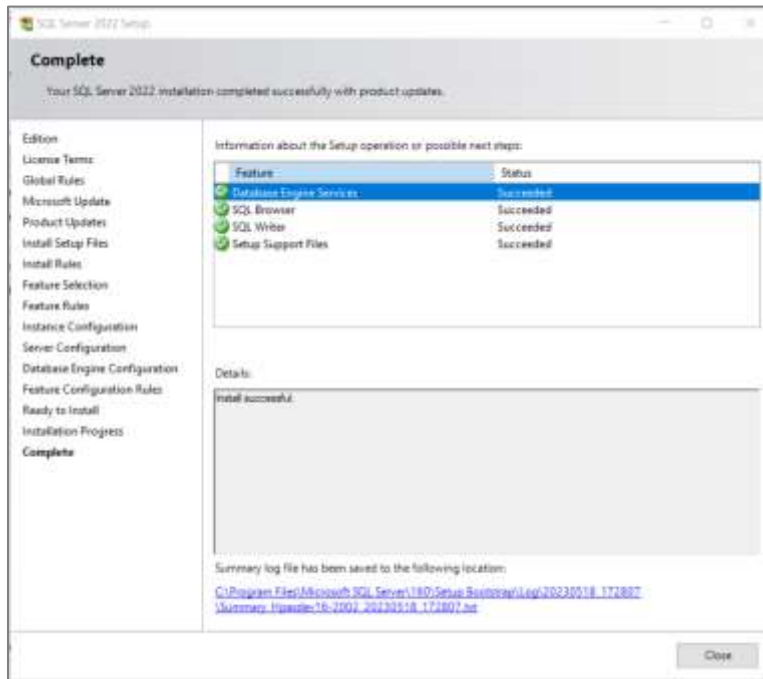
19. Click **Next** on the Server Configuration dialog box to open the Database Engine Configuration dialog box.



20. Configure applicable settings for your site on the Account Provisioning and Data Directories tabs and click **Next** to display the Error Reporting dialog box.
21. Select one an option if you want to send error reports and feature usage information to Microsoft and click **Next** to open the Installation Configuration Rules dialog box.
22. Verify that no installation configuration rule checks failed and click **Next** to open the Ready to Install dialog box.
23. Verify that detailed SQL Server installation configuration information in the Ready to install SQL Server area is correct and, if correct, click **Install**. The Installation Progress dialog box appears.



24. Click **OK** to any system message that appears and open the Complete dialog box.



25. Verify that information in the following locations is correct in the saved Summary log and click **Close**.

- Information about the Setup operation or possible next steps area
- Supplemental Information area

Installing Third-Party Servers

This section describes how to set up each of the SQL servers. The SQL server names from the graphic in [Minimum Supported Configuration](#) have been used in this section.

Installing the Ingenix iCES server

If you plan to use Install Ingenix® iCES™, install that product now. Refer to the Ingenix server installation guides provided with the Ingenix software for detailed installation instructions.

Install the following components:

- iCES Install Client Notes
- iCES Install Server Notes
- iCES Install WebServer Notes

NOTE: Installation of the Ingenix XML Translator version 1.0 or higher is required on each server where the HPA Core application is installed.

Installing the Impact Pro server

If you plan to use Impact Pro™, install that product now. Refer to the Impact Pro installation information provided with the Impact Pro software for detailed installation instructions.

Install the following components:

- Impact Pro Processing Engine Installation Guide.pdf
- Impact Pro Reporting System Installation Guide.pdf

IV Database Restoration and Update

This chapter includes the following sections:

- [Initiating a New Client Database](#)
- [Upgrading an Existing Client Database](#)
- [Automated Database Upgrade](#)

Initiating a New Client Database

This section describes how to use the Microsoft® SQL Server® Management Studio utility to restore the HPA databases.

Rather than create new databases from scratch, restore the HPAdatabase backup files shipped with each HPA major release to the SQL Servers.

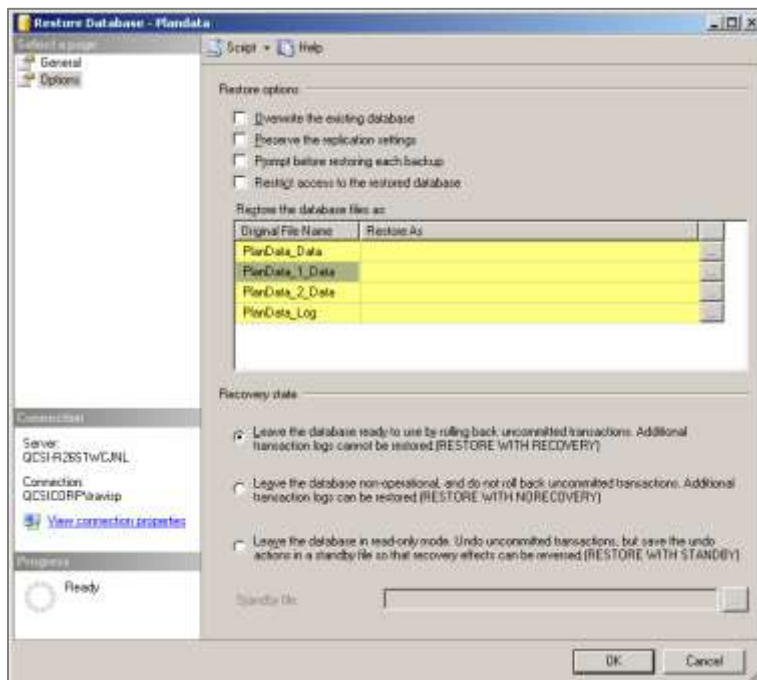
Restoring the HPA Databases

You must restore each of the following databases:

- MessageStore
- PlanData
- PlanDocument
- PlanEvent
- PlanIntegration
- QCSIDB
- HPA_System_Manager: You must restore this database on all servers where the preceding databases reside. For more information, see [Restoring the HPA System Manager Database](#).

To restore the HPA databases:

1. Create a folder for the database that you are restoring.
2. Select **Start | All Programs | Microsoft SQL Server | SQL Server Management Studio**.
3. Right-click **Databases** in the Object Explorer area, then click **Restore Database** to open the Restore Database dialog box.



4. Type the name in the To database field. (In this example, the **PlanData** database is restored.)

5. Select **From device** and click ... (ellipsis) and complete the following actions on the Specify Backup dialog box.
 - 5.1 Click **Add** to use the Locate Backup File dialog box to browse to the location of the .bak database file you wish to restore and select it.
 - 5.2 Click **OK**.
 - 5.3 Click **OK** in the Specify Backup dialog box to close.
6. Select a backup set in the Select the backup sets to restore area.
7. Select **Options** in the left pane.
9. Change the drive and folder name in the Restore As column to specify the folder created in [step 1](#) of this process. An example of the first entry is—C:\HPA Database Folders\PlanData\PlanData.mdf.
10. Click **OK** to start the restore. The restore progress can be tracked in the lower left area of the Restore Database dialog box. A message appears when completed.
11. Click **OK** to proceed.

Repeat [these steps](#) for each database (MessageStore, PlanDocument, PlanEvent, PlanIntegration, HPA_System_Manager).

Updating the Tempdb, Model, and QCSIDB Databases

After restoring each of the HPA databases, update the Tempdb, model, and QCSIDB databases with the Administrator user-defined data type.

To update the Tempdb, model, and QCSIDB databases:

1. Select **File | Open | File** in the SQL Server Management Studio menu bar.
2. Browse to the location of the New Client Only Scripts folder, select the HPA Create UDT.sql file, and click **Open** in the Open File dialog box.
3. Select **tempdb** from the database list.
4. Select **Query | Execute** (or click the Red exclamation point in the toolbar) to run the query.

Tip! If successful, "Command(s) completed successfully" displays in the Messages area.

5. Select **model** from the database list.
6. Select **Query | Execute** (or click the Red exclamation point in the toolbar) to run the query.

Tip! If successful, "Command(s) completed successfully" displays in the Messages area.

7. Select **File | Open | File**.
8. Select the Linked NT `Login.sql` file and click **Open** in the Open File area to display the query.
9. Select **QCSIDB** from the database list.
10. Select **Edit | Find**, then **Replace | Quick Replace** to open the Find and Replace dialog box and complete the following actions.
 - 10.1 Type `<DOMAIN\ACCOUNT>` in the Find what box.
 - 10.2 Type the username created in [Configuring the Required Service Account](#) in the Replace with box.
 - 10.3 Click **Replace All** and **OK** in the "occurrences were replaced" message box.
 - 10.4 Close the Find and Replace dialog box.
11. Select **Query | Execute** (or click the Red exclamation point in the toolbar) to run the query.

Updating the HPA Service Account Database Permissions

NOTE: The default HPA Service Account has limited access unless security permission is enabled, as described in this section.

You must update the HPA Service Account permissions in the following databases:

- MessageStore
- Model
- PlanData
- PlanDocument
- PlanEvent
- PlanIntegration
- QCSIDB
- Tempdb

To update the HPA Service Account database permissions:

Refer the following link: <https://gwproductengineering.jfrog.io/artifactory/mms-cms-cef-generic-prod/HPA%20Service%20account%20roles%20and%20permissions%20on%20SQL%20Server.zip> and execute **HPA ServiceAccount SQL Role and Grants** script in the database and refer READ ME file to verify the Windows Service account is assigned to database role of “QNXTServiceAccount”

SQL Encryption

SQL encryption is the process of encrypting data using a symmetric key or password. It is a valuable tool to ensure security by restricting access and allows decrypting using corresponding symmetric key or password.

SQL encryption requires Database Master Key (DMK) with a password to protect certificates and symmetric keys. The default password for the master key in the plandata schema backup database is listed below. The password should be unique for each client, so change the master key password after the database is restored. See Create database master key, a topic in the [Encrypt a Column of Data article](#), if needed.

Enable Support for Identical Keys Across Multiple Environments

Drop old and create a existing symmetric key with Identity_value.

```

/***** IDENTICAL KEY CONFIG *****/
USE ###YOURDATABASE###;
DROP SYMMETRIC KEY CA_ENCSYMKEY ;

CREATE SYMMETRIC KEY [CA_ENCSYMKEY] WITH
KEY_SOURCE = 'CA_ENCSYMKEY', --This value can never change, encrypted data loss
is eminent
ALGORITHM = AES_256,
IDENTITY_VALUE = 'CA_ENCSYMKEY' --This value can never change, encrypted data
loss is eminent
ENCRYPTION BY CERTIFICATE [CA_ENCCERT];
/***** SPRINT-139-START *****/

```

NOTE: Existing clients upgrading from Release 23 to Release 27 or 28 **must** run the R27_SQL Encryption And Decryption Tool.zip encryption conversion tool in JFrog if banking information has been stored in Administrator.

These clients should **not** run the Release 28 one-time update script (1_HPA 5.0.28.00 PlanData_OnetimeScript.sql) against their plandata db, as this was already completed for the Release 27 upgrade.

Work with the Engineering team when installing an upgrade.

Safe Keeping of Secrets

Use the storage vault to back up the master key (##MS_DatabaseMasterKey##).

Additional steps should be taken to backup the secrets using SQL operation. The backup file should be collected and stored in the same vault.

NOTE: When backing up secrets from the SQL server, a backup operation password is needed to encrypt the backup file. This backup operation password “ENCRYPTION BY PASSWORD” should be safeguarded as well because it is needed when restoring secrets.

```
--Backup master key for symmetric encryption
/***** BACKUP SECRETS *****/
USE ###YOURDATABASE##;
OPEN MASTER KEY DECRYPTION BY PASSWORD = 'WhpSZmNmyJyKdzUm3uJe';    ---Must have
password to backup master key

BACKUP MASTER KEY TO FILE = 'c:\temp\exportedmasterkey'    -- location on the SQL server
    ENCRYPTION BY PASSWORD = 'sd092735kjn$&adsg';    --Encryption password should be
    different from master key
GO
/***** BACKUP SECRETS *****/

--Restore master key for symetric encryption
/***** RESTORE SECRETS *****/

USE ###YOURDATABASE##;
GO
RESTORE MASTER KEY
    FROM FILE = 'c:\temp\exportedmasterkey'
    DECRYPTION BY PASSWORD = 'sd092735kjn$&adsg'    -- NOTE: the value is the
    "Encryption By Password" from backup operation above
    ENCRYPTION BY PASSWORD = 'WhpSZmNmyJyKdzUm3uJe';
GO
/***** RESTORE SECRETS *****/
```

Updating Report Views

To update Report Views:

1. Create linked servers as needed.

NOTE: A linked server is a way of authorizing another SQL server to run queries against your databases. For more information, refer to the [Microsoft Library Linked Servers page](#).

2. Determine if your QCSIDB database and PlanData database are on the same SQL server for each PlanData database (for example, production, test, or reporting).

NOTE: If the QCSIDB database is on a different SQL server, use the `ViewLinkedServer.sql` script provided to create a linked server. The script must be run on the SQL server where the PlanData database resides because the script links the other server to the PlanData SQL server.

3. Edit the `ReportViews.sql` script provided to alter the following views:
 - RptQRoleVw
 - RptQUserVw
4. Change the path to the table based on the servername.databasename.owner.tablename format.

Tip! Specific instructions and examples are included in the script.

- Run the updated `ReportViews.sql` script on the PlanData database.

Managing Database Properties

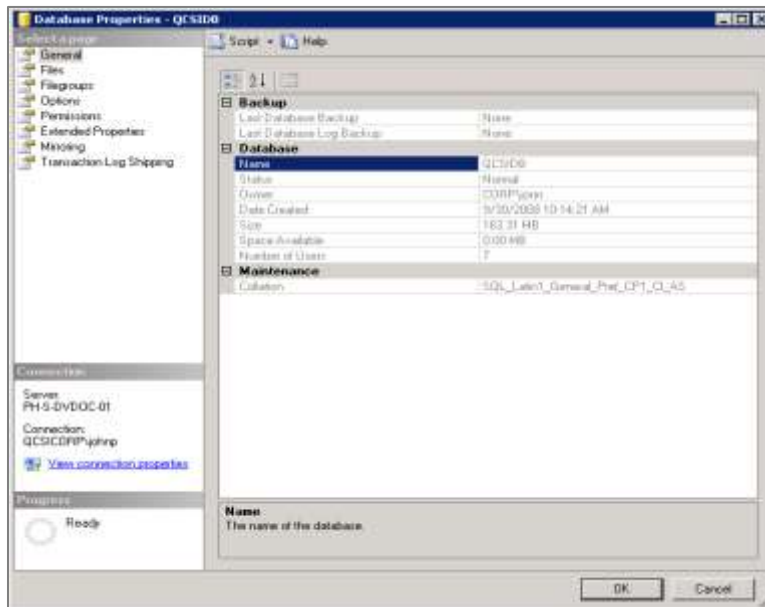
The last configuration operation you must do for each database is to manage the database properties. For each database, you must do the following:

- Resize the database
- Configure the default file group for each database
- Confirm the selected options

Updating QCSIDB Properties

To update the QCSIDB properties:

- Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
- Right-click the applicable QCSIDB database and select **Properties** to open the Database Properties dialog box.



- Select **Files** in the Select a page area and change the Initial Size (MB) column value for qcsidb_Data to at least 200 MB in the Database files area.



- Select **Filegroups** to verify there is only one file group.
- Select **Options**, then **Full** in the Recovery Model list to set options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced
Recovery area	Page Verify to CHECKSUM

6. Click **OK** to save the changes.

Updating PlanData Properties

To update the PlanData properties:

1. Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
2. Right-click the name of your PlanData database and select **Properties**. The Database Properties dialog box appears.
3. Click **Files** in the Select a page area.
4. Resize files as needed referring to the following recommended settings:
 - 1000 MB for the Plandata_Data file
 - 5000 MB for Plandata_1_Data file
 - 5000 MB for the Plandata_2_Data file and all remaining files in the PlanData file group
5. Select **Filegroups**.

Tip! The PlanData database has multiple file groups. PlanData file groups should be the default file group.

6. Select **Options**, then **Full** in the Recovery Model list to set options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced
Recovery area	Page Verify to CHECKSUM

7. Click **OK** to save the changes.

Updating PlanDocument Properties

To update PlanDocument properties:

1. Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
2. Right-click the name of your PlanDocument database and select **Properties**. The Database Properties dialog box appears.
3. Click **Files** in the Select a page area.
4. Resize files as needed.

NOTE: Gainwell Technologies recommends a setting of at least 200 MB for Plandocument_Data1 file.

5. Select **Filegroups**.

Tip! The PLANDOC file group is selected by default.

6. Select **Options**, then **Full** in the Recovery Model list to set options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced

7. Click **OK** to save the changes.

Updating Plan Integration Properties

To update PlanIntegration properties:

1. Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
2. Right-click the name of your PlanIntegration database and select **Properties**. The Database Properties dialog box appears.
3. Click **Files** in the Select a page area.
4. Resize files as needed.

NOTE: Gainwell Technologies recommends a setting of at least 200 MB for planintegration_data file and 10000 MB for the planintegration_data2 file.

5. Select Filegroups.

Tip! The PROCESSLOG file group should be selected by default.

6. Select **Options**, then **Full** in the Recovery Model list to set options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced
Recovery area	• Page Verify to CHECKSUM

7. Click **OK** to save the changes.

Updating PlanEvent Properties

To update PlanEvent properties:

1. Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
2. Right-click the name of your PlanEvent database and select **Properties**. The Database Properties dialog box appears.
3. Click **Files** from the Select a page area.
4. Resize files as needed.

NOTE: Gainwell Technologies recommends a setting of at least 200 MB for aspnetdb file and 10000 MB for the aspnetdb_log file.

5. Select **Options**, then **Full** in the Recovery Model list to set options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced
Recovery area	• Page Verify to CHECKSUM

6. Click **OK** to save the changes.

Updating MessageStore Properties

To update MessageStore properties:

1. Expand the Databases file in the Object Explorer area of SQL Server Management Studio.
2. Right-click the name of your MessageStore database and select **Properties**. The Database Properties dialog box appears.
3. Click **Files** in the Select a page area.
4. Resize files as needed.

NOTE: Gainwell Technologies recommends a setting of at least 500 MB for the `ers_message_store_data` file and at least 200 MB for the `ers_message_store_Log` file.

5. Select Filegroups.

Tip! The MessageStore database has only one file group, so PRIMARY is the default file group.

6. Select **Options**, then **Simple** in the Recovery Model list to select options listed in the following table.

In the...	Set...
Automatic area	• Auto Create Statistics to True • Auto Update Statistics to True • Auto Update Statistics Asynchronously to True
Miscellaneous area	• Arithmetic Abort Enabled to True • Parameterization to Forced
Recovery area	• Page Verify to CHECKSUM

7. Click **OK** to save the changes.

Upgrading an Existing Client Database

NOTE: This procedure must be performed before using the HPA Automated Database Upgrade utility to upgrade client databases to this release of HPA.

Before beginning the upgrade process, you must do the following:

1. Make sure the HPA Service Account database security permissions are configured as described in [Updating the HPA Service Account Database Permissions](#).
2. Shut down all HPA services running on any HPA application servers associated with the application environment being upgraded. These services include:
 - QCSI Master Service
 - QCSI Orphan Retry Service
 - HPA Connect Inbound Receiver Service
 - HPA QEnterprise Distribution Hub
 - HPA QEnterprise Execution Unit
 - HPA Web Application
 - HPA View Business Services

Migrating a HPA Database to a Newer Version of SQL Server

There are two main options for migrating a HPA application database to a newer version of SQL Server. You can detach the database from the current SQL Server version and attach it to the newer SQL Server version or backup the current database, and restore it to a newer SQL Server version.

Choosing between the two options depends on the upgrade path, hardware available, and recovery considerations. Use the following general rules deciding which method to use:

- Detach method
 - Use when you plan to uninstall an earlier SQL Server version and install a newer SQL Server version in place, upgrade the SQL Server version in place, or when physically moving the database off of the earlier SQL Server version to the newer SQL Server version and there is no need to retain a copy of the database on the older SQL Server version.
 - Detach method is faster, as restore time is not incurred.
 - Less server and disk resources are needed.
- Backup/Restore method
 - Use this method when you want to keep a copy of the database on the older version SQL server instance as a fallback server.
 - Slower than the Detach method, because of the copy and restore time on the database.
 - More server and disk resources are needed.

Detach Process Overview

The following is a detach process overview:

1. Back up the databases to be migrated.
2. Use the `sp_detach_db` command to disconnect the databases from the current server.
3. If uninstalling or upgrading the SQL Server version in place, stop the SQL Server Service for the version to be upgraded.
4. If uninstalling the current SQL Server version uninstall the old SQL Server version and install the new SQL Server version.
5. If upgrading SQL Server versions in place, upgrade the SQL Server instance to the new SQL version.
6. Attach the HPA databases using the `sp_attach_db` command.

Backup/Restore Process Overview

The following is a back up and restore process overview:

1. Back up the databases to be migrated.
2. Copy the database backup files to a location accessible by the new SQL Server instance.
3. Restore the database backups using the normal restore method.

After the databases have been relocated to the new SQL Server version, do the following for each database:

1. Set the database compatibility level to the value associated with the new SQL Server version.
2. In the database properties, confirm the database Page Verify setting is **CHECKSUM** for each database.
3. Confirm the Auto Create Statistics database property is **True**.
4. Confirm the Auto Update Statistics database property is **True**.
5. Confirm the Auto Update Statistics Asynchronously setting is **True**.
6. Rebuild all the indexes.
7. If the databases were moved and an HPA version upgrade was not performed, all stored procedures should be recompiled using `sp_recompile` and all views refreshed using `sp_refreshview`.

See [Managing Database Properties](#) for instructions on setting database properties.

Updating QCSIDB DBAlias Table Information

Regardless of the method used to upgrade an HPA application database from one SQL Server version to another, make sure the correct Server, QDatabase, and DbPlatformType values are associated with the database in any DBAlias records in the QCSIDB database pointing to the databases on the old server.

To update the DbPlatformType:

1. Using Microsoft SQL Server Management Studio or a similar SQL tool, use the following SQL command to update the DbAlias table information for the database(s) that have been upgraded:

```
UPDATE dbo.dbAlias
SET Server = '<newSQLservername>',
-- if the database name changed uncomment the following line:
--QDatabase = '<newdatabasename>', DbPlatformType = <value>
WHERE Server = '<oldSQLservername>'
-- always compare with the old database name value
--even if the database name did not change
AND QDatabase = '<olddatabasename>'
```

2. Replace values enclosed in angle brackets (<>) with the applicable values for your site.
3. Replace `<newSQLservername>` with the server name or the server name\instance name of the SQL Server instance that holds the upgraded database. If the database name was changed during the upgrade process, replace `<newdatabasename>` with the new name of the database and uncomment the line indicated in the UPDATE command.
4. Replace `<value>` with one of the following DbPlatformType values:
 - If the database is now on an SQL Server 2005 instance, replace `<value>` with **64**.
 - If the database is now on an SQL Server 2008 or above instance, replace `<value>` with **128**.
5. To ensure that only the DbAlias records specific to the upgraded database are changed, make sure to replace the following values with correct values:
 - Replace `<oldSQLservername>` with the server name or server name\instance name of the SQL Server that housed the database prior to being upgraded.
 - Replace `<olddatabasename>` with the name (or prior name if the name of the database was changed during the upgrade process) of the database upgraded.

Updating Database Properties

Verify all HPA database properties are set to match the properties and options.

Updating Tempdb and the SQL Server Model Database

Make sure the latest HPA data types are defined in the Tempdb and Model databases. See [Updating the Tempdb, Model, and QCSIDB Databases](#).

Manually Upgrading the PlanEvent Database for HPA 5.0

Before upgrading the PlanEvent database, make sure that you have performed the required manual upgrade procedures for the HPA version from which you are upgrading. See [HPA Upgrades](#).

In HPA 5.0, the PlanEvent database configuration must be expanded to accommodate new service transaction tables.

Preparing the PlanEvent Database for a Manual Upgrade

To prepare the database for upgrade, a manually executed SQL script file must be customized and executed.

To prepare all HPA PlanEvent databases for upgrade to HPA 5.0 prior to running the DBUpgrade Wizard:

1. Select **File | Open | File** in SQL Server Management Studio to open the Open File dialog box.
2. Browse to and select the `HPA <version> Update PlanEvent.sql` file.
3. Click **Open** to display the query in the right pane.
4. Select the applicable PlanEvent database from the database drop-down list.
5. Select **Edit | Find and Replace | Quick Replace** to open the Find and Replace dialog box.
6. Type `$$$$` in the Find what box.
7. Type the name of the applicable PlanEvent database in the Replace with box.
8. Click **Replace All**.
9. Click **OK** in the system message dialog box.
10. Type `^^^^` in the Find what box.
11. Type the device\file name prefix to use for additional PlanEvent physical database files to be created during the database upgrade in the Replace with box. For example, `D:\HPA DB Files\PlanEvent`.
12. Click **Replace All**.
13. Click **OK** in the system message dialog box and close the Find and Replace dialog box. The updates are reflected in the query text.
14. Select **Query | Execute** (or click the red exclamation point in the toolbar) to execute the query.

Data Conversion

This section describes how HPA data is converted during the upgrade to HPA 5.0.

Member Medicare Attribute Conversion

During the PlanData database upgrade to HPA 5.0, HPA converts Member Medicare Attribute information from the MemberAttribute table into the new Member CMS tables. All data converted is left unmodified and is retained in its original form in the MemberAttribute table.

Any error, warning, or informational messages generated during the data conversion is stored in the `conv_Q<version>CvtMemberAttrLog_yyyymmdd_hhmmss` conversion table. The conversion table is in the PlanData database.

This data conversion reads specific MemberAttribute information from Attributes in the `AttributeGroup = GRP201` and converts data into the MemberCmsFactor table.

The following general rules are employed during the conversion of MemberAttribute data:

- It is possible that multiple MemberAttribute records, associated with one `Attributeld`, can be valid for the same effective period. In this case only one MemberAttribute record is converted. The HPA database upgrade selects a record using the following sort order:
 - highest effective date value
 - highest termination date value
 - highest last update value
 - lowest attribute value
- The first record found in the sort order above is converted and all other records are ignored.
- New MemberCmsFactor records are only written for a Member when that Member has at least one MemberAttribute record for the set of `Attributeld` values associated with the new table.
- When the source MemberAttribute values contain character data values that are too long to fit into the destination column, the values are truncated before they are stored in the MemberCmsFactor table.
- MemberAttribute records with a blank in TheValue column are not converted for `attributeId = CMSMED0012`.
- MemberAttribute records with a blank in the Value column are converted to **000** for `attributeId = CMSMED0048`.

The following tables describe the source and destination locations for each of the MemberCmsFactor table records populated during this conversion.

MemberCmsFactor Table

Column Name	Attributeld	Notes
MemId		Convert data from the MemberAttribute.MemId column.
Factor	Varies based on Factor	See the Attribute Mapping Grid Table below.
EffDate		Convert data from the MemberAttribute.EffDate column.
TermDate		Convert data from the MemberAttribute.TermDate column.
Value		Convert data from the MemberAttribute.TheValue column.

Attribute Mapping Grid Table

Factor	Factor Description	Attributeld	Attribute Description	Notes
ESO	Employer Subsidy Override	CMSMED0012	Employer Subsidy Override	A blank value in the MemberAttribute.TheValue column is not converted.
LISL	Low Income Subsidy Level	CMSMED0048	Part D Subsidy Level	Convert if value is in Qualifier table where QualifierType is CMS_MEM_LISL. A blank value in the MemberAttribute.TheValue column is automatically converted to 000.

MemberCms Rx Data Conversion

During the PlanData database upgrade to HPA 5.0, HPA automatically converts Rx information contained in the MemberCms table into the new MemberCmsRx table. At the end of the conversion process the data stored in the converted columns is removed from the MemberCms table.

Before the data conversion, the database upgrade process copies the MemberCms data into the `conv_Q<version>_MemberCms_yyyymmdd_hhmmss` table. This table is located in the PlanData database.

The data conversion reads all records in the MemberCms table with a non-blank Rxld value and copies the Memld, RxBin, RxPcn, RxGroup and Rxld column data to the corresponding columns in the new MemberCmsRx table. All new records written to the MemberCmsRx table have the Status column set to **P**, EffDate column set to **01/01/1980**, and TermDate column set to **12/31/2078**.

After the data is copied to the MemberCmsRx table the RxBin, RxPcn, RxGroup, and Rxld columns are removed from the MemberCms table.

When the HPA 5.0 PlanData database upgrade is complete, verify that the MemberCms Rx data was correctly converted into the MemberCmsRx table. After the data has been verified the `conv_Q<version>_MemberCms_yyyymmdd_hhmmss` table can be deleted from the PlanData database.

MemberCms Data Conversion

During the PlanData database upgrade to HPA 5.0, HPA automatically converts Medicare Late Enrollment Penalty and Low Income Subsidy Amount data contained in the MemberCms table into the MemberCmsFactor table. When the conversion process is complete, the data stored in the converted columns is removed from the MemberCms table.

Before the data conversion, the database upgrade process copies the MemberCms data into the `conv_Q<version>_MemberCms_yyyymmdd_hhmmss` table. This table is located in the PlanData database. This copy process is performed at the same time the MemberCmsRx copy process is performed to minimize the number of times the MemberCms table must be read.

This data conversion reads all records in the MemberCms table with a non-NULL value in the Lis or Lep columns. All new records written to the MemberCmsFactor table have the EffDate column set to **01/01/1980** and TermDate column set to **12/31/2078**. Any data that cannot be converted into MemberCmsFactor records include a message written to the `conv_Q<version>CvtMemberCmsLog_yyyymmdd_hhmmss` table.

The following tables describe the source and destination locations for each of the MemberCmsFactor table records populated during this conversion.

MemberCmsFactor Table

Column Name	Notes
MemId	Convert data from the MemberCms.MemId column.
Factor	Either LIS or LEP depending upon which MemberCms table column the source data was copied from. See the Attribute Mapping Grid Table below.
EffDate	Set to 01/01/1980.
TermDate	Set to 12/31/2078.
Value	Convert data from the MemberCms table column (Lis or Lep). See the Attribute Mapping Grid Table below.

Attribute Mapping Grid Table

Factor	Factor Description	MemberCms Table Column	Notes
LIS	Low Income Subsidy Amount	Lis	All non-NULL values are converted.
LEP	Late Enrollment Penalty	Lep	All non-NULL values are converted.

After copying the data to the MemberCmsFactor table the Lis and Lep columns are removed from the MemberCms table.

When the HPA 5.0 PlanData database upgrade is complete, verify that the MemberCms Lis and Lep data was correctly converted into the MemberCmsFactor table. After verifying the data, the following tables can be deleted from the PlanData database:

- `conv_Q<version>_MemberCms_yyyymmdd_hhmmss`
- `conv_Q<version>CvtMemberCmsLog_yyyymmdd_hhmmss`

MemberCmsEnroll Data Conversion

As part of the PlanData database upgrade to HPA 5.0, HPA automatically converts Premium Withhold Option data in the MemberCmsEnroll table into the MemberCmsFactor table. When the conversion process is complete, data stored in the converted columns is removed from the MemberCmsEnroll table.

Before the data conversion, the database upgrade process copies all of the MemberCmsEnroll data into a table named `conv_Q<version>_MemberCmsEnroll_yyyymmdd_hhmmss`. This table is in the PlanData database.

The data conversion reads all records in the MemberCmsEnroll table with a value in the PremiumWithhold column and that are associated with an EnrollKeys Internal enrollment segment record. All new records written to the MemberCmsFactor table have the EffDate and TermDate columns set to the values found in the associated EnrollKeys table record. Any data that cannot be converted into MemberCmsFactor records include a message written to a conversion log table named `conv_Q<version>CvtMemberCmsEnrollLog_yyyymmdd_hhmmss`.

The following table provides the source and destination locations for the MemberCmsFactor table records populated during this conversion.

MemberCmsFactor Table

Column Name	Notes
MemId	Convert data from the MemberCmsEnroll.MemId column.
Factor	Set to PWO .
EffDate	Convert data from the EnrollKeys. EffDate column associated with the MemberCmsEnroll record.
TermDate	Convert data from the EnrollKeys. TermDate column associated with the MemberCmsEnroll record.
Value	Convert data from the MemberCmsEnroll. PremiumWithhold column.

After the data is copied to the MemberCmsFactor table, the PremiumWithhold column is removed from the MemberCmsEnroll table.

When the HPA 5.0 PlanData database upgrade is complete, verify that MemberCmsEnroll PremiumWithhold data was correctly converted into the MemberCmsFactor table.

After data has been verified, the following tables can be deleted from the PlanData database:

- `conv_Q<version>_MemberCmsEnroll_YYYYMMDD_HHMMSS`
- `conv_Q<version>CvtMemberCmsEnrollLog_YYYYMMDD_HHMMSS`

MemberCmsFactor Data Conversion

The PlanData database upgrade to HPA 5.0 automatically converts Medicaid ID information in the MemberCmsFactor table into the new MemberMedicaid table. When the conversion process is complete, data stored in converted MemberCmsFactor records is removed from the MemberCms table.

Before the data conversion, the database upgrade process copies all of the MemberCmsFactor records with Factor column value **MCAID** into a table named

`conv_Q<version>_MemberCmsFactor_YYYYMMDD_HHMMSS`. This table is in the PlanData database.

The data conversion process reads all records in the MemberCmsFactor table with Factor column value **MCAID** and copies the MemId, Value, EffDate, and TermDate column data to MemId, MedicaidId, EffDate, and TermDate columns in the new MemberMedicaid table.

When the HPA 5.0 PlanData database upgrade is complete, verify that Member Medicaid ID data was correctly converted from the MemberCmsFactor table into the new MemberMedicaid table. Once the data has been verified, the `conv_Q<version>CvtMemberCmsFactorLog_YYYYMMDD_HHMMSS` table can be deleted from the PlanData database.

Updating ACom3 Status

Use Microsoft SQL Server Management Studio or a similar SQL tool to execute the `acom3status_update.sql` database scripts on the designated database.

NOTE: This script should be run only if you are exporting ACom3 payments.

During the initial run of the ACom3 Payment Export, any invoice records that exist in the version of HPA that you are upgrading from that have not been previously sent to ACom3 are exported to ACom3 unless you select not to process them.

Gainwell Technologies has included a conversion script (`acom3status_update.sql`) that updates the status of all existing invoice records. If the ACom3 status is blank and the invoice status is PAID, the conversion script updates the status of these payments to SENT. Because the records have a status of SENT, the application does not consider these records the first time you run an ACom3 Payment Export.

Updating Report Views

After upgrading the database, update Report Views. To update Report Views:

1. Create linked servers, as needed.

For each PlanData database (production, test, reporting, etc.), determine if your QCSIDB database and PlanData database are on the same SQL server. If QCSIDB database is on a different SQL server, use the `ViewLinkedServer.sql` script provided to create a linked server.

The script must be run on the SQL server where the PlanData database resides because the script links the other server to the PlanData SQL server.

NOTE: A linked server is a way of authorizing another SQL server to run queries against your databases.

2. Edit the `ReportViews.sql` script provided to alter the following views:
 - RptQRoleVw
 - RptQUserVw
3. Change the path to the table based on the `servername.databasename.owner.tablename` format. Specific instructions and examples are included in the script.
4. Run the updated `ReportViews.sql` script on the PlanData database.

Automated Database Upgrade

To run the DBUpgrade wizard successfully and automatically upgrade the databases, the following prerequisites must be satisfied:

- The database must have been previously updated to HPA 3.4.
- The HPA_System_Manager database must reside on all servers that house databases to be upgraded **and must be** at the current version for the upgrade.
- The spq_ExecUpgrade procedure must reside in each database to be upgraded.
- You should have Service Account equivalent rights on the server and be DBO in each DB.
- Run the 07 HPA 5.0 `Create_SPQ_ExecUpgrade.sql` script against all databases being upgraded using the Automated Database Upgrade tool.

Restoring the HPA_System_Manager Database

The automated database upgrade process is designed to upgrade the PlanData, QCSIDB, PlanIntegration, and PlanDocument databases. Before you run the DBUpgrade wizard, a copy of the HPA_System_Manager database must be restored on each server where the PlanData, QCSIDB, PlanIntegration, PlanEvent, and PlanDocument databases reside (see [Restoring the HPA Databases](#) for information on how to restore databases). The database must be named HPA_System_Manager on each server.

For example, if the PlanData and QCSIDB databases reside on the Production1 server, and the PlanIntegration and PlanDocument databases reside on the Production2 server, you would restore two copies of the HPA_System_Manager database. One copy must be restored on the Production1 server, and another on the Production2 server. The copy located on each server manages upgrades for individual databases on that server.

Once you have copies of HPA_System_Manager on all servers where databases to be upgraded reside, the HPA_System_Manager is included in future schema and reference data updates, and should not need to be recreated.

Using the DBUpgrade Wizard

This section describes how to update the databases on your server. The database upgrade is delivered in one executable (.exe) file and is installed using an InstallShield Wizard.

Creating the Stored Procedure

Prior to first use of the DBUpgrade Wizard, the stored procedure SPQ_ExecUpgrade must be created in each database to be upgraded. Run the 07 HPA <version> `Create_SPQ_ExecUpgrade.sql` script in each database to be upgraded to create the stored procedure.

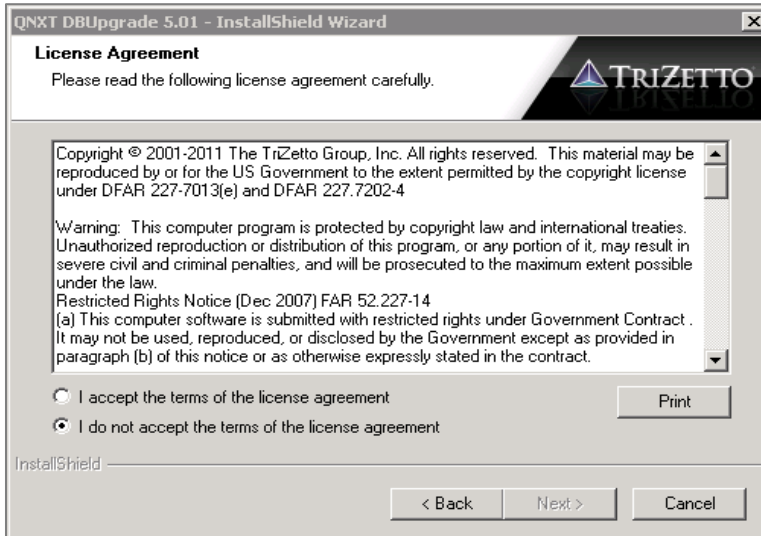
To run the script:

1. Right-click the database to be upgraded and select **New Query** in SQL Server Management Studio.
2. Select **Open – file** in the HPA installation package from the main menu.
3. Open the script from the folder location and click **Execute**.

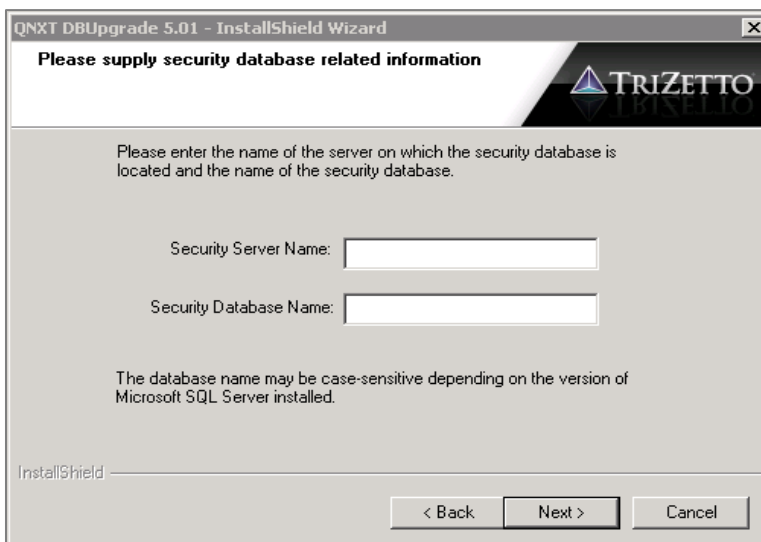
Running the Wizard

To run the DBUpgrade Wizard:

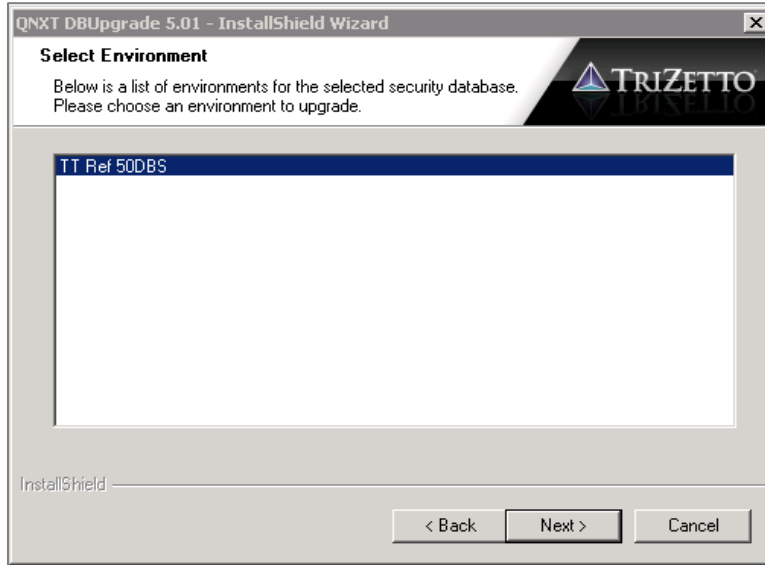
1. Browse to the HPA Installation Package and select the `HPA_DBUpgrade 5.0.exe` installation file.
2. Right-click the setup file and click **Run as administrator**. The InstallShield Wizard starts and the Welcome page appears.
3. Click **Next** to display the License Agreement page.



4. Review the license agreement, select **I accept the terms of the license agreement**, then click **Next**.
5. Complete the following action on the next page.
 - 5.1. Type the name of the SQL Server on which the security database resides as the Security Server Name.
 - 5.2. Type the name of the security database as the Security Database Name.
 - 5.3. Click **Next**.

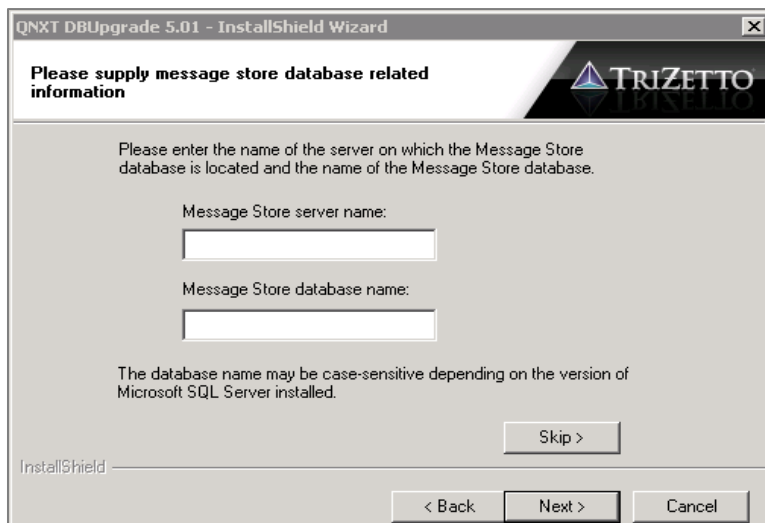


6. Select the environment for the database you want to upgrade and click **Next** on the Select Environment page.

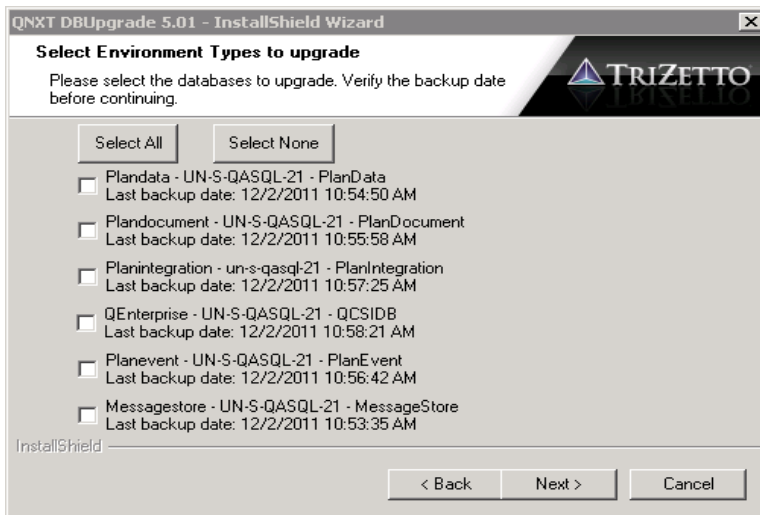


7. Supply Message Store serve and database names and click **Next**.

Tip! Click **Skip** to bypass upgrading the Message Store database.



8. Select the databases you want to upgrade and click **Next**.

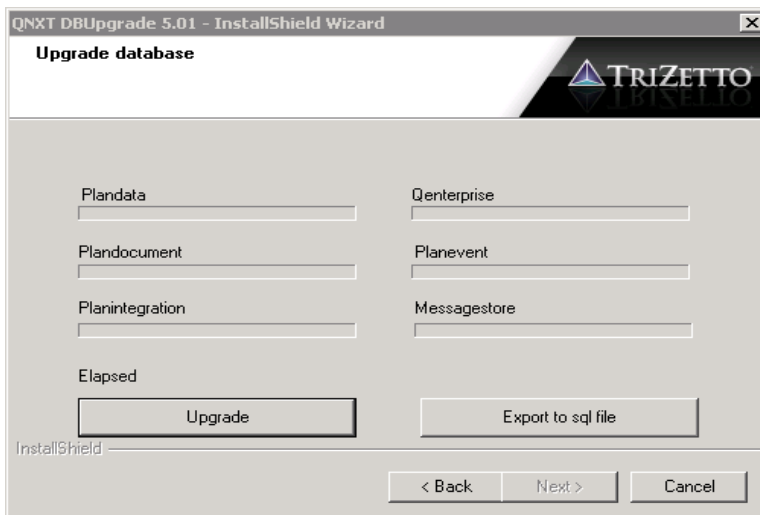


9. Select **Upgrade** or **Export to sql file** as described below and click **Next**:

- **Upgrade:** InstallShield automatically begins the upgrade. The progress indicator for each selected database displays. After all databases are updated, the error/verify dialog box displays.
- **Export to sql file:** InstallShield generates an SQL script that updates each database on the server from the out of date version to the target version. For example, if the DBUpgrade Wizard is run on a server that has HPA 3.4 SP02 databases installed, the application creates an SQL script to update databases from 3.4 SP02 to 5.0. Retrieve the scripts that are generated from the location chosen during the install and apply them in the traditional method using the SQL Server Query Analyzer.

The SQL Execution Errors dialog box displays errors that occurred during the database upgrade. An error screen displays for each selected database.

NOTE: This dialog box appears only if you selected **Upgrade** on the Upgrade database dialog box and errors exist.



10. Click **Finish** to complete the database installation upgrade.

NOTE: Depending on the HPA version from which you are upgrading, the DBUpgrade EXE may appear in the **Programs and Features** dialog box. As of HPA 5.0, the HPA Upgrade EXE file has been updated so you can install the latest version of the .exe directly. You do not need to uninstall a 5.0 version of the .exe file. Also, after it is installed, the .exe file no longer appears in the Programs and Features dialog box.

Troubleshooting the Automated Database Upgrade

While every effort has been made to test and validate the upgrade scripts in the HPA_System_Manager database, there is a possibility that errors may occur in the upgrade scripts. The standard HPA upgrade process was augmented by breaking the upgrade scripts down to the most granular level possible. A validation step was added that is executed after each change is applied. This script upgrade model allows for two possible types of errors:

- **SQL Execution Errors:** These are physical errors at the SQL server level. The error is reported in the ErrorMessage field of the UpgradeProcessDetail table. Change_ID and Sub_change fields in the table link back to the actual statement fired in the Schema_changes table.

If errors are reported by the process, find the errors in the table and compare them with the actual SQL statement in the Schema_changes table.

- **Verification Errors:** These are errors where the actual SQL statement ran successfully, but with unexpected results on the SQL server. After every schema change statement is processed on the server, a check verifies the change was successfully completed. These results are recorded in the UpgradeProcessDetail tables in the Verify_Cond and Verify_Result fields.
 - The Verify_Cond field represents the *desired* output of the verification SQL statement.
 - The Verify_Result field represents the *actual* result of the statement as it was run against the database being upgraded.

If these values do not match, the ChangeVerified field in UpgradeProcessDetail is **N** and the error is reported back to the installation GUI. If these types of errors are encountered, compare the verification_sql, the Verify_Result, and the Verify_cond with the actual results of the SQL Statement run against the database being upgraded.

Uninstalling a Previously Installed Database Upgrade Wizard

This section only applies to clients who have installed a previous version of the Database Upgrade wizard. If you do not have a previous version of the Database Upgrade wizard installed, proceed to DBUpgrade Wizard.

NOTE: Depending on what version of HPA you are upgrading from, the DBUpgrade EXE may appear in the Programs and Features dialog box. As of HPA 5.0, the HPA Upgrade EXE file has been updated so you can install the latest version of the .exe directly. You do not have to uninstall a 5.0 version of the .exe file. After it is installed, the .exe file no longer appears in the Programs and Features dialog box.

To uninstall the Database Upgrade wizard:

1. Click **Start – Control Panel** to display the Control Panel dialog box.
2. Click **Programs and Features** to display the Programs and Features dialog box.
3. Select **DBUpgrade** from the list and click **Uninstall**.
4. Complete the uninstall process.

Database Upgrade Architecture

The following sections describe the database upgrade architecture.

HPA_System_Manager

The HPA_System_Manager database houses all of the schema changes, and tracks the results of all upgrades. A separate instance of this database needs to reside on each server that houses databases that you intend to upgrade. See [Restoring the HPA System Manager Database](#) for more information.

Database Tables

The following are HPA_System_Manager database tables:

- **Schema_changes:** This table is the heart of the process and contains specific schema update scripts that represent a subset of the work required to update a database from an older version of HPA to the most current.
- **UpgradeProcess:** This is a header table that tracks the general progress of the upgrade. When the upgrade process is started by calling SPQ_GetConnInfo, a process ID is automatically created through an identity column, which is passed back to the caller. That process ID is used to call stored procedures that exist downstream and track the status of the entire upgrade process.
- **UpgradeProcessDetail:** This is a detail-level tracking table that holds results of each individual upgrade script.
- **Change_map:** This is a staging table for the upgrade scripts required for a particular upgrade. Changes are pulled from the schema_changes table, sorted into the correct order, and staged here for SPQ_ExecUpgrade to pull from.
- **Version_Map:** This is a cross-walk from HPA application version to the corresponding schema version.
- **Schema_Map:** The schema-map tracks all possible schema iterations and houses a 'required change' field. By traversing this field in reverse order, the process is able to apply only the changes that are needed for the particular upgrade being performed.

Stored Procedures

The following are HPA_System_Manager stored procedures:

- **Spq_GetConnInfo:** Confirms that the target database exists and that it has been updated to a schema version that is recognized by the HPA_System_Manager database which manages the upgrade.
- **Spq_BuildMap:** Builds a list of schema changes required to get the database from the current schema version to the version required by the version of HPA being installed.
- **SPQ_ExecUpgrade:** Takes each individual schema change and applies it to the database. Progress is written back to the UpgradeProcessDetail table.
- **SPQ_PercentComplete:** Tracks the percentage complete based on the number of changes to be applied. Due to the fact that some changes take longer than others, you should not rely on this to gauge how long the upgrade will take.

Post Database Upgrade

This section describes the tasks that are completed after the database upgrade.

Remove Data Conversion Copies and Log Tables

During the PlanData database upgrade to HPA 5.0, HPA creates the following:

- Conversion tables containing copies of the original data. All data conversion tables created during the upgrade are prefixed with conv_Q<version>.
- Tables containing log messages related to records that could not be converted. Review all of the conversion information provided in [Data Conversion](#). This section contains details about the data converted, the names of the data copy tables and the names of the data conversion log tables.

After confirming that all data conversions ran successfully during the database upgrade process, remove these conversion tables from your PlanData database.

V SQL Reporting Services

This chapter includes the following topics:

- [Installing SQL Reporting Services](#)
- [Configuring SQL Reporting Services Installation](#)
- [Testing the SQL Reporting Services Installation](#)

Installing SQL Reporting Services

The SQL Reporting Services component allows HPA to generate reports from SQL Server.

NOTE: Reporting Services software is included with SQL Server. It is not included in the HPA installation package. If you are upgrading from a previous version of Microsoft® SQL Server® Reporting Services, refer to the Microsoft website at [http:// www.microsoft.com](http://www.microsoft.com) for specific upgrade information before completing the following instructions. After you have completed all prerequisite tasks, you can install the Microsoft Reporting Services Server.

This chapter defines settings required by HPA that differ from a standard Microsoft installation.

To install SQL Reporting Services:

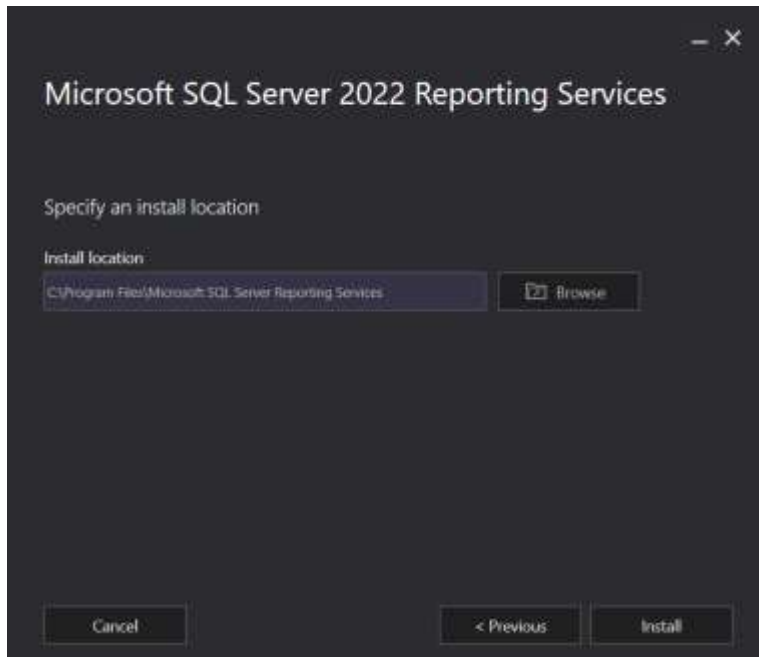
1. Access the SQL Server Reporting Services.
2. Right-click the setup file and click **Run as administrator**.

NOTE: If the Windows Server installation does not have the prerequisite versions of Microsoft .NET Framework or Microsoft Windows® installed, the installer upgrades these components automatically. If this occurs, allow the updates, restart your server, then relaunch the SQL Server installation program.

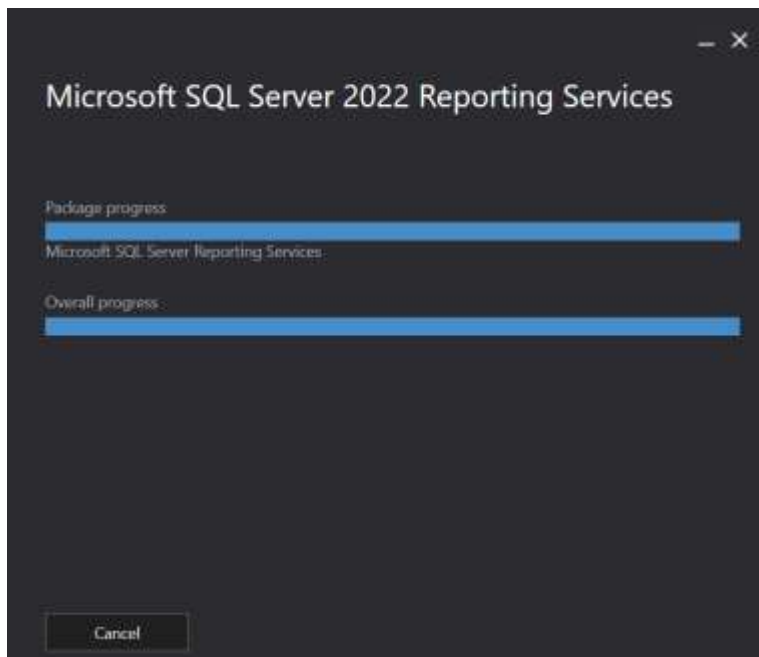
3. Click **Install Reporting Services** when the Microsoft SQL Server 2022 Reporting Services window appears.



4. Enter your product key and click **Next**. The **License Terms** dialog box appears.
5. Select the **I accept the license terms** option and click **Next**. The Install Database Engine dialog box appears. Click **Next**.
6. **Specify an Install location** dialog box appears. Click **Install**.



The Installation Progress dialog box appears.



After the setup process ends, the Complete dialog box appears.



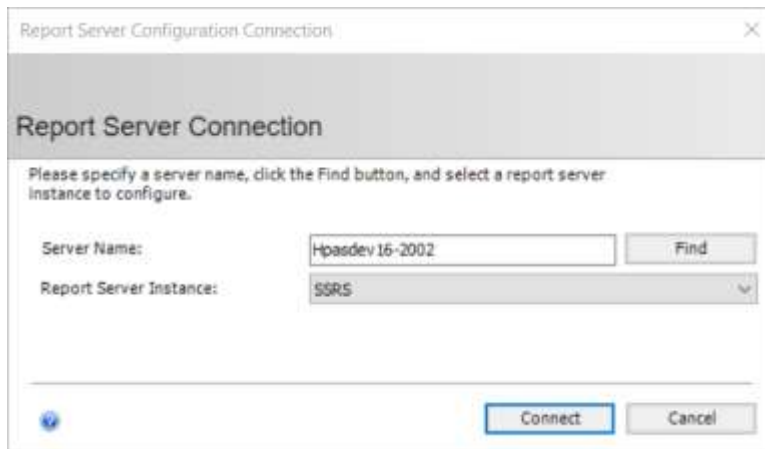
7. Click **Close** once Setup is complete.

Configuring SQL Reporting Services

After the Reporting Services software has been installed, it must be configured. To configure SQL Reporting Services:

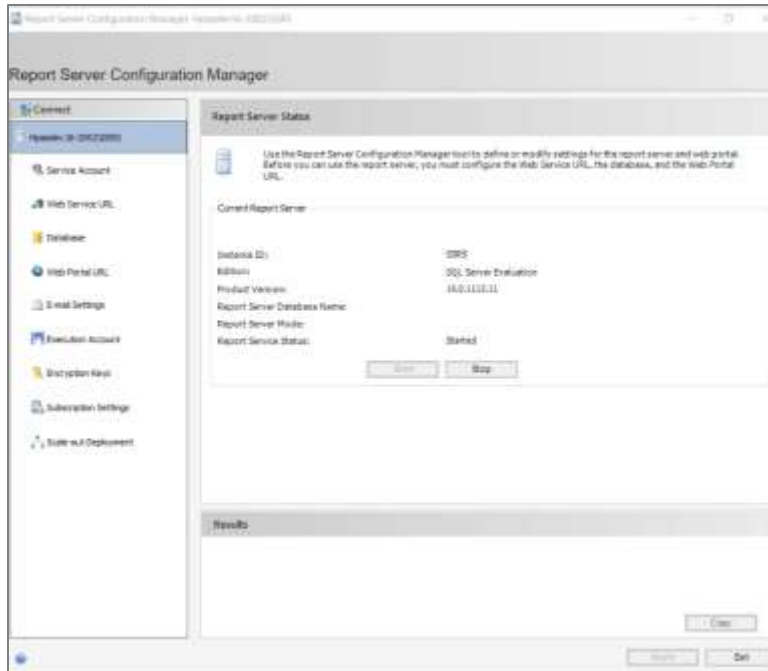
1. Select **Start | All Programs | Microsoft SQL Server Reporting Services | Report Server Configuration Manager**.

The Reporting Services Configuration Connection dialog box appears.

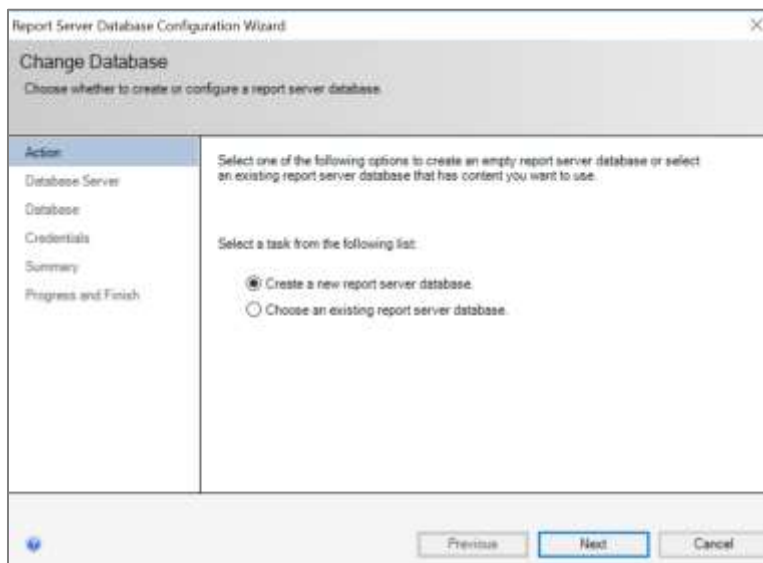


2. Click **Connect**.

The Reporting Services Configuration Manager dialog box appears.



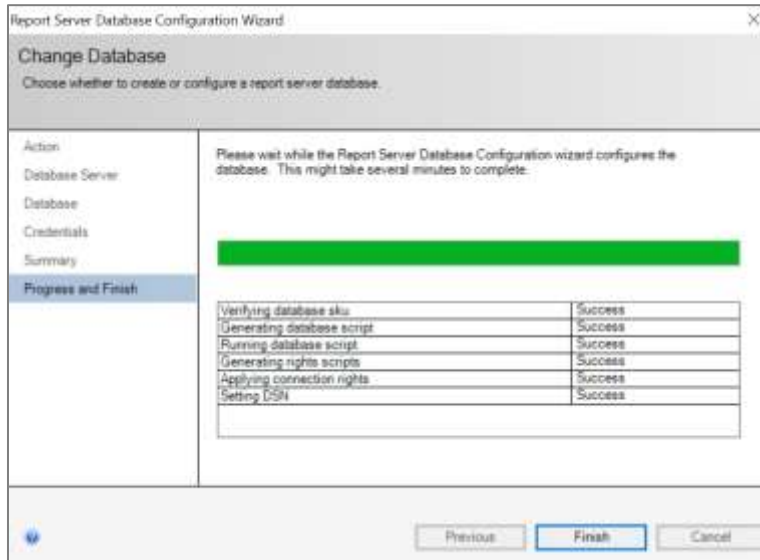
3. Select **Database** in the left pane and click **Change Database**. The Report Server Database Configuration Wizard appears.
4. Verify that the **Create a new report server database** is selected on the Action step and click **Next**. The Database Server page appears.



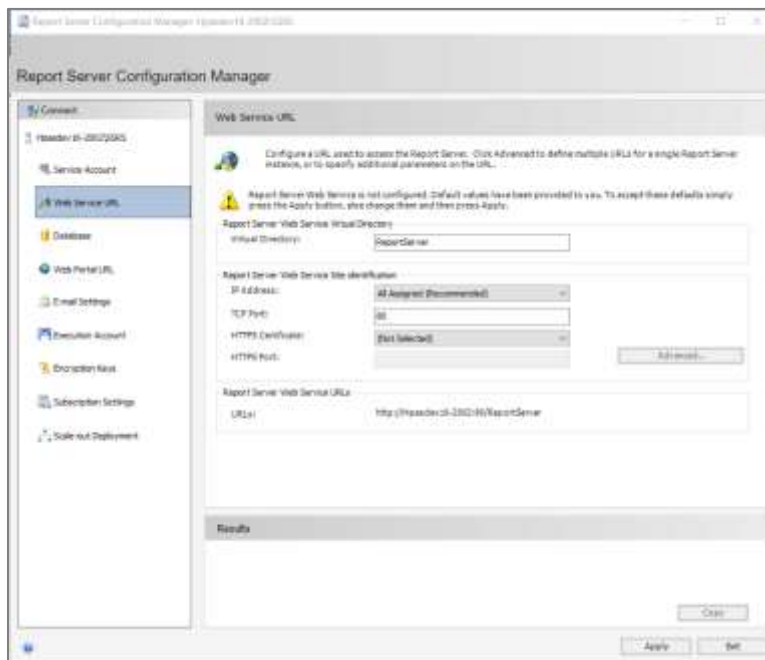
5. Enter credentials for an account that has authority to create databases on the desired server, click **Test Connection** to verify successful connection, and click **Next**. The Change Database page appears.

6. Click **Next** to accept default values. The Credentials page appears.
7. Modify credentials SQL Server Reporting Services uses to connect to the SQL Server Instance and Reporting Services databases as necessary and click **Next**.

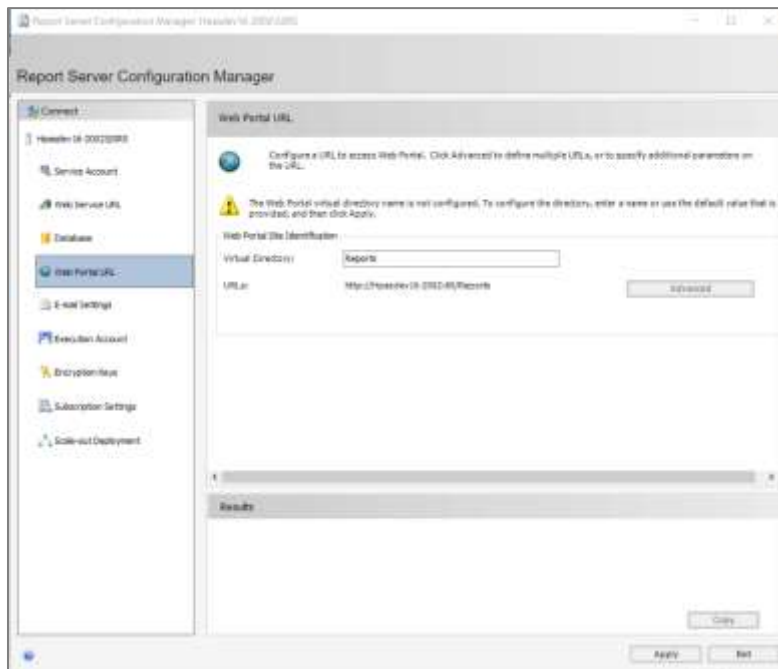
8. Ensure that **TCP Port 80** is not blocked by server firewall software and click **Next**. The Progress and Finish page appears.



9. Verify that all Report Server Database Configuration wizard steps complete successfully and click **Finish**.
10. Select **Web Service URL** in the left pane of the Reporting Services Configuration Manager dialog box. The Web Services URL area appears.



11. Set Virtual Directory to **ReportServer**, then click **Apply**.
12. Select **Report Manager URL** in the left pane of the Reporting Services Configuration Manager dialog box. The Report Manager URL appears.



13. Set Virtual Directory to **Reports**, then click **Apply**.
14. Click **Exit** to close the SQL Server Reporting Services Configuration Manager.

Testing the SQL Reporting Services Installation

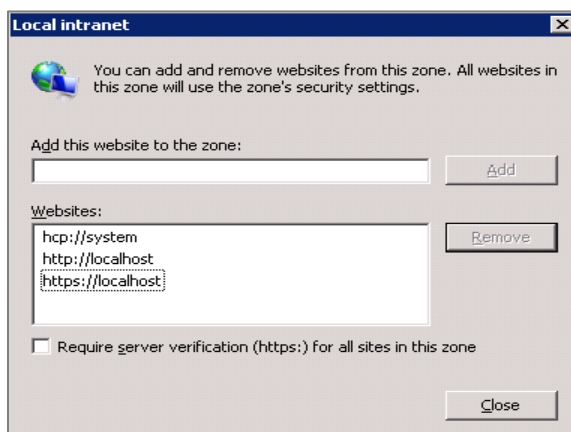
After installing SQL Reporting Services, test the application to make sure it is installed correctly.

To test the SQL Reporting Services installation:

1. Open an instance of **Internet Explorer**.
2. Click the Tools drop-down arrow near the upper-right corner of the browser, then click **Internet Options**. The Internet Options dialog box appears with the General tab is selected by default.
3. Click the Security tab.

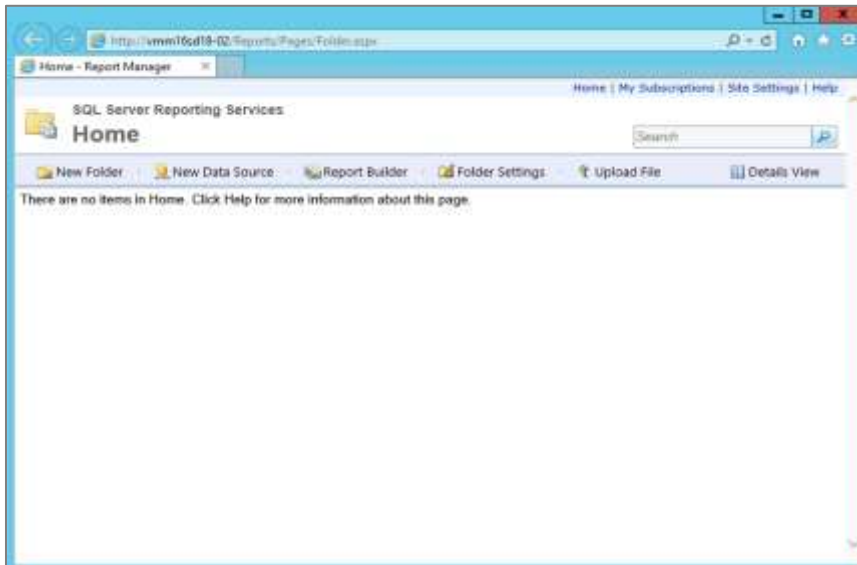


4. Select **Local intranet** from the Select a zone to view or change security sections area, then click **Sites**. The Local intranet dialog box appears.



5. Type the name of the server(s) where SQL Server Reporting Services is located in the Add this website to the zone box, then click **Add**. The new server name displays in the Websites area.
6. Click **Close**.
7. Click **OK** to close the Internet Properties dialog box.

8. Enter the reporting services server name and name of the report manager virtual directory created in the [Report Manager URL step](#) in the Internet Explorer address bar. The SQL Server Reporting Services page displays if the installation was successful.



The QCSI Model Reports folder is installed during the HPA installation. You can view all of the QCSI model reports listed after HPA Reports has been installed.

Installing the Picklist Table

After you confirm that Microsoft Reporting Services is installed, use Microsoft SQL Server or a similar SQL tool to execute the following database script on the Report Server database:

```
01 tlReportPickList Reference Data.sql
```

Refer to Microsoft SQL Server documentation for detailed information about executing database scripts.

VI HIPAA Processing Server Installation and Configuration

HIPAA processing has two installation options, with BizTalk or without BizTalk, using Connect API.

- Installation Option with BizTalk Server
- Installation Option with [Connect API Server](#)

This chapter includes the following BizTalk installation-related topics:

- [BizTalk Server Requirements and Configuration](#)
- [Installing BizTalk Server](#)
- [Configuring BizTalk Server to Run in 64-Bit Mode](#)
- [Verifying the Receiver Location is Disabled](#)
- [Creating a Second Host](#)
- [Creating a Second Host Instance of BizTalk Server](#)
- [Modifying SQL Adapter from 64-Bit to 32-Bit](#)
- [Restarting the Host Instances](#)
- [Enabling Received Locations](#)
- [Confirming that All Orchestrations Have Started](#)
- [Verifying the Processes in Windows Task Manager](#)

BizTalk Server Requirements and Configuration

This chapter describes how to install and configure Microsoft® BizTalk Server®. A local administrator account must be added prior to BizTalk Server install and configuration. This account is used to setup both the BizTalk Server services.

SQL Server must be installed before you can install BizTalk Server. For more information, see [SQL Server Installation and Configuration](#).

If using Business Activity Monitoring (BAM) Tools, you have configured SQL Server Shared Features as described in step 13 of [Installing SQL Server](#).

64-Bit BizTalk Server Considerations

NOTE: HPA 5.0 supports BizTalk Server Enterprise or BizTalk Server Standard 64-bit software only. BizTalk Server 32-bit software is not supported in this version.

BizTalk Server Enterprise Edition allows clients to run BizTalk engines in 64-bit processes, but does not allow all processes to run in 64-bit.

An article at the following link describes applications supported in the 64-bit version of BizTalk server:
<https://msdn.microsoft.com/en-US/library/aa560166.aspx>.

The BizTalk EDI Runtime (formerly the HIPAA Accelerator) uses the SQL Adapter which is not supported by 64-bit.

NOTE: Running the FTP adapter, SQL adapter, POP3 adapter, and MIME Decoder on 64-bit host instances is not supported.

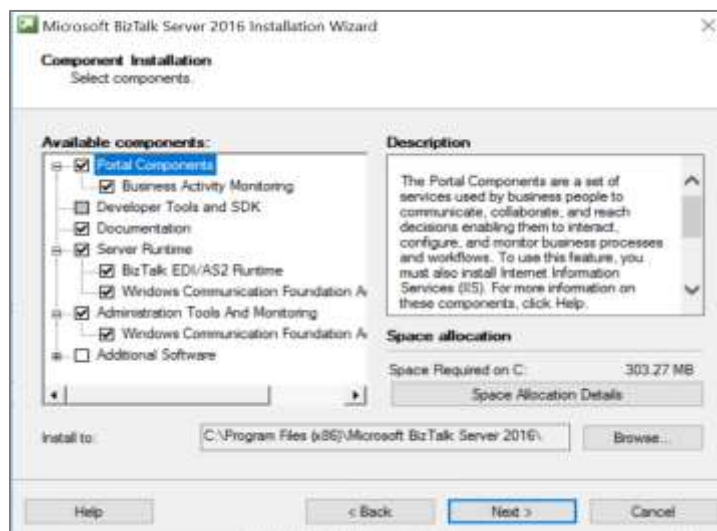
Installing BizTalk Server

To configure BizTalk Server for HPA:

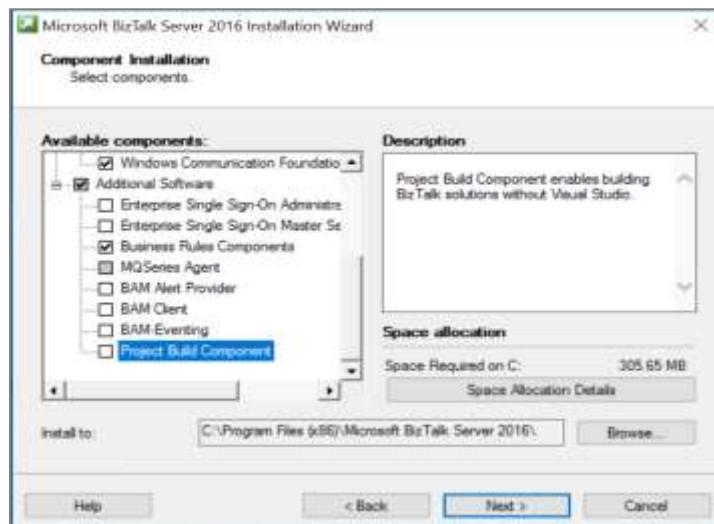
1. Launch the BizTalk Server installation program.
2. Right-click the setup file, then click **Run as administrator**.

NOTE: This escalates your security level to give you limited administrative access. If you are a member of the local administrators' group, you can do this without entering administrator credentials.

3. Initiate the BizTalk Server Installation Wizard and complete installation steps as directed in the wizard.
4. Make the following selections in the Available components area on the Component Installation (Select components) dialog box.
 - Business Activity Monitoring



- Additional Software
- Business Rules Components and other options required for installation (options may vary for each installation)



5. Click **Next** to continue the installation as directed through the wizard.

After BizTalk Server is successfully installed, download and install all available BizTalk Server Hotfixes and updates from the Microsoft Support website at <https://msdn.microsoft.com/en-us/biztalk/ff513699.aspx>.

If using the Value-Based Benefits (VBB) application with HPA:

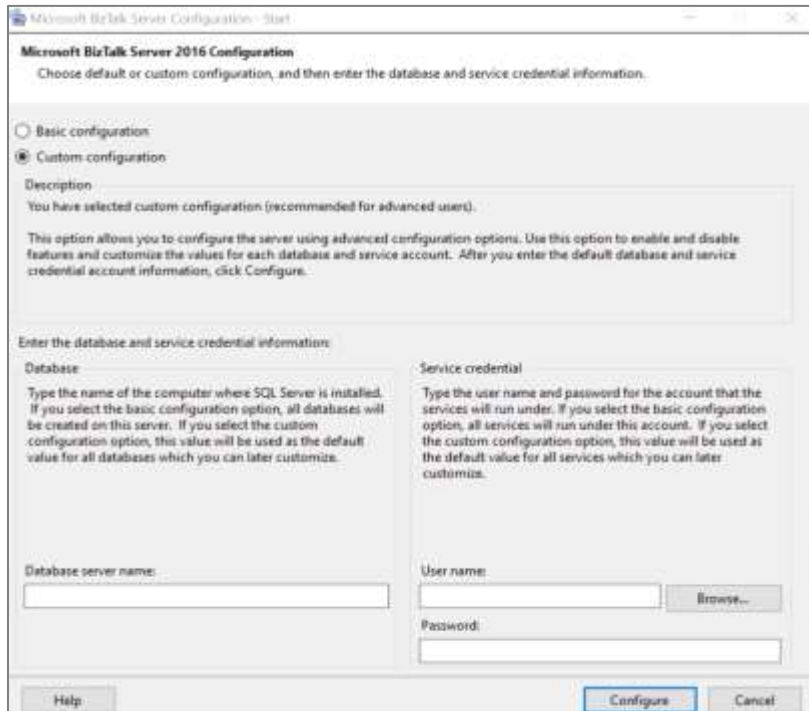
- Download, install, and configure the Microsoft BizTalk ESB Toolkit from <http://www.microsoft.com/download/en/details.aspx?displaylang=en.&id=5632>.
- Install Windows PowerShell from <http://technet.microsoft.com/en-us/windowsserver/bb512919>.

IMPORTANT: See toolkit installation instructions at [http://msdn.microsoft.com/en-us/library/ee384248\(BTS.10\).aspx](http://msdn.microsoft.com/en-us/library/ee384248(BTS.10).aspx) before running the ESB Toolkit Configuration wizard.

Configuring BizTalk Server to Run in 64-Bit Mode

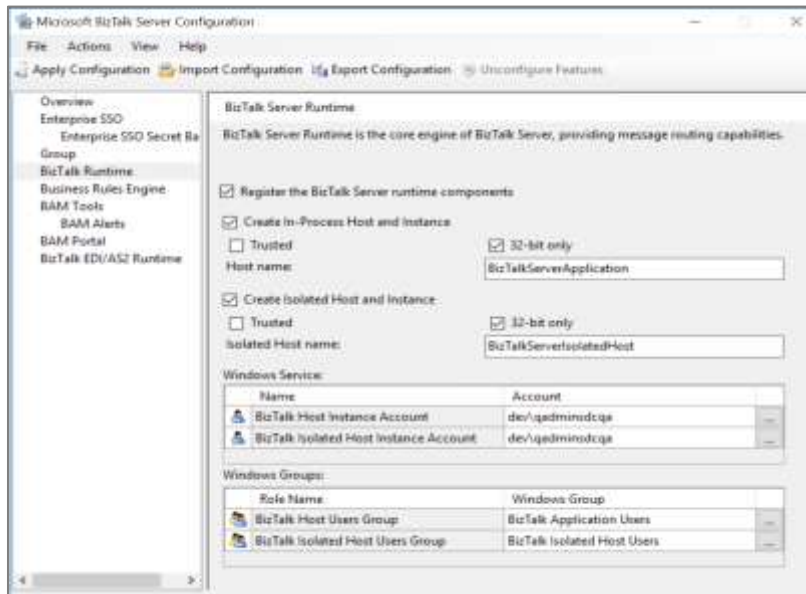
This section describes how to configure the BizTalk Server to run in the 64-bit mode. If you installed BizTalk Server Standard Edition, see [Restarting the Host Instances](#).

1. Select **Start | All Programs | Microsoft BizTalk Server**.
2. Click **Run as Administrator**.



3. Take the following action when the Microsoft BizTalk Server Configuration - Start dialog box appears.
 - 3.1 Select **Custom Configuration**.
 - 3.2 Type the name of the database server name, user name, and password.
 - 3.3 Click **Configure**.
4. Select **Enterprise SSO** to configure the enterprise SSO.
5. Click **Group** to configure the group.

6. Click **BizTalk Runtime**, then take the following action.



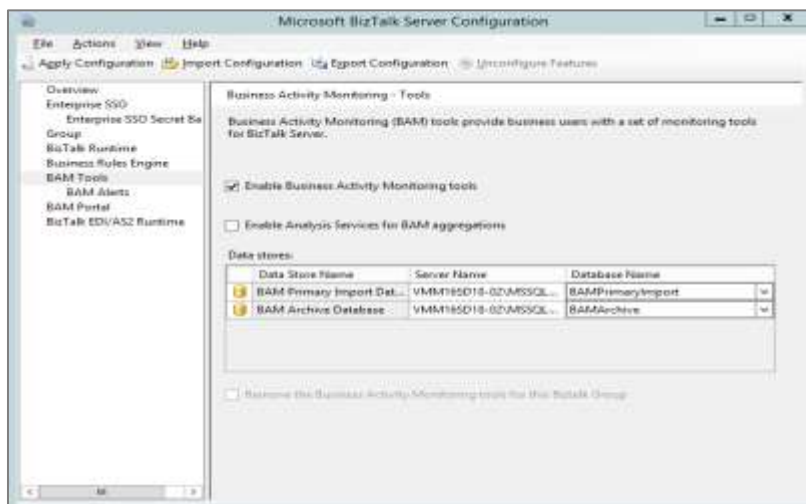
- 6.1 Select the Register the BizTalk Server runtime components check box.
- 6.2 Select Create In-Process Host and Instance.
- 6.3 Clear the 32-bit only check box.

NOTE: If you configure BizTalk for 32-bit only, you cannot use 64-bit functionality unless you reinstall BizTalk server. (You cannot change the option later.)

- 6.4 Type the name of the 64-bit host in the Host name box.
- 6.5 Select the Create Isolated Host and Instance check box.
- 6.6 Clear the 32-bit only check box.
- 6.7 Begin Configure the accounts in the Windows Service area.

7. Click **Business Rules Engine** and configure the business rules engine.
8. Click **BAM Tools**.

NOTE: If using Business Activity Monitoring (BAM) tools, you must configure SQL Server Shared Features as described in step 13 of [Installing SQL Server](#).



9. Select the Enable Business Activity Monitoring tools checkbox.

-
- The screenshot shows the Microsoft BizTalk Server Configuration console. The left-hand tree view is expanded to 'BAM Portal'. The main pane displays the 'Business Activity Monitoring - Portal' configuration page. It includes instructions to select 'Enable BAM Portal' in the Help menu. Below this, there are sections for 'Web service accounts' and 'Windows groups'. The 'Web service accounts' section contains a table with two accounts: 'BAM Management Web Service user' and 'BAM Application Pool Account', both pointing to 'atftouycl\spatocmms'. The 'Windows groups' section contains a table with two groups: 'BAM Portal Users' and 'Everyone'. At the bottom, there is a 'BAM Portal Web Site' dropdown menu set to 'Default Web Site'.
- Microsoft BizTalk Server Configuration
- File Actions View Help
- Apply Configuration Import Configuration Export Configuration Unconfigure Features
- Overview
Enterprise SSO Secret Store
Group
BizTalk Runtime
Business Rules Engine
BAM Tools
BAM Alerts
BAM Portal
BizTalk EDI/AS2 Runtime
- Business Activity Monitoring - Portal
- Select Enable BAM Portal on this computer to enable this feature. For assistance with this feature, select Help.
- ☒ Enable BAM Portal
- Web service accounts:
- | Name | Account |
|---------------------------------|---------------------|
| BAM Management Web Service user | atftouycl\spatocmms |
| BAM Application Pool Account | atftouycl\spatocmms |
- Windows groups:
- | Role Name | Windows Group |
|------------------|---------------|
| BAM Portal Users | Everyone |
- BAM Portal Web Site: Default Web Site

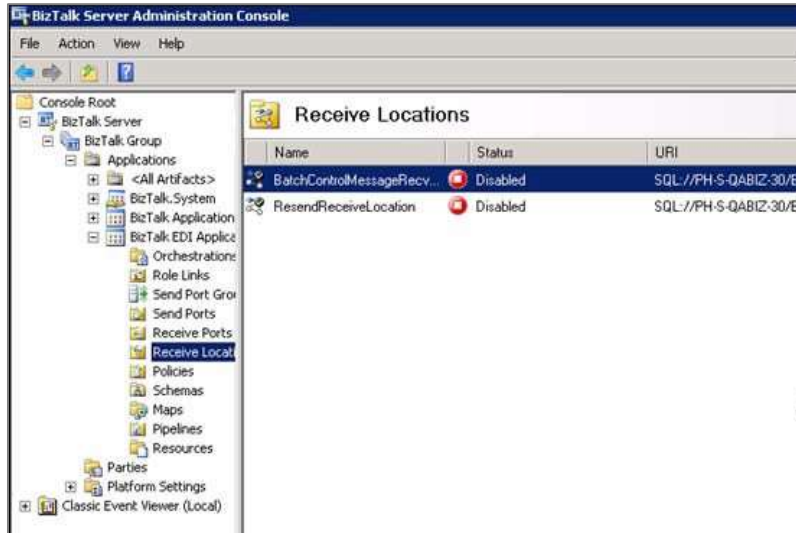
-
- The screenshot displays the Microsoft BizTalk Server Configuration console. The left-hand navigation pane lists various components, with 'BizTalk EDI/AS2 Runtime' highlighted. The main right-hand pane is titled 'BizTalk Electronic Data Interchange and AS2'. It contains a paragraph explaining that the service supports sending and receiving messages using the EDI or AS2 protocol. Below this, there are four configuration options, all of which are checked with a checkmark in a box:
- ☒ Enable BizTalk EDI/AS2 Runtime on this computer
 - ☒ Enable BizTalk EDI for this BizTalk Group
 - ☒ Enable BizTalk AS2 for this BizTalk Group
 - ☒ Enable BizTalk EDI/AS2 Runtime Status Reporting for this BizTalk Group
- At the bottom of the right pane, there is an unchecked option:
- ☐ Remove BizTalk EDI, AS2, and Status Reporting functionality from this BizTalk Group

-
- 77

Verifying the Receive Location is Disabled

After the BizTalk Server configuration is complete, verify that the Receive Location is disabled.

1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration**, then click **Run as Administrator**.
3. In the left pane, select **Console Root | BizTalk Server Administration | BizTalk Group | Applications | BizTalk EDI Application | Receive Location**.
4. Verify the BatchControlMessage receive location is disabled.

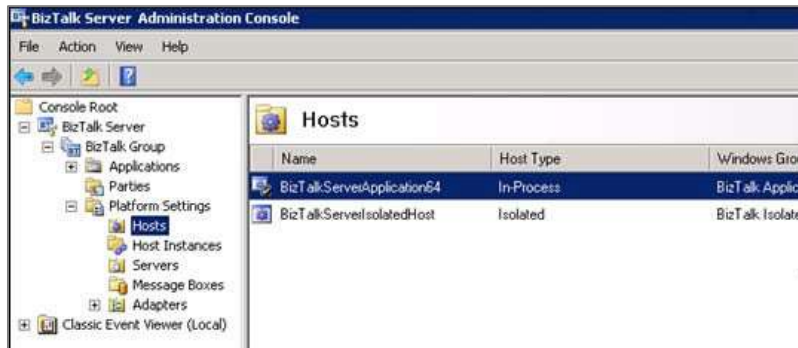


5. Select **Start | All Programs | Administration Tools | Event Viewer**. The Event Viewer dialog box appears.
6. In the left pane, Select **Event Viewer (Local) | Windows Logs | Application**.

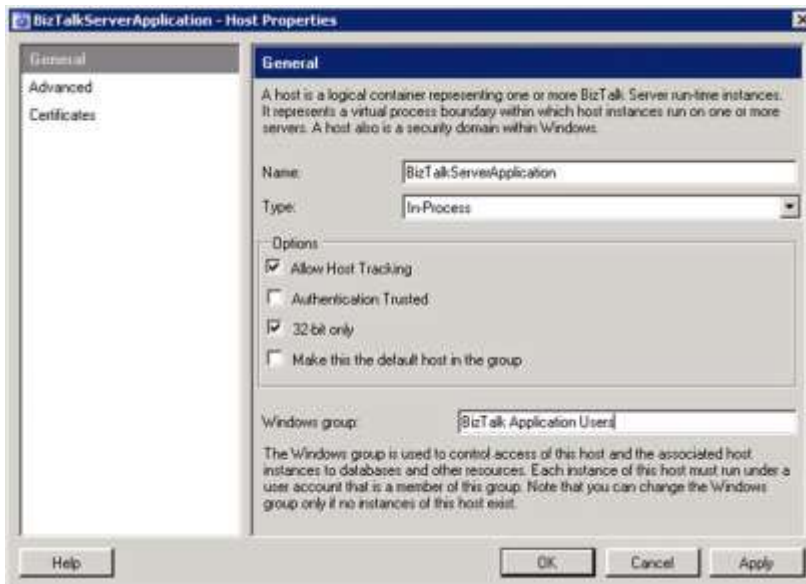
Creating a Second Host

You must create a second (32-bit) host to support the BizTalk EDI Applications. To create a second host:

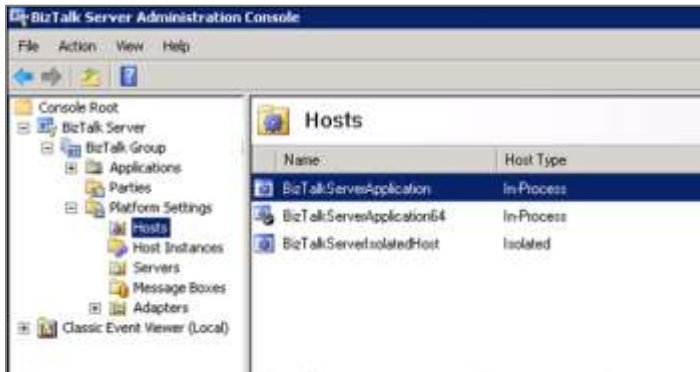
1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console**, then click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, Select **Console Root | BizTalk Server Administration | BizTalk Group | Platform Settings | Hosts**.



4. Right-click **Hosts**.
5. Select **New | Host**.
6. Complete the following action on the BizTalkServerApplication - Host Properties dialog box.
 - 6.1 Type **BizTalkServerApplication** in the Name box.
 - 6.2 Select **In-Process** in the Type list.
 - 6.3 Select Allow Host Tracking and 32-bit only in the Options area.
 - 6.4 Type **BizTalk Application Users** in the Windows group box.
 - 6.5 Click **Apply**, then **OK**.



7. Navigate to the Hosts folder and verify that two hosts have been defined.



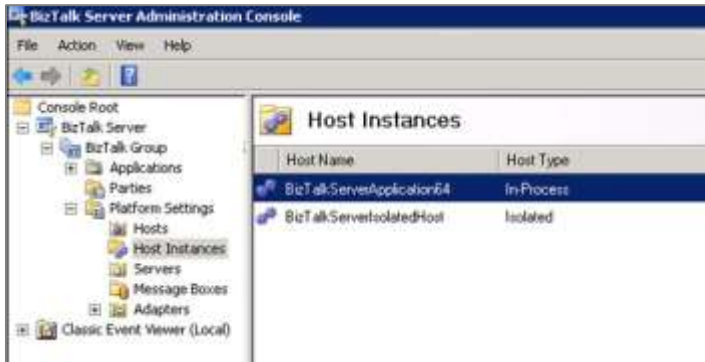
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Creating a Second Host Instance of BizTalk Server

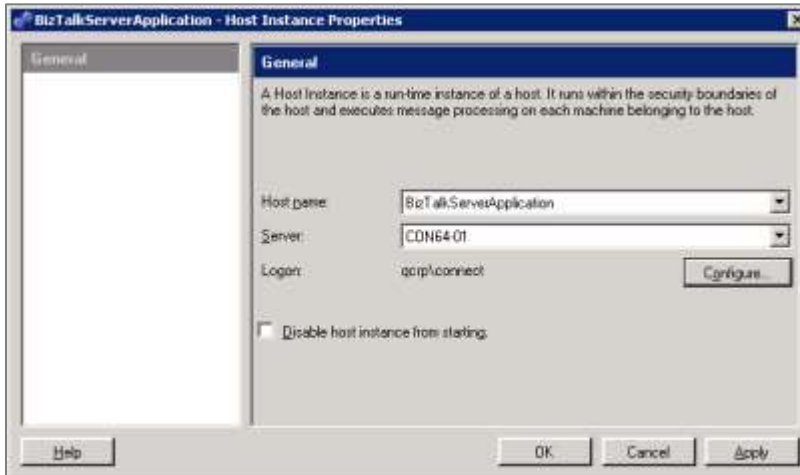
After the 32-bit Host is created, you must create a second Host Instance of BizTalk Server.

To create a second Host Instance of BizTalk Server:

1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console**, then click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, Select **Console Root | BizTalk Server Administration | BizTalk Group | Platform Settings | Host Instances**.



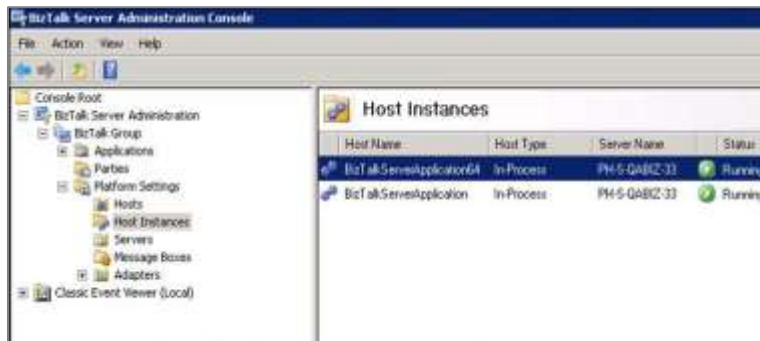
4. Right-click **Host Instances**.
5. Select **New | Host Instance**. The BiztalkServerApplication - Host Instance Properties dialog box appears.



6. Select a Host Name and Server, then click **Configure**. The Logon Credentials dialog box appears.



7. Type a Logon ID and Password, then click **OK**.
8. Click **Apply** on the BiztalkServerApplication - Host Instance Properties dialog box, then click **OK**.
9. Verify that 64-bit and 32-bit instances of BizTalk are running in the Host Instances area.

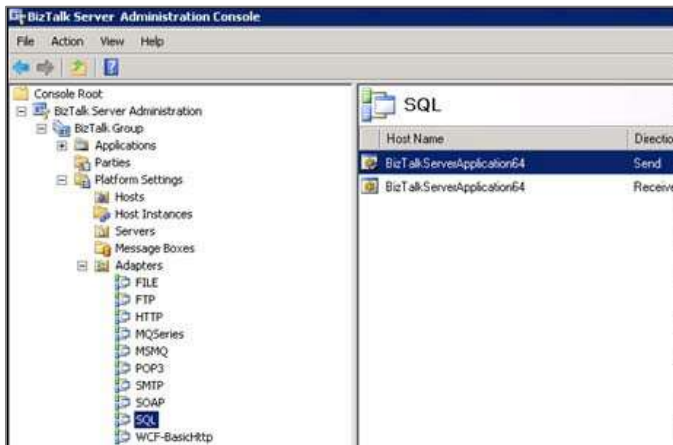


Modifying SQL Adapter from 64-Bit to 32-Bit

NOTE: This is required only if you are installing the BizTalk Server Developer or Enterprise versions. BizTalk Server Standard is 32-bit only.

To modify SQL Adapter from 64-bit to 32-bit:

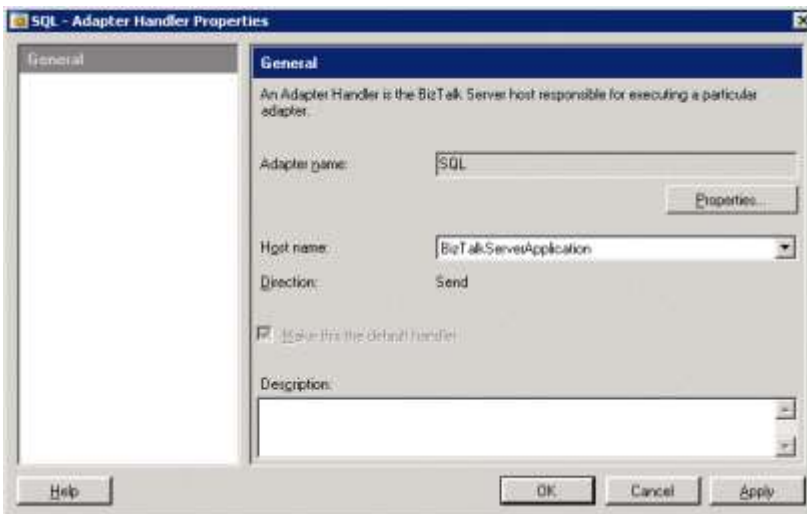
1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console** and click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, Select **Console Root | BizTalk Server Administration | BizTalk Group | Platform Settings | Adapters | SQL**.



NOTE: Send and receive Hosts default to the 64-bit process.

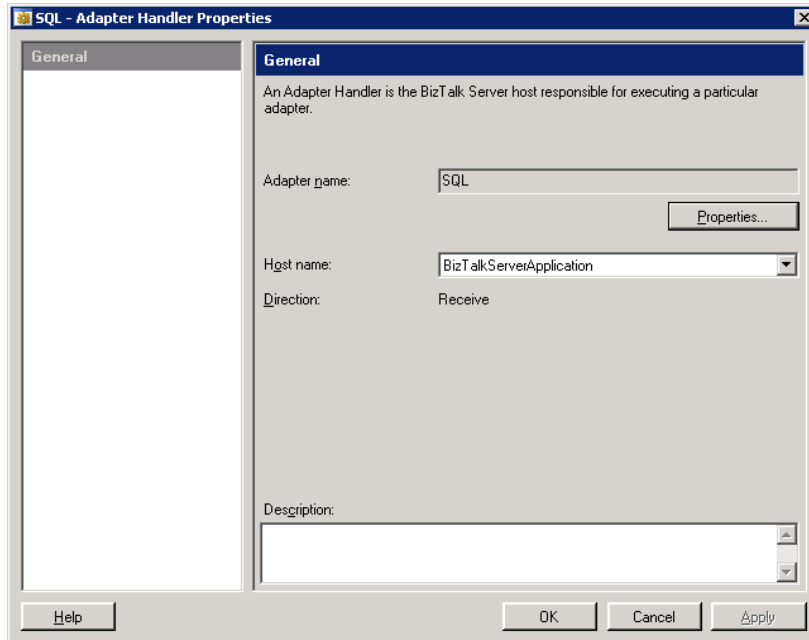
4. Double-click the Send Host, then complete the following actions on the SQL - Adapter Handler Properties dialog.
 - 4.1 Select **BizTalkServerApplication** in the Host name list.
 - 4.2 Click **Apply**, then **OK**.

Tip! You may need to click **OK** to close a system message box.



5. Double-click the Receive Host, then complete the following actions on the SQL - Adapter Handler Properties dialog.
 - 5.1 Select **BizTalkServerApplication** in the Host name list.
 - 5.2 Click **Apply**, then **OK**.

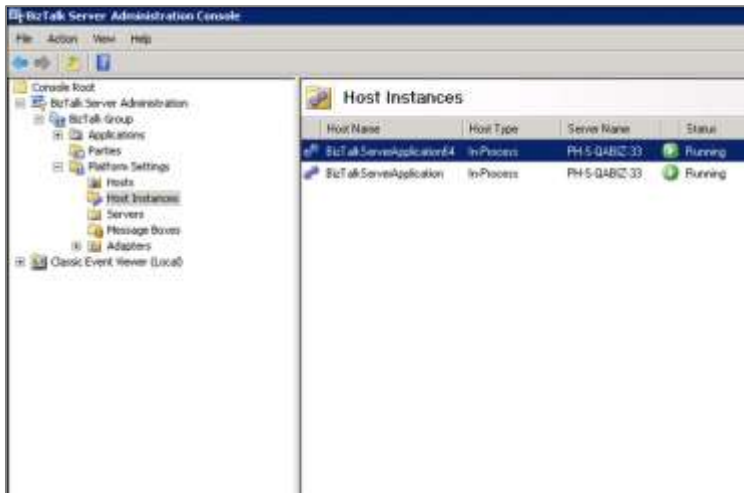
Tip! You may need to click **OK** to close a system message box.



Restarting the Host Instances

You must restart a 64-bit and a 32-bit host instance. To restart the host instance:

1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console**, then click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, Select **Console Root | BizTalk Server Administration | BizTalk Group | Platform Settings | Host Instances**.



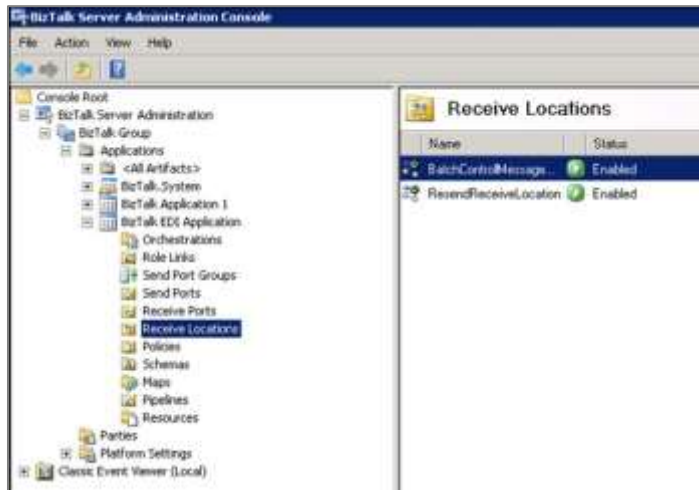
4. Right-click the 32-bit host instance, then select **Restart**.
5. If one exists, Right-click the 64-bit host instance and select **Restart**.

The host instances are now restarted.

Enabling Receive Locations

You must enable a 64-bit and a 32-bit receive location. To enable a 64-bit and 32-bit receive location:

1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console**, then click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, select **Console Root | BizTalk Server Administration | BizTalk Group | Applications | BizTalk EDI Application | Receive Locations**.



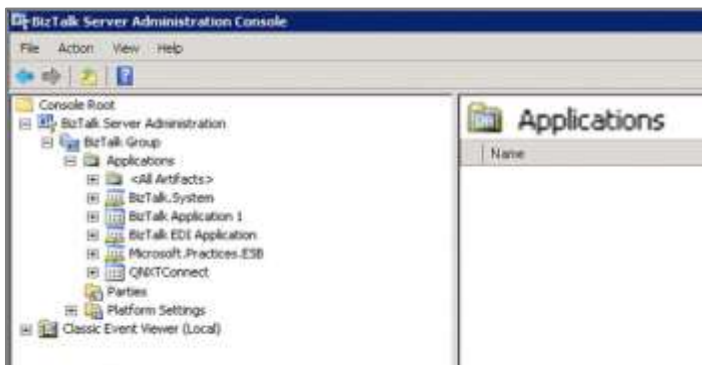
4. Right-click the 32-bit receive location, then select **Enable**.
5. If one exists, right-click the 64-bit receive location and select **Enable**.

The receive location(s) are started.

Confirming that All Orchestrations Have Started

To confirm that all the Orchestrations have started:

1. Select **Start | All Programs | Microsoft BizTalk Server | BizTalk Server Administration**.
2. Right-click **BizTalk Server Administration Console**, then click **Run as Administrator**. The Biztalk Server Administration Console dialog box appears.
3. In the left pane, select **Console Root | BizTalk Server Administration | BizTalk Group | Applications**.



NOTE: The list of applications differs for each installation.

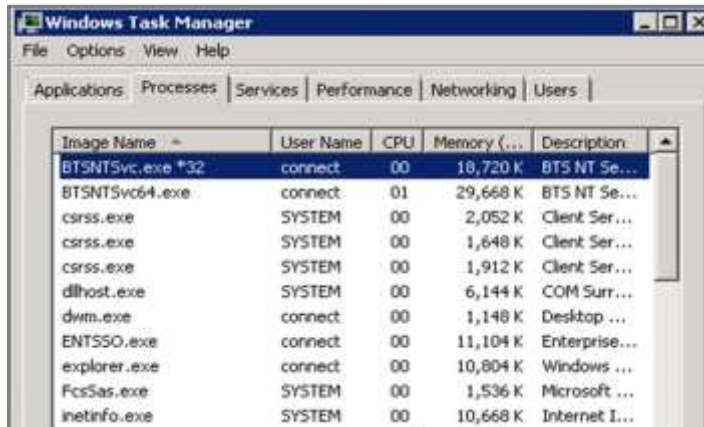
4. Expand the first application and select the **Orchestrations** folder.

Tip! If no orchestration is started, right-click the orchestration and click **Start**.

5. Repeat steps 3 and 4 for all applications in the Applications folder.

Verifying the Processes in Windows Task Manager

1. Open Task Manager.



2. Select the Processes tab.
3. Verify that 32-bit and 64-bit services appear in the list.

VII QFramework Installation

This chapter includes the following topics:

- [QFramework Requirements](#)
- [Installing QFramework](#)
- [Verifying QFramework and QFramework Services Installations](#)
- [QEnterprise Services](#)
- [QEnterprise Configuration](#)

QFramework Requirements

NOTE: Previous versions of HPA component installation programs included logic to notify users if an HPA component was being installed without correct prerequisites in place. Currently, HPA 5.0 component installation wizards **do not** automatically check for installation prerequisites. This is a known issue and will be resolved in a future change package. Make sure that all prerequisites are in place before installing each HPA component. For prerequisite information, see "Requirements" sections in applicable HPA component chapters.

New Installations

Following are QFramework requirements for new installations:

- You must have the Microsoft .NET Framework installed before you can install QFramework. See [Core Installation Requirements](#) for a complete list of software applications that must be installed and configured.
- QFramework and the HPA product suite require that all of the core reference databases are restored and accessible from the server where QFramework is being installed. Required core reference databases are PlanData, QCSIDB, PlanDocument, PlanIntegration, and MessageStore. See [Restoring the HPA Databases](#) for restoration instructions.

Existing Installations

Following are QFramework requirements for existing installations:

- You must have Microsoft .NET Framework installed before you can install QFramework. See [Core Installation Requirements](#) for a complete list of software applications that must be installed and configured.
- Prior to installing the newest version of QFramework, you must perform an upgrade to all installed HPA product databases related to the environment being upgraded. See [Upgrading an Existing Client Database](#) for more information.
- Before starting the upgrade process, ensure all HPA services running on HPA application servers associated with the application environment being upgraded have been shut down. These services include:
 - QCSI Master Service
 - QCSI Orphan Retry Service
 - HPA Connect Inbound Receiver Service
 - HPA QEnterprise Distribution Hub
 - HPA QEnterprise Execution Unit
 - HPA Web Application
 - HPA View Business Services

Installing QFramework

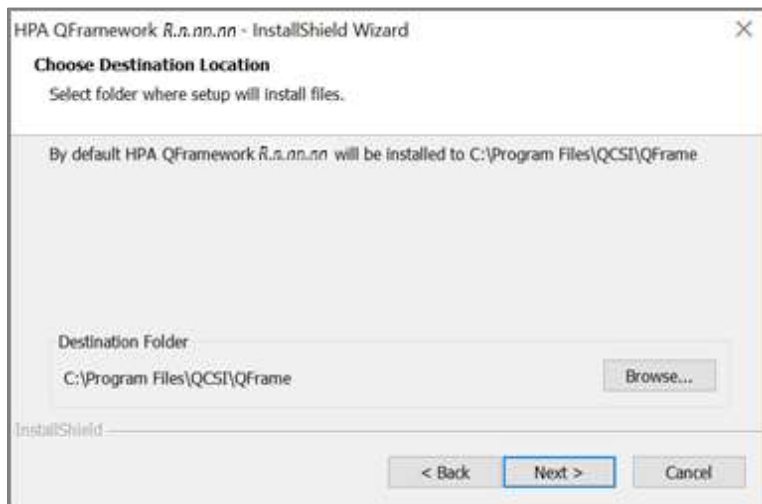
This section describes how to install QFramework on an Execution Unit (EU) server and Distribution Hub (DH) server. Repeat this process for each installation, making the applicable selection in the Select Features step.

If using a Micro-Dyn product with HPA, make sure it is installed before installing QFramework. For information about installing and configuring Micro-Dyn products see [Micro-Dyn Preparation and Setup](#).

To install QFramework:

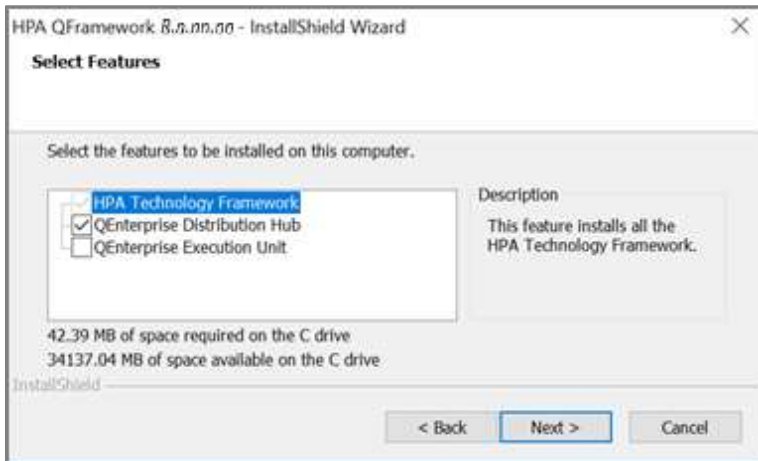
1. Browse to the HPA Installation Package and run the `HPA_QFramework_<version>_x64.exe` installation file.
2. Click **Next** when the InstallShield Wizard starts and the Welcome page appears.
3. After reviewing the license agreement, select **I accept the terms of the license agreement** and click **Next**.
4. Accept the default destination folder or click **Browse** to select a different folder for the installation and click **Next** on the Choose Destination Location window.

IMPORTANT: The destination folder must be the same for all HPA products installed.



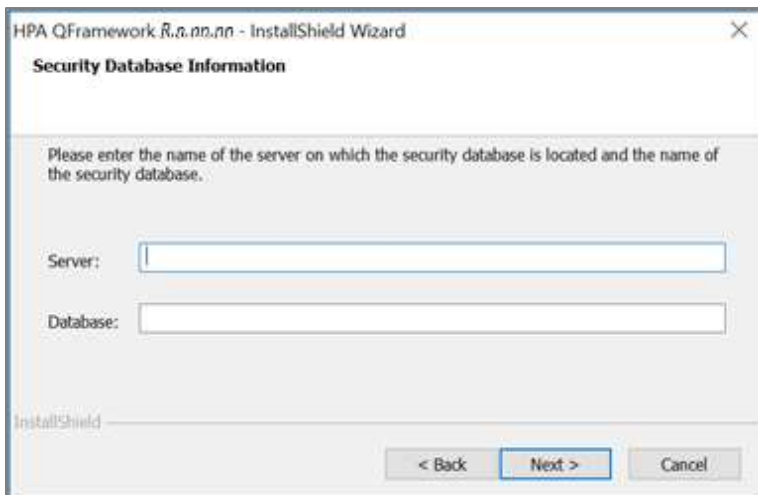
5. Complete the following actions on the Select Features window.
 - 5.1 Select **QCSI Technology Framework** to install QFramework
 - 5.2 Choose **one** QEnterprise application based on the following note.

NOTE: Select **QEnterprise Distribution Hub** or **QEnterprise Execution Unit**. These applications cannot be installed on the same server. **QEnterprise Distribution Hub** is selected by default. If needed, repeat the installation process on a separate server to install the other application.

5.3 Click **Next**.

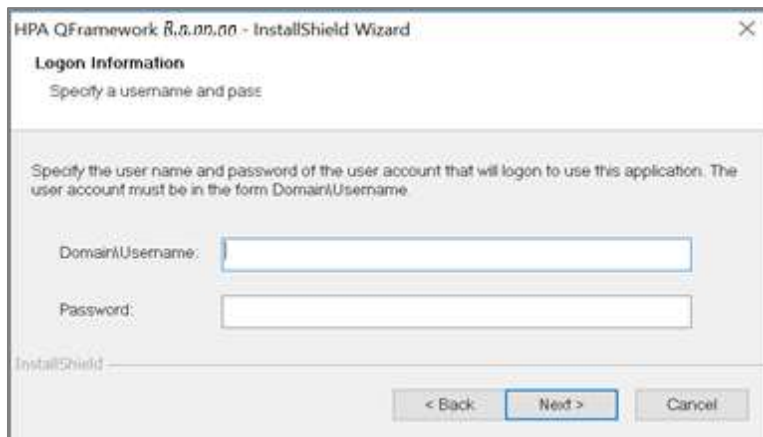
6. Complete the following actions on the Security Database Information window.

6.1 Type the name of the server on which the security database is located in the Server box.

6.2 Type the name of the security database in the Database box and click **Next**.

7. Type the name and password for the domain account with system administration privileges and click **Next** on the Logon Information window.

NOTE: The user name must be entered in the domain\username format.



HPA QFramework R.0.00.00 - InstallShield Wizard

Logon Information

Specify a username and password

Specify the user name and password of the user account that will logon to use this application. The user account must be in the form Domain/Username.

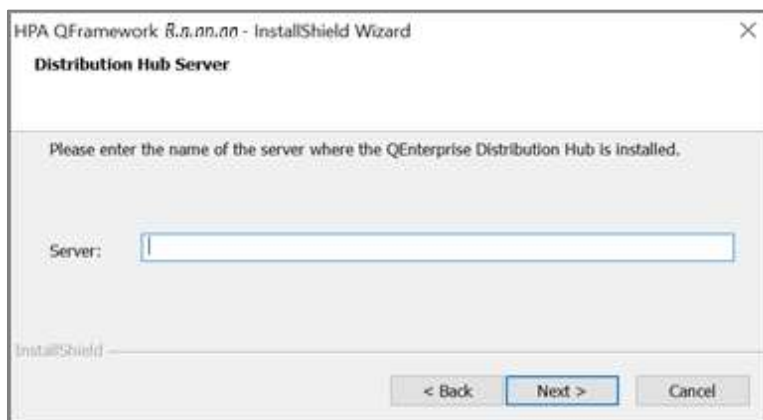
Domain/Username:

Password:

InstallShield

< Back Next > Cancel

8. Type the name of server where the QEnterprise Distribution Hub is installed and click **Next** on the Distribution Hub Server window.



HPA QFramework R.0.00.00 - InstallShield Wizard

Distribution Hub Server

Please enter the name of the server where the QEnterprise Distribution Hub is installed.

Server:

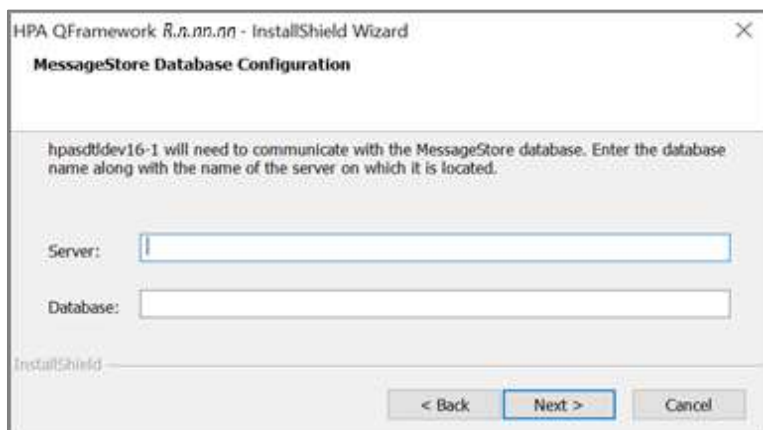
InstallShield

< Back Next > Cancel

9. Complete the following actions on the MessageStore Database Configuration window.

NOTE: The MessageStore Database stores all messages and transactions until they can be successfully processed by an Execution Unit.

- 9.1 Type the name of the server on which the MessageStore is located in the Server box.
- 9.2 Type the MessageStore database name in the Database box and click **Next**.



HPA QFramework R.0.00.00 - InstallShield Wizard

MessageStore Database Configuration

hpsdtdev16-1 will need to communicate with the MessageStore database. Enter the database name along with the name of the server on which it is located.

Server:

Database:

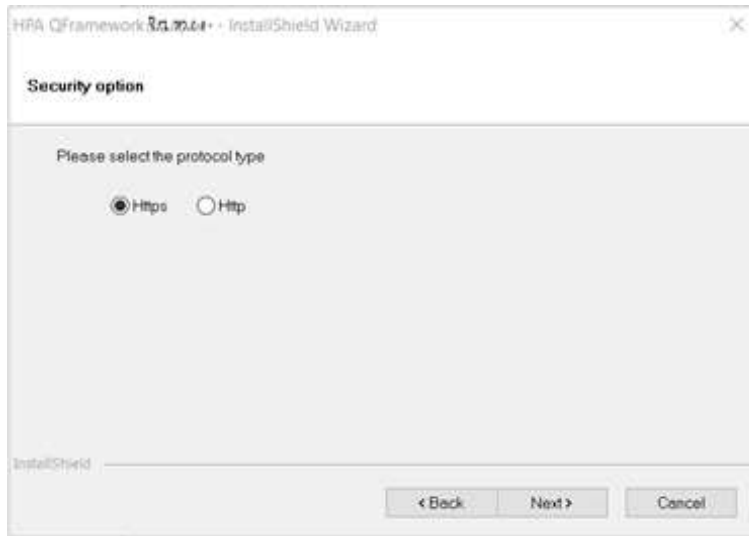
InstallShield

< Back Next > Cancel

10. Select the protocol type on the Security option window.

NOTE: Protocol type should be same for all installers (QFramework, HPA and HPA Connect).

If Https is enabled, SSL needs be configured on the server where QFramework is installed (See [Configure SSL Certificate](#)).



11. Click **Next** to start the installation when the Start Copying Files window appears.

The progress indicator shows the installation status.

12. After the installation is complete, click **Finish** to exit the wizard.

Install QFramework on all Distribution Hub and Execution Unit servers as needed before proceeding.

NOTE: QFramework and QEnterprise applications (see [QEnterprise Services](#)) are usually installed and uninstalled together. If you uninstall QFramework, the entire QFramework product (including QEnterprise) is uninstalled.

Verifying QFramework and QFramework Services Installations

After installing QFramework and the related services, verify that the programs are installed correctly as described below.

To verify that QFramework is installed:

1. Go to **Start | Control Panel | Programs and Features**, then verify that the QFramework program is in the Programs list.
2. Complete the following steps to verify that QCSI Master Service and QEnterprise Distribution Hub/QEnterprise Execution Unit services are installed:
 - 2.1 Select **Start | All Programs | Administrative Tools | Services**.
 - 2.2 Verify that the services are in the Programs list.

QEnterprise Services

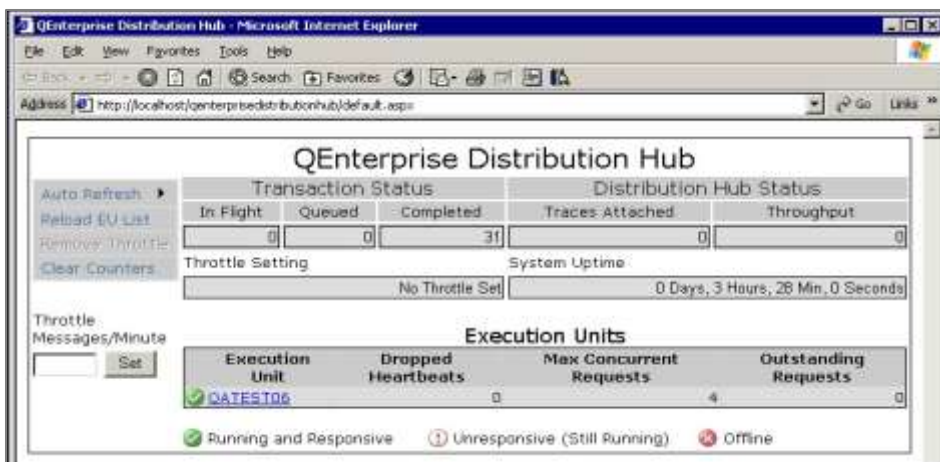
QEnterprise services consist of a Distribution Hub and an Execution Unit. This section describes these services. If QEnterprise was not installed as part of the QFramework installation described in [Installing QFramework](#), proceed to the [HPA Installation](#).

QEnterprise Distribution Hub

There can only be one QEnterprise Distribution Hub for each QEnterprise installation, but the hub can be an IIS Load Balanced cluster consisting of many machines. The main logic of the Distribution Hub is implemented as a .NET web application.

The Distribution Hub (DH) physical implementations are represented by a Windows Service **and** supporting Web Application and are the center of activity for HPA's Service Oriented Architecture.

It is important that the service and website remain functioning to prevent an individual database or message type event that would cause the Distribution Hub to stop or fail. The Distribution Hub is designed to send detailed information relevant to its activity to the Trace tool provided by the installation.



The QEnterprise Distribution Hub service allows you to track and perform some manual configuration of the message/transaction processing. This service allows for manual Throttle configuration (to slow down/speed up processing based upon system configuration and performance limits). It allows for real-time tracking of all transaction statuses and allows for monitoring of all applicable Execution Units.

The following tables list descriptions for fields that appear on the Distribution Hub Web Service interface.

Auto Refresh Area

Field	Description
Auto Refresh	Click to set the number of minutes the system waits before this page is refreshed.
Reload EU List	Click to reload the list of messages sent to the Execution Unit.
Remove Throttle	Clears the configured value in the Throttle Setting field.
Clear Counters	Clears the data in fields on this page.

Transaction Status Area

Field	Description
In Flight	Number of messages currently sent to the Execution Unit.
Queued	Number of messages the Distribution Hub has waiting to distribute to the Execution Unit.
Completed	Number of messages sent to the Execution Unit during this session. Throttle SettingNumber of messages per minute the Distribution Hub is configured to send to the Execution Unit.

Distribution Hub Status Area

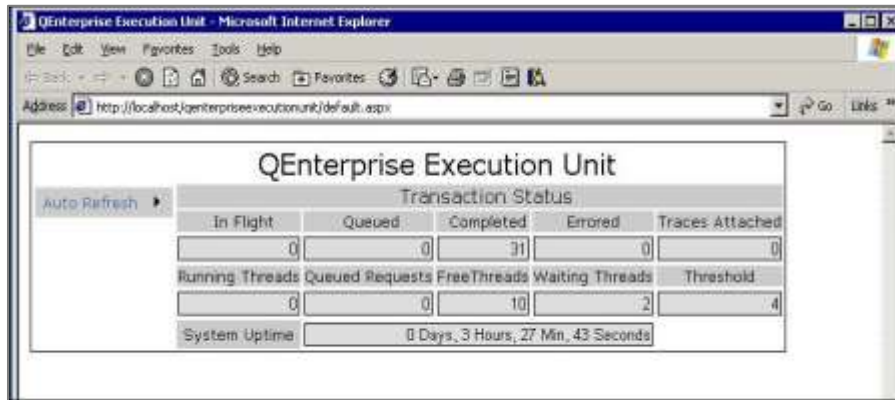
Field	Description
Traces Attached	Number of QEnterprise Trace instances that are attached to the Distribution Hub.
Throughput	Number of messages processed per minute. This is a dynamic number and changes as the number of messages per minute changes/
System Uptime	Amount of time Distribution Hub has been running continuously for this session.

Execution Units Area

Field	Description
Execution Unit	Name of the Execution Units attached to the Distribution Hub. Dropped Heartbeats Number of heartbeats sent to the Execution Unit that did not receive a response from the Execution Unit.
Max Concurrent Requests	Maximum number of concurrently executing threads during this session
Outstanding Requests	Number of messages sent to the Execution Unit that are waiting to process.

QEnterprise Execution Unit

Messages are processed by a QEnterprise Execution Unit, which is implemented as a .NET web application. First, the .ASP page receives a post that contains the work item's transaction ID. The Execution Unit retrieves the payload data from the message store.



Running the Execution Unit Web Service allows for verification that the Execution Unit is online and that the services are running correctly. This allows for real-time monitoring of transaction processing. The following is a description of the fields that appear on the Execution Unit Web Service interface.

Auto Refresh Area

Field	Description
Auto Refresh	Sets the number of minutes the system waits before this page is refreshed. Type a value, then click Set Refresh Interval .

Transaction Status Area

Field	Description
In Flight	Number of messages being processed on the Execution Unit.
Queued	Number of messages the Execution Unit is waiting to process.
Completed	Number of messages processed during this session.
Errored	Number of messages that could not be processed.
Traces Attached	Number of QEnterprise Trace instances attached to the Execution Unit.
Running Threads	Number of threads currently running.
Queued Requests	Requests queued and waiting to be processed.
Free Threads	Threads not participating in the thread pool.
Waiting Threads	Threads waiting for work items.
Threshold	Max number of threads allowed to run in the thread pool.
System Uptime	Amount of time Distribution Hub has been running continuously for this session.

QEnterprise Message Structure

The Asynchronous Simple Message has no message structure, so QEnterprise does not have to wrap the application-defined message. All states that flowed with the message in the QEnterprise header are now stored in `message_store`. Message ID, timeouts, call method, and other options are passed to the distribution server as a query string on the HTTP POST. This allows the distribution server to avoid parsing the contents of the payload unless transforming it.

QEnterprise Execution Models

The following are QEnterprise execution models:

- **Synchronous Simple Message:** Also referred to as Fastcall, is a QEnterprise call that bypasses the message store unless logging or tracing is enabled and passes the job and message ID via HTTP POST to the Distribution Hub. Once a job is received, the hub posts an HTTP POST, which contains the message ID, to an available Execution Unit for processing.
- **Asynchronous Simple Message:** After receiving a job request, the Distribution Hub submits the job to the SQL server for storage. Once the Execution Unit receives the HTTP POST from the hub, it uses the message ID to retrieve the job from the SQL server. Upon completion of all tasks associated with the job, the Execution Unit generates an HTTP POST to notify the Distribution Hub, which subsequently sends a reply to the client through a prearranged Windows socket.

QEnterprise Configuration

This section discusses how to configure QEnterprise once it has been installed. QEnterprise is installed during the QFramework installation process and involves configuring the Distribution Hub and Execution Units.

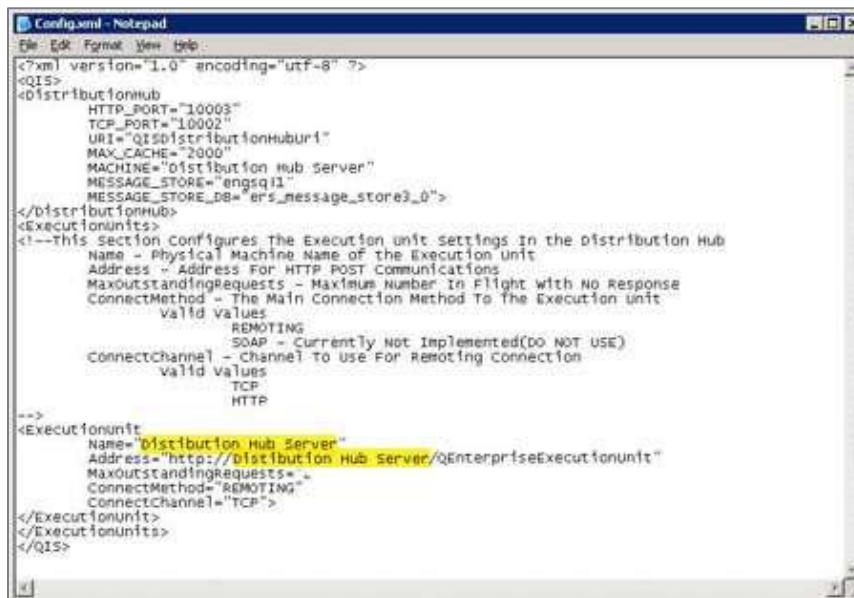
Configuring the Distribution Hub

The first step to configuring QEnterprise is to configure the Distribution Hub, which includes renaming the first Execution Unit on the Distribution Hub and adding additional Execution Units to the Distribution Hub. This section describes those processes and how to remove an Execution Unit from a Distribution Hub.

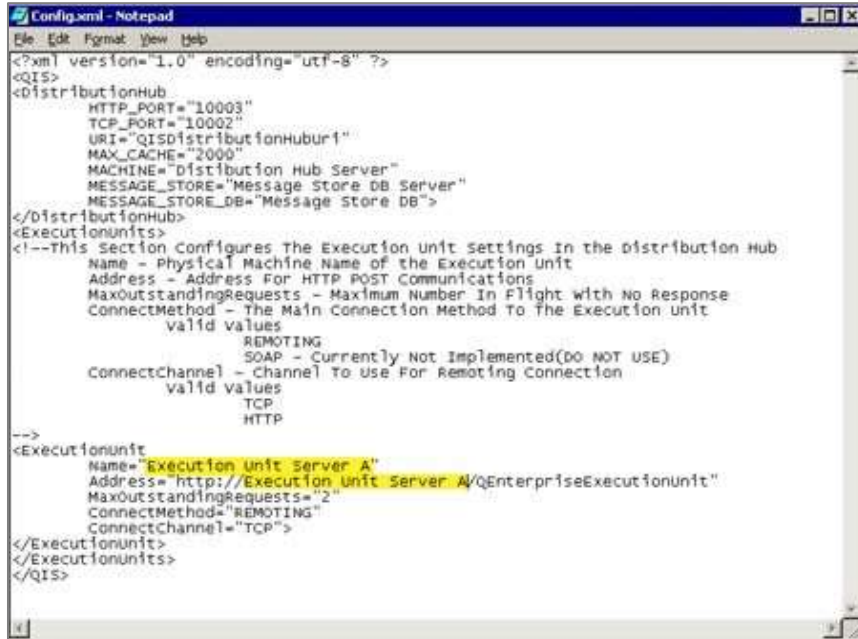
Renaming the Execution Unit on the Distribution Hub

By default, the QIS Distribution Hub config.xml page names the first Execution Unit with the Distribution Hub server's name. You must change that name before configuring QEnterprise.

1. Open Microsoft Windows Explorer and navigate to
C:\Program Files\QCSI\QFrame\Services\Distribution Hub\Web
2. Find and open `Config.xml` in a text editor such as Microsoft Windows Notepad.
3. Locate the section of the code that begins as:
==>
<Executionunit.
4. Select the name of the Distribution Hub server, as shown in the following example:



- Replace the Distribution Hub Server name with the name of the Execution Unit in the name value and address value. (The following example shows the generic name Execution Unit Server A.)



```

<?xml version="1.0" encoding="utf-8" ?>
<QIS>
<DistributionHub>
  HTTP_PORT="10003"
  TCP_PORT="10002"
  URI="QISDistributionHuburi"
  MAX_CACHE="2000"
  MACHINE="Distribution Hub Server"
  MESSAGE_STORE="Message Store DB Server"
  MESSAGE_STORE_DB="Message Store DB"
</DistributionHub>
<ExecutionUnits>
<!--This section configures the Execution unit settings in the Distribution Hub
  Name - Physical Machine Name of the Execution Unit
  Address - Address For HTTP POST Communications
  MaxOutstandingRequests - Maximum Number In Flight with No Response
  ConnectMethod - The Main Connection Method To The Execution unit
  Valid values
    REMOTING
    SOAP - Currently Not Implemented(DO NOT USE)
  ConnectChannel - Channel To Use For Remoting Connection
  Valid values
    TCP
    HTTP
-->
<ExecutionUnit
  Name="Execution Unit Server A"
  Address="http://Execution Unit Server A/QEnterpriseExecutionUnit"
  MaxOutstandingRequests="2"
  ConnectMethod="REMOTING"
  ConnectChannel="TCP">
</ExecutionUnit>
</ExecutionUnits>
</QIS>

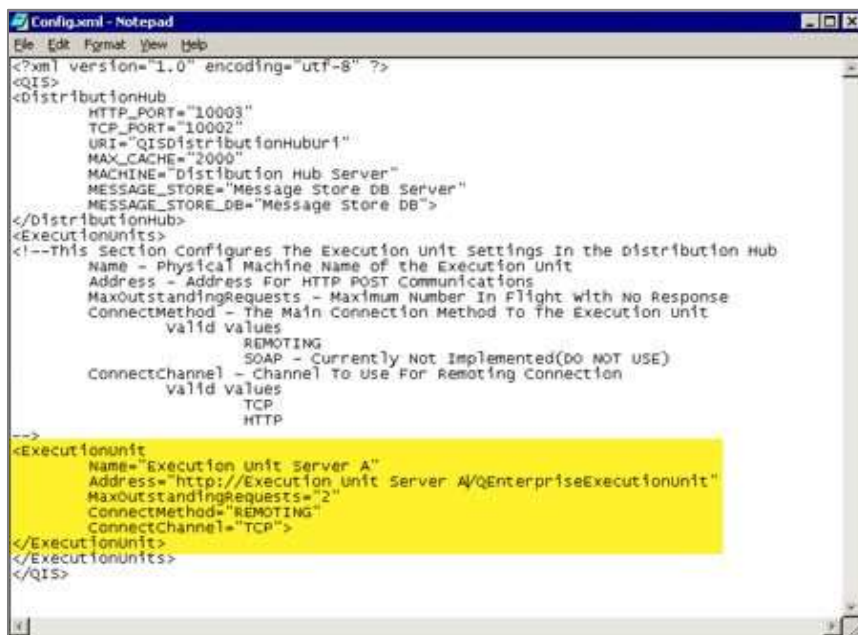
```

- Select **File | Save** to save the changes made to the **Config.xml** file.

Adding Multiple Execution Units

To edit a QEnterprise Distribution Hub (DH) configuration file:

- Open Microsoft Windows Explorer and navigate to
\\Program Files\\QCSI\\QFrame\\Services\\Distribution Hub\\Web.
- Find and open **Config.xml** in a text editor such as Microsoft Windows Notepad.



```

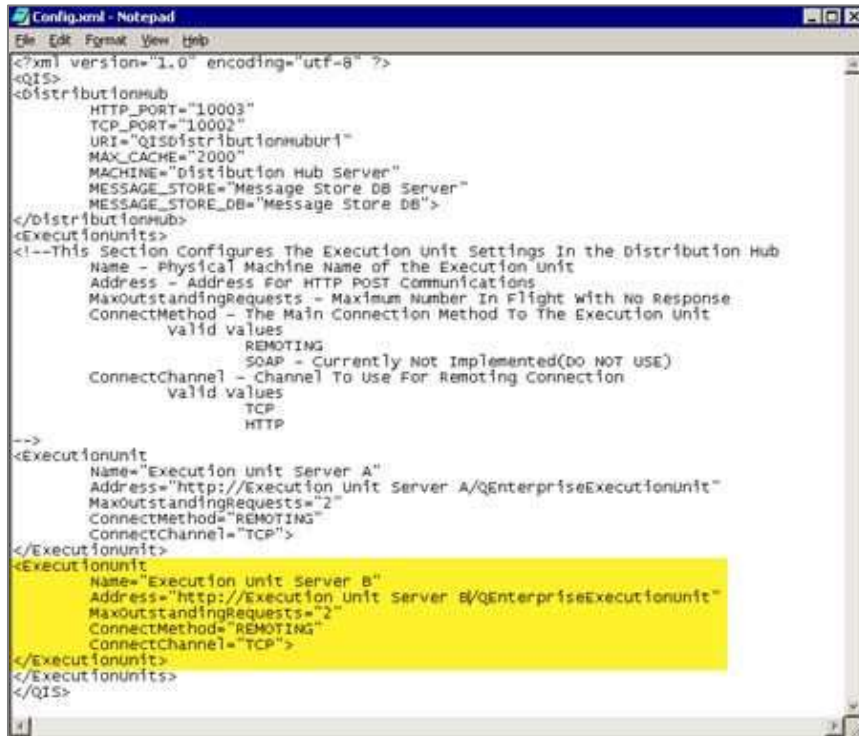
<?xml version="1.0" encoding="utf-8" ?>
<QIS>
<DistributionHub>
  HTTP_PORT="10003"
  TCP_PORT="10002"
  URI="QISDistributionHuburi"
  MAX_CACHE="2000"
  MACHINE="Distribution Hub Server"
  MESSAGE_STORE="Message Store DB Server"
  MESSAGE_STORE_DB="Message Store DB"
</DistributionHub>
<ExecutionUnits>
<!--This section configures the Execution unit settings in the Distribution Hub
  Name - Physical Machine Name of the Execution Unit
  Address - Address For HTTP POST Communications
  MaxOutstandingRequests - Maximum Number In Flight with No Response
  ConnectMethod - The Main Connection Method To The Execution unit
  Valid values
    REMOTING
    SOAP - Currently Not Implemented(DO NOT USE)
  ConnectChannel - Channel To Use For Remoting Connection
  Valid values
    TCP
    HTTP
-->
<ExecutionUnit
  Name="Execution Unit Server A"
  Address="http://Execution Unit Server A/QEnterpriseExecutionUnit"
  MaxOutstandingRequests="2"
  ConnectMethod="REMOTING"
  ConnectChannel="TCP">
</ExecutionUnit>
</ExecutionUnits>
</QIS>

```

3. Locate the section of the code that begins as:

```
-->
<Executionunit>
```
4. Copy the complete code segment starting with `<Executionunit>` and ending with `</Executionunit>`.
5. Paste the copied code segment between `</Executionunit>` and `</Executionunits>`.
6. Edit the new code section to reflect the applicable name.

NOTE: In the following example, the new name became **Execution Unit Server B**.



```

Config.xml - Notepad
File Edit Format View Help
<?xml version="1.0" encoding="utf-8" ?>
<QIS>
  <DistributionHub
    HTTP_PORT="10003"
    TCP_PORT="10003"
    URI="QISDistributionHuburl"
    MAX_CACHE="2000"
    MACHINE="Distribution Hub Server"
    MESSAGE_STORE="Message Store DB Server"
    MESSAGE_STORE_DB="Message Store DB"
  </DistributionHub>
  <ExecutionUnits>
    <!--This Section Configures The Execution Unit Settings In the Distribution Hub
    Name - Physical Machine Name of the Execution Unit
    Address - Address For HTTP POST Communications
    MaxOutstandingRequests - Maximum Number In Flight with No Response
    ConnectMethod - The Main Connection Method To The Execution Unit
    Valid values
      REMOTING
      SOAP - Currently Not Implemented(DO NOT USE)
    ConnectChannel - Channel To Use For Remoting Connection
    Valid values
      TCP
      HTTP
    -->
    <ExecutionUnit
      Name="Execution unit Server A"
      Address="http://Execution Unit Server A/QEnterpriseExecutionUnit"
      MaxOutstandingRequests="2"
      ConnectMethod="REMOTING"
      ConnectChannel="TCP">
    </ExecutionUnit>
    <ExecutionUnit
      Name="Execution Unit Server B"
      Address="http://Execution Unit Server B/QEnterpriseExecutionUnit"
      MaxOutstandingRequests="2"
      ConnectMethod="REMOTING"
      ConnectChannel="TCP">
    </ExecutionUnit>
  </ExecutionUnits>
</QIS>
  
```

7. Select **File | Save** to save the modified `Config.xml` file.

The QEnterprise Distribution Hub service must be restarted for changes to be loaded into the Distribution Hub.

Removing an Execution Unit from a Distribution Hub

To remove an EU from the Distribution Hub:

1. Open Windows Explorer and navigate to
`\\Program Files\QCSI\QFrame\Services\Distribution Hub\Web.`
2. Find and open `Config.xml` with Microsoft Windows Notepad.
3. Locate the section of the code that begins as:
`-->`
`<Executionunit`
4. Select and delete the code segment starting with `<Executionunit` and ending with
`</ Executionunit>`.
5. Repeat step 3 until applicable Execution Units have been removed from the `Config.xml` file.
6. Select **File | Save** to save the changes made to the file.

The QEnterprise Distribution Hub service must be restarted for changes to be loaded into the Distribution Hub.

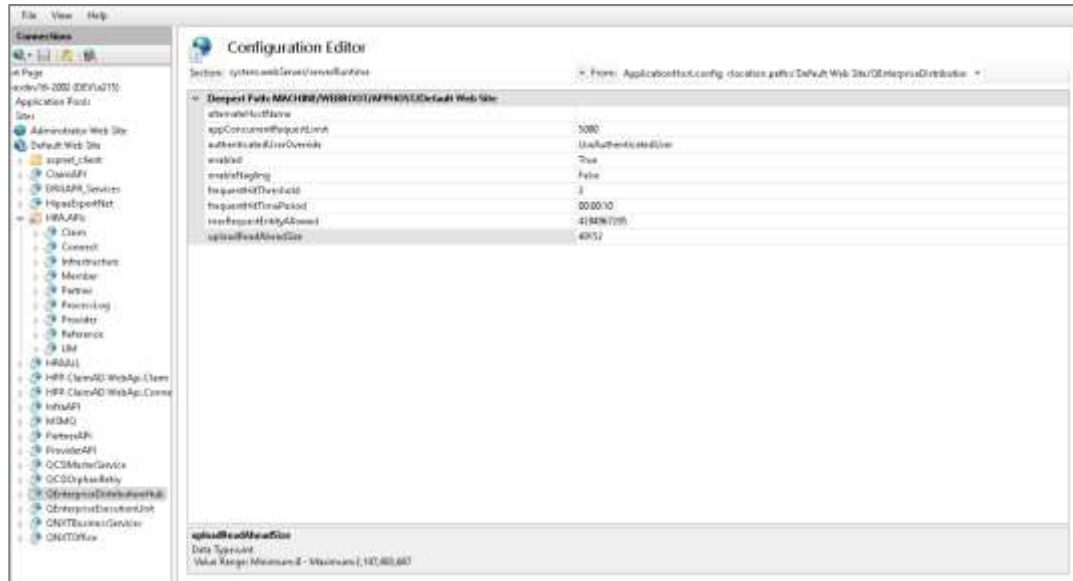
Configure UploadReadAheadSize for QEnterprise Distribution Hub in IIS

The **UploadReadAheadSize** property establishes the number of bytes a Web server will read into a buffer. The default value is 49152 (49KB). The value must be between 0 and 2147483647.

To process large files in **Connect 6020_275** transaction, UploadReadAheadSize limit should be increased in IIS as per the request size. Recommended value is 1073741824(1GB).

To change the value of UploadReadAheadSize:

1. Open IIS Manager
2. Select **QEnterpriseDistributionHub** application.
3. Double click "Configuration Editor."
4. Select system.webServer and then serverRuntime.
5. Modify the uploadReadAheadSize value (specify in bytes).
6. Click "Apply"



Configuring the Execution Unit

After you have configured the Distribution Hub, you can now configure each Execution Unit.

Dedicate QEnterprise Execution Unit to a Specific TransactionType

QEnterprise installs the `Config.xml` file on the server with a series of HIPAA Transaction type messages included under a single Execution Unit. This is the starting point for customers to customize the XML. This section describes editing the XML code, so only the messages processed by a specific EU are handled by that EU. Other EUs have their own `Config.xml` file and may process a different set of transaction types.

NOTE: Do not edit a `Config.xml` file by deleting the unwanted messages because it causes the loss of message code lines and makes them unavailable for future customization.

Gainwell Technologies recommends that you complete the following steps to comment out messages not currently needed for the given EU, thereby retaining those code lines for future used, if needed.

Editing an Execution Unit Configuration File

To edit an Execution Unit XML configuration file:

1. Open Windows Explorer and navigate to `\\Program Files\QCSI\QFrame\Services\Execution Unit\Web`.
2. Open `Config.xml` with Microsoft Windows Notepad.
3. Locate the code section that begins with `<Messages>`.

4. Cut the messages and paste them below the `<Messages>` tag.

```

<?xml version="1.0" encoding="utf-8" ?>
<QIS>
  <ExecutionUnit
    HTTP_PORT="10001"
    TCP_PORT="10000"
    URI="QISExecutionUnit1"
    MACHINE="Execution Unit Server"
    MESSAGE_STORE="Message Store DB Server"
    MESSAGE_STORE_DB="Message store DB">
    <Messages>
      <Message LogType="AdjustClaim"/>
      <Message LogType="ADV_CAP_BATCH"/>
      <Message LogType="ADV_CAP_PROC"/>
      <Message LogType="AuthSave"/>
      <Message LogType="AuthValidation"/>
      <Message LogType="AutoPCAssign"/>
      <Message LogType="Biztalk-270"/>
      <Message LogType="Biztalk-271"/>
      <Message LogType="Biztalk-276"/>
      <Message LogType="Biztalk-277"/>
      <Message LogType="Biztalk-278"/>
      <Message LogType="Biztalk-278e"/>
      <Message LogType="Biztalk-278e1"/>
      <Message LogType="Biztalk-820"/>
      <Message LogType="Biztalk-820e"/>
      <Message LogType="Biztalk-834"/>
      <Message LogType="Biztalk-834e"/>
      <Message LogType="Biztalk-835"/>
      <Message LogType="Biztalk-837"/>
      <Message LogType="Biztalk-837e"/>
      <Message LogType="CalcBenefit"/>
      <Message LogType="CalcCobSavings"/>
      <Message LogType="CapProvProc"/>
      <Message LogType="CapRestartProc"/>
      <Message LogType="CheckZip"/>
      <Message LogType="ClaimAdjud"/>
      <Message LogType="ClaimAutoAdjust"/>
      <Message LogType="ClaimExport"/>
      <Message LogType="ClaimFileLoad"/>
      <Message LogType="ClaimImport"/>
      <Message LogType="ClaimImportBulk"/>
      <Message LogType="ClaimMassAdjud"/>
      <Message LogType="ClimResetSvcUnit"/>
      <Message LogType="CreateRefund"/>
    </Messages>
  </ExecutionUnit>
</QIS>

```

5. Complete the following steps to comment out messages for unnecessary transaction types.
 - 5.1 Create a new line by pressing **Enter** at the end of the last transaction to be managed by the Execution Unit.
 - 5.2 Type `<!--` on the new line to comment out code that follows it.
 - 5.3 Locate the end Messages tag: `</Messages>`.
 - 5.4 Add another new line by pressing **Enter** at the end of the line above the `</Messages>` tag.
 - 5.5 On the new line, type `-->` to mark the end of the commented-out lines.

```

MESSAGE_STORE="Message Store DB Server"
MESSAGE_STORE_DB="Message store DB">
<Messages>
  <Message LogType="AdjustClaim"/>
  <Message LogType="ADV_CAP_BATCH"/>
  <Message LogType="ADV_CAP_PROC"/>
  <Message LogType="AuthSave"/>
  <Message LogType="AuthValidation"/>
  <Message LogType="AutoPCAssign"/>
  <!--
  <Message LogType="Biztalk-270"/>
  <Message LogType="Biztalk-271"/>
  <Message LogType="Biztalk-276"/>
  <Message LogType="Biztalk-277"/>
  <Message LogType="Biztalk-278"/>
  <Message LogType="Biztalk-278e"/>
  <Message LogType="Biztalk-278e1"/>
  <Message LogType="Biztalk-820"/>
  <Message LogType="Biztalk-820e"/>
  <Message LogType="Biztalk-834"/>
  <Message LogType="Biztalk-834e"/>
  <Message LogType="Biztalk-835"/>
  <Message LogType="Biztalk-837"/>
  <Message LogType="Biztalk-837e"/>
  -->
  <Message LogType="CalcBenefit"/>
  <Message LogType="CalcCobSavings"/>
  <Message LogType="CapProvProc"/>
</Messages>

```

6. Save changes made to the `Config.xml` file.
7. Restart the QEnterprise Execution Unit service, then the QEnterprise Distribution Hub service to complete the changes.

Configuring the Web configuration files for DH and EU

1. Open Microsoft Windows Explorer and navigate to
`C:\Program Files\QCSI\QFrame\Services\Distribution Hub\Web`
2. Find and open **web.config** file in a text editor such as Microsoft Windows Notepad.
3. Locate the section of the code that begins as:
`<allow roles`
4. Uncomment the section highlighted in the image below and update the DomainUsers group. This setting is applicable to EU web.config file as well.
5. Save the file.



```
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <system.web>
    <system.codedom>
      <compilers>
        <compiler language="c#;cs;csharp" type="Microsoft.CodeDom.Providers.DotNetCompilerPlatform.CSharpCodeProvider, Microsoft.CodeDom.Providers.DotNetCompilerPlatform, Version=4.5.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35" />
        <compiler language="vb;vbs;visualbasic;vbscript" extension=".vb" warningLevel="4" compilerOptions="/langversion:default /nowarn:41008,41009,41010" />
      </compilers>
    </system.codedom>
    <system.webServer>
      <security>
        <requestFiltering>
          <requestLimits maxAllowedContentLength="1073741824" />
        </requestFiltering>
      </security>
    </system.webServer>
    <location path="default.aspx">
      <system.web>
        <authorization>
          <!--<allow roles="DEV\HPASServersAdmin" />-->
          <deny users="*" />
        </authorization>
      </system.web>
    </location>
  </system.web>
</configuration>
```

VIII HPA Installation

This chapter includes the following topics:

- [HPA Requirements](#)
- [Installing HPA](#)
- [Configuring the ICD 10 Translator](#)

HPA Requirements

The following sections describe HPA requirements.

NOTE: Previous versions of HPA component installation programs included logic that notified users if an HPA component was being installed without correct prerequisites in place. Currently, HPA 5.0 component installation wizards do not automatically check for installation prerequisites. This is a known issue and will be resolved in a future change package. Make sure that all prerequisites are in place before installing each HPA component. For prerequisite information, see “Requirements” sections in applicable HPA component chapters.

HPA Installation Prerequisites

The following requirements must be in place before installing HPA:

- All core installation requirements and important prerequisites. See [Requirements](#) for more information.
- QFramework - See [Installing QFramework](#) for installation requirements.

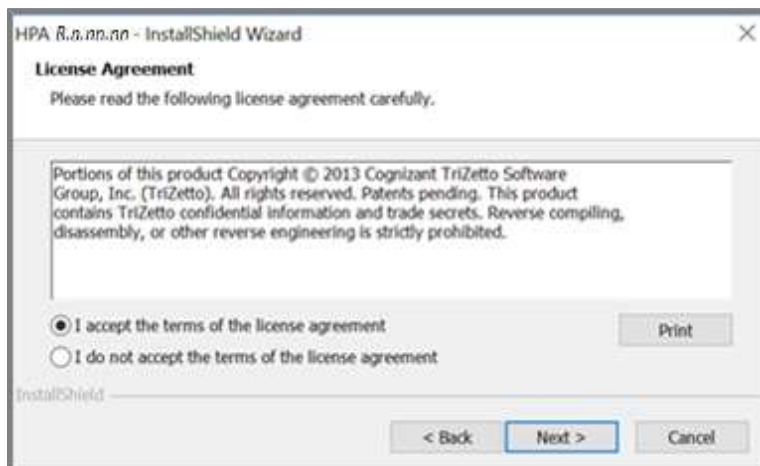
HPA Optional Requirements

If using Micro-Dyn components with HPA, you must install them before installing HPA. See [Micro-Dyn Preparation and Setup](#) for more information.

Installing HPA

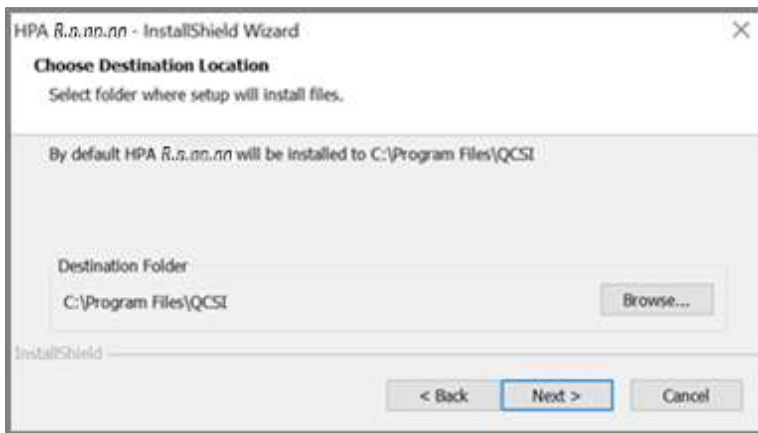
After all prerequisites have been loaded and configured, you can install HPA. To install the HPA core application:

1. Browse to the HPA Installation Package, then run the `HPA_<version>_x64.exe` installation file.
2. Click **Next** on the Welcome window when the InstallShield Wizard starts.
3. After reviewing the license agreement, select **I accept the terms of the license agreement** and click **Next** on License Agreement window.



4. Accept the default destination folder or click **Browse** to select a different folder for the installation and click **Next** on the Choose Destination Location window.

NOTE: The destination folder must be the same for all HPA products installed.

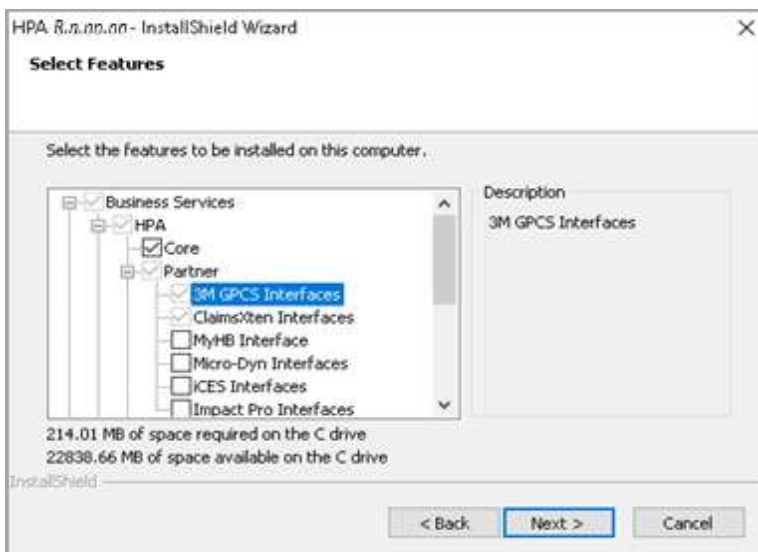


NOTE: Business Services must be installed on all servers where HPA Connect will be installed. Otherwise, the HPA Connect installation will fail.

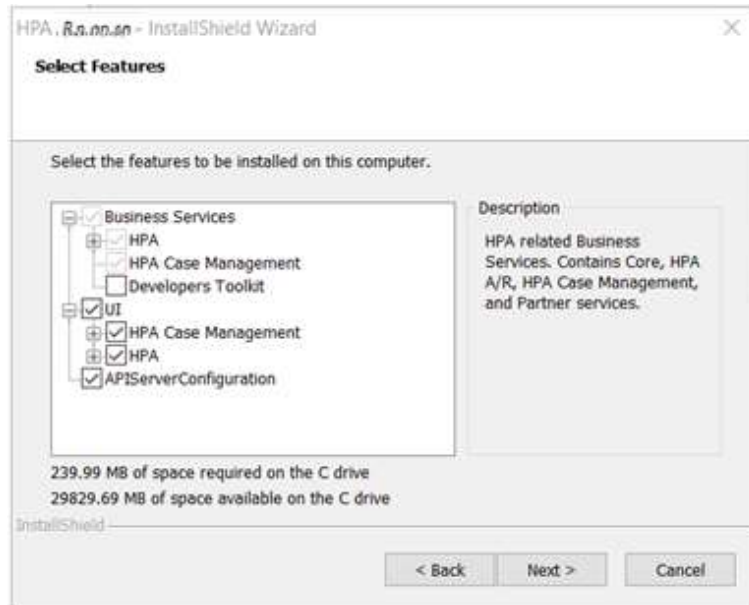
5. Select the features to install and click **Next** on the Select Features window.

IMPORTANT: You must activate a separate application license before you can install the **MyHB Interface**, **Micro-Dyn Interfaces**, **iCES Interfaces**, and **Impact Pro Interfaces**.

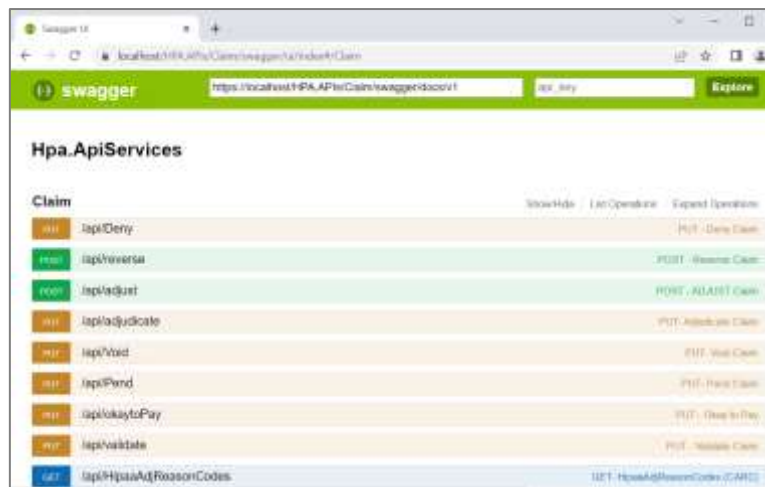
- If you select **Impact Pro Interfaces**, see the [Impact Pro Interface step](#) for configuration options.
 - If you select **Developers Toolkit**, the software development kit (SDK) is deployed in `C:\Program Files\QCSI\SDK`.
 - For all other options selected, continue to the next step
-



- 5.1 Select the feature **API Server configuration** on Select Features window to install HPA APIs.

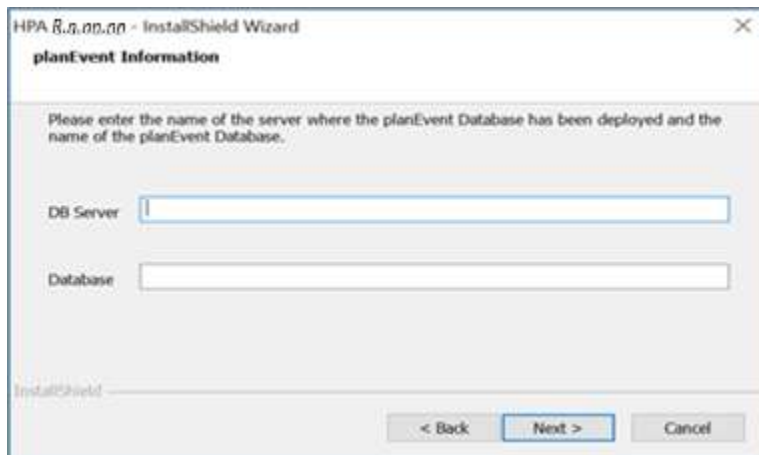


Verify SSL is configured (See [Configure SSL Certificate](#)) and test HPA Claim API service to see if it responds on using SSL connection. In a browser URL bar paste the link below:
<https://localhost/HPA.APIs/Claim/swagger/ui/index>. The browser should display a UI like the one below.



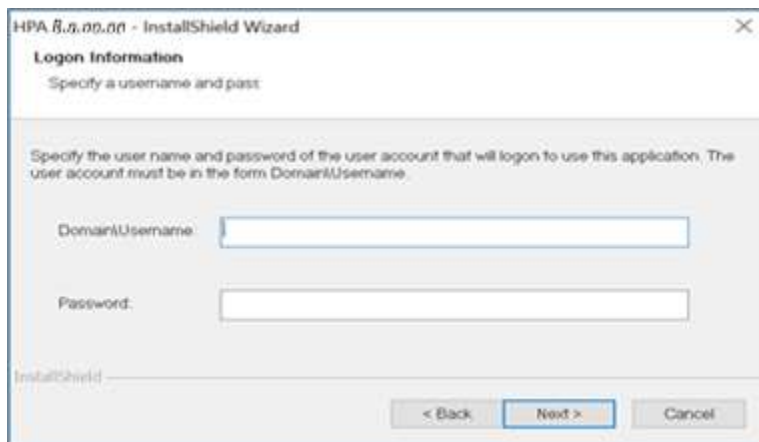
6. Complete the following actions on the planEvent Information window.
 - 6.1 Type the name of the server where the planEvent database has been deployed in the DB Server box.

6.2 Type the name of the planEvent database in the Database box and click **Next**.



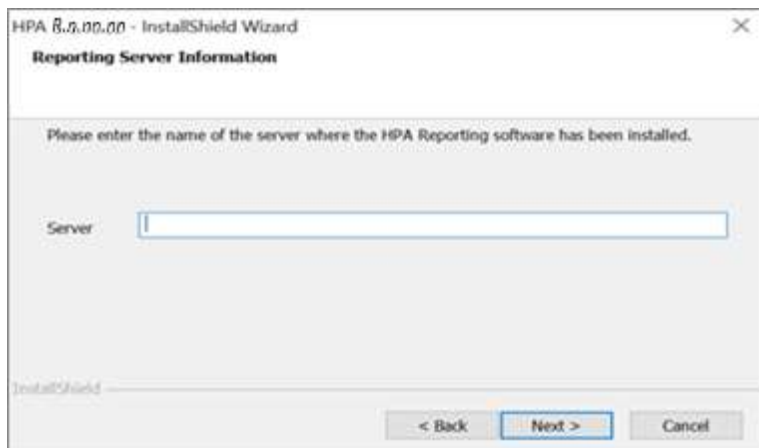
7. Type the user name and password for the domain account with system administration privileges and click **Next** on the Logon Information window.

NOTE: The user name must be entered in the domain\username format.



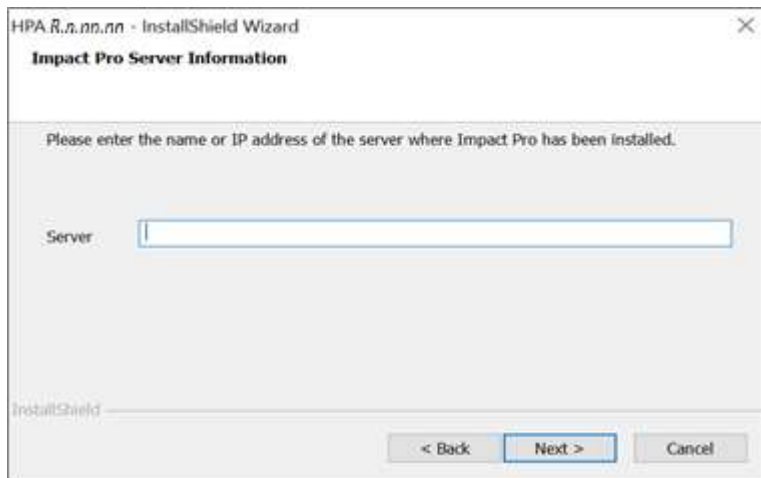
8. If the Reporting Server Information window appears, type the name of the server where the HPA Reporting software is installed and click **Next**.

NOTE: The following window displays only if you selected the UI/HPA option from the Select Features list. See the [Select Features step](#).



- If the Impact Pro Server Information window appears, type the name of the Impact Pro Server and click **Next**.

NOTE: The following window displays only if you selected Impact Pro Interface from the Select Features list. See the [Select Features step](#).

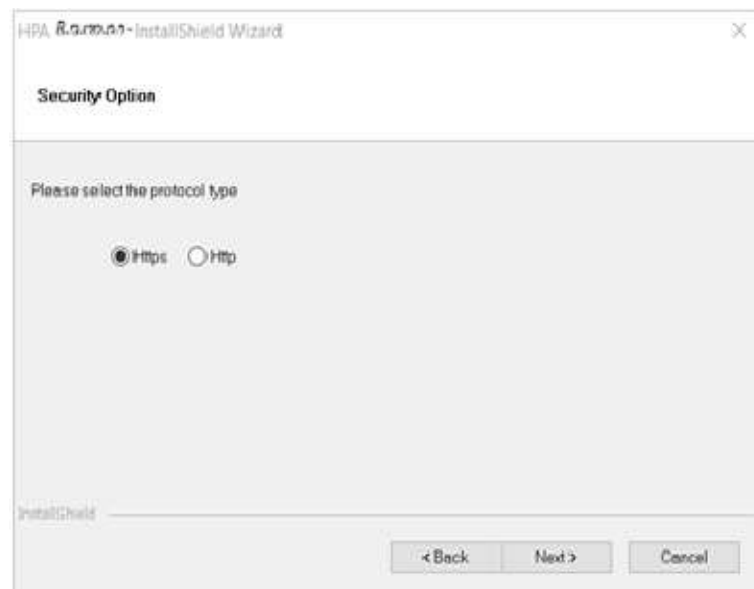


- Select the protocol type on the Security option window.

NOTE: Protocol type should be same for all installers (QFramework, HPA and HPA Connect)

If Https is enabled, SSL needs to be configured on the server where HPA is installed (See [Configure SSL Certificate](#))

Https protocol should be used while configuring any external products in the web.config file.



- Review installation details and click **Next** on the Start Copying Files window to start the installation.

A progress indicator shows the installation's status.

- After the installation is complete, click **Finish** to exit the wizard.

After installing QFramework and HPA, you can configure the ICD-10 translator for your system. See [Configuring the ICD-10 Translator](#).

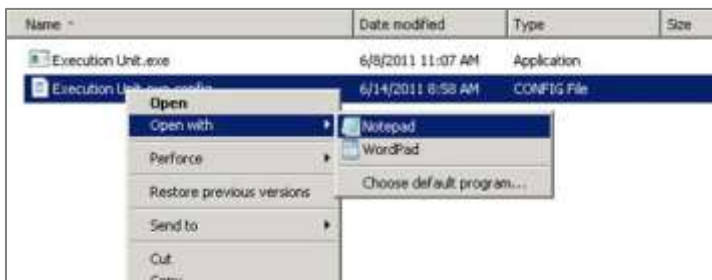
Configuring the ICD-10 Translator

You must configure the Execution Unit.exe.config and Web.config files to communicate with the IHMF (Integrated Healthcare Management Foundation) 1.52 ICD10 Translator manager so that information can be passed back and forth between the ICD-10 Translator and HPA.

Configuring the Execution Unit.exe.Config File

To configure the Execution Unit.exe.Config file:

1. Right-click **Start**, then select **Windows Explorer**.
2. Select **Program Files | QCSI | QFrame | Services | Execution Unit**.
3. Right-click the `Execution Unit.exe.config` file and select **Open with | Notepad**.



4. After the file opens in Notepad, locate the IHMF ICD10 Translator.

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="" />
<add key="IHMF_ICD10_MappingSet" value="" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

5. Add the Service Address value: `http://YourIPAddress/ICDTMTranslateService?wsdl`

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

6. Add the Mapping Set value: `HPA01`

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="QNXT01" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

7. Add the Protocol value: `HTTP`

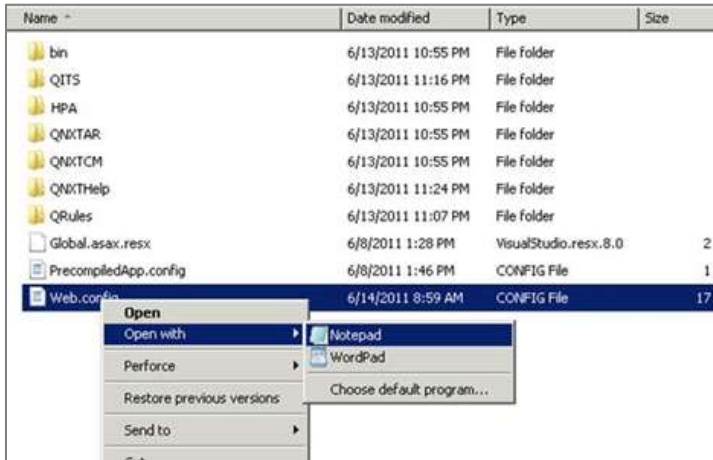
```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="QNXT01" />
<add key="IHMF_ICD10_Protocol" value="HTTP" /><!-- Expected Values: HTTP, TCP -->
```

8. Save and close the file.

Configuring the Web.exe.Config File

To configure the Web.exe.Config file:

1. Right-click **Start**, then select **Windows Explorer**.
2. Select **Program Files | QCSI | QNXT | Web**.
3. In the right pane, right-click the **Web.config** file and select **Open with | Notepad**.



4. After the file opens in Notepad, locate the IHMF ICD10 Translator..

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="" />
<add key="IHMF_ICD10_MappingSet" value="" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

5. Add the Service Address value: `http://YourIPAddress/ ICDTMTranslateService?wsdl`

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

6. Add the Mapping Set value: `QNXT01`

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="QNXT01" />
<add key="IHMF_ICD10_Protocol" value="TCP" /><!-- Expected Values: HTTP, TCP -->
```

7. Add the Protocol value: `HTTP`

```
<!-- IHMF ICD10 Translator -->
<add key="IHMF_AppServer" value="http://YourIPAddress/ICDTMTranslateService?wsdl" />
<add key="IHMF_ICD10_MappingSet" value="QNXT01" />
<add key="IHMF_ICD10_Protocol" value="HTTP" /><!-- Expected Values: HTTP, TCP -->
```

8. Save and close the file.

IX HPA Connect Installation

This chapter includes the following topics:

- [HPA Connection Requirements](#)
- [Installing HPA Connect on a BizTalk Server](#)
- [Verifying the HPA Connect Installation](#)
- [Verifying BizTalk Server Configuration Tools](#)
- [Installing HPA Connect Business Services](#)
- [Verifying Business Services Server Tools](#)
- [Configuring HPA Connect Orphan Retry](#)
- [Installing HPA Connect HIPAA API](#)
- [Configuring HPA Connect](#)

HPA Connect Requirements

The following HPA components must be installed and configured before installing HPA Connect:

- QFramework (see [QFramework Installation](#))
- HPA (see [HPA Installation](#))
- HPA Business Services (see [Installing HPA Connect Business Services](#))

If you are installing HPA Connect on a BizTalk Server, you must have Microsoft SQL Server installed.

If you are installing the SQL and BizTalk servers on two separate servers, confirm domain access. BizTalk Server requires servers to be on a domain (a “workgroup” will NOT work). There are services configured to run under a domain account that must exist on all servers (including SQL Servers housing the BizTalk Server and HPA Connect databases).

Installing HPA Connect on a BizTalk Server

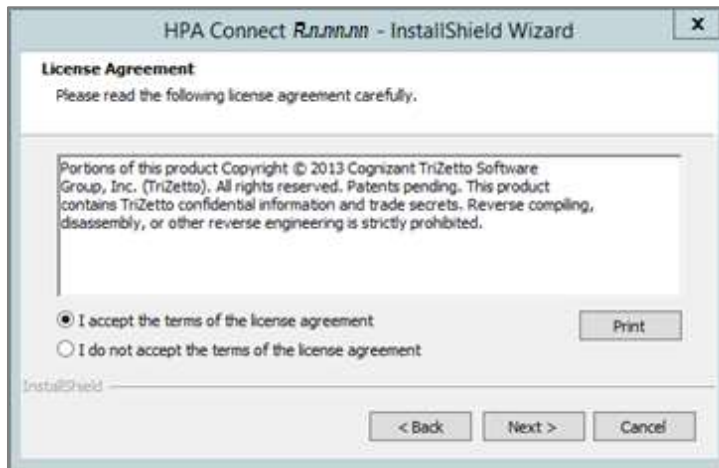
If you are a current HPA client and you already have trading partners configured, you can import the file with the trading partner information during installation process. See [step 8](#).

To install HPA Connect on a BizTalk server:

1. Browse to the HPA Installation Package and run the `HPA_Connect_<version>_x64.exe` installation file.

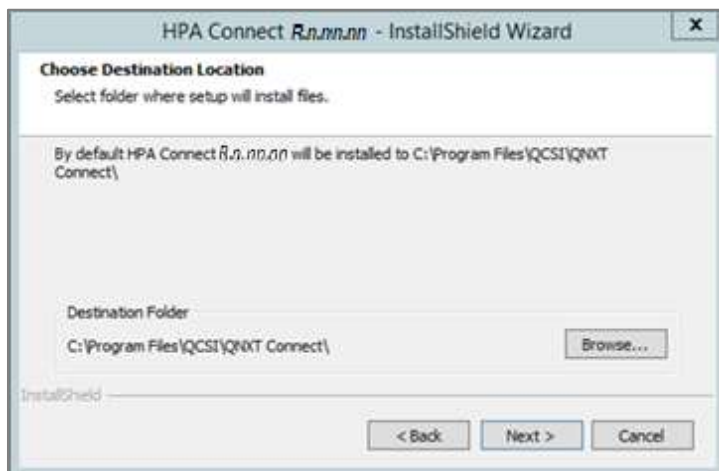
The InstallShield Wizard starts and the Welcome window appears.

2. Click **Next**.
3. Review the license agreement and select **I accept the terms of the license agreement**, then click **Next** on the License Agreement window.



4. Accept the default destination folder or click **Browse** to select a different folder for the installation, then click **Next** on the Choose Destination Location window.

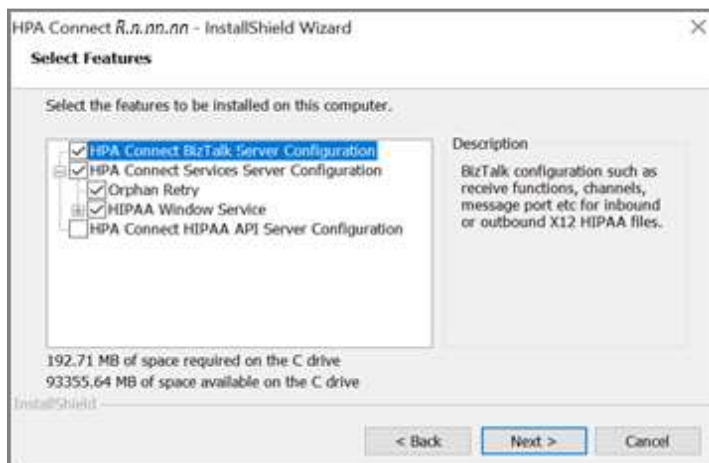
NOTE: The destination folder must be the same for all HPA products installed.



5. Complete the following actions on the Select Features window.
 - 5.1 Select **HPA Connect BizTalk Server Configuration** and **HPA Connect Services Configuration**.

NOTE: This installation recommends multiple servers. For more information, see [Installing HPA Connect Business Services](#).

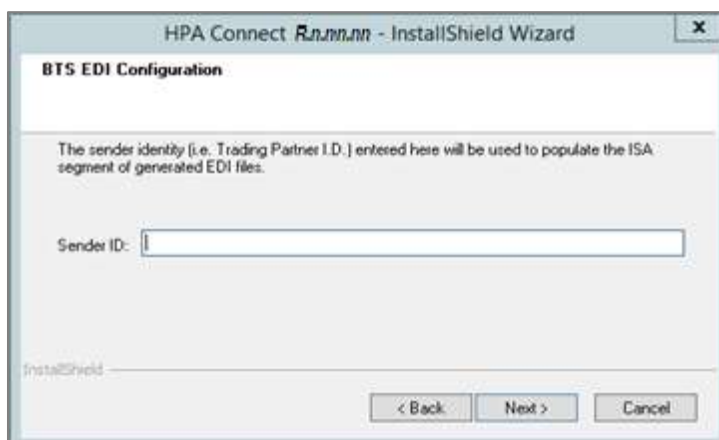
5.2 Click **Next**.



6. Complete the following action on the COM+ Identity Security Dialog window.
 - 6.1 Type the domain user name (must be an administrator) in the User name box.
 - 6.2 Type the password in the Password box.
 - 6.3 Click **Next**.



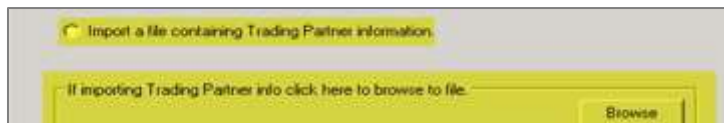
7. Type your trading partner identity in the Sender ID box and click **Next** on the BTS EDI Configuration window.



8. Refer to the following notes and select the appropriate option for your installation, then click **Next** on the Import BTS Configuration window.



NOTE: If you select **Import a file containing Trading Partner information**, you must import the file that you want to import, as shown below.

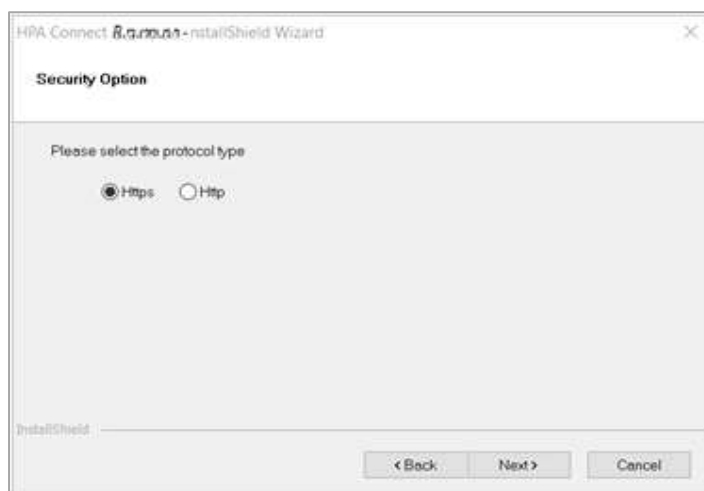


IMPORTANT: If you are a current HPA client with existing trading partner configurations (e.g., alias, receive port, receive location for the 837 and/or alias, receive port, receive location, send port for the 835), you must export the file before it can be imported. For more information, see the BizTalk Server documentation.

9. Select the protocol type on the Security option window.

NOTE: Protocol type selected should be same for all installers (**QFramework, HPA and HPA Connect**)

If Https is enabled, SSL needs to be configured on the server where HPA is installed (See [Configure SSL Certificate](#))



10. Review installation details and click **Next** on the Start Copying Files window to start the installation.

The progress indicator shows the installation status.

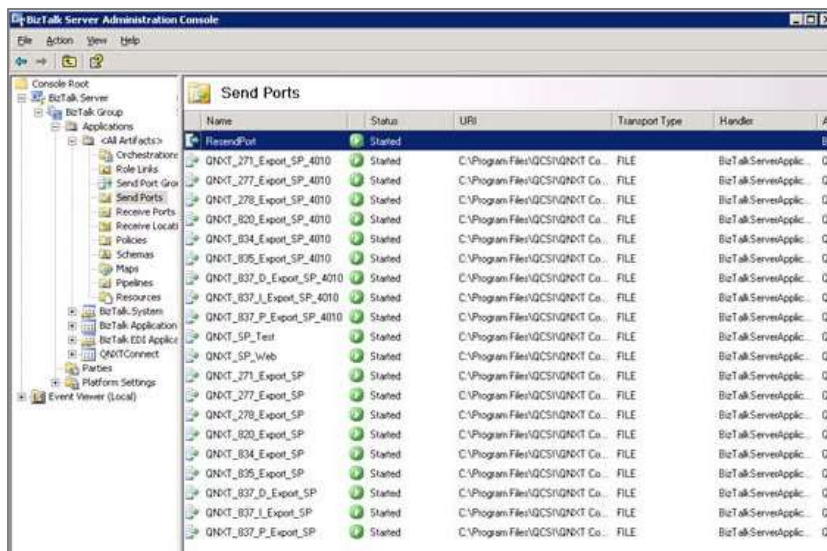
11. After the installation is complete, click **Finish** to exit the wizard.

Verifying the HPA Connect Installation

You can verify the installation of HPA Connect by viewing the Send Ports section in the Microsoft BizTalk Administrative Console.

To verify that HPA Connect is installed:

1. Select **Start | All Programs | Microsoft Biztalk Server | Biztalk Server Administration**.
2. In the left pane, select **Console Root | BizTalk Server Administration | BizTalk Group | Applications | <All Artifacts> | Send Ports**.



3. Verify that HPA Connect send ports are available and started.

Verifying BizTalk Server Configuration Tools

If the BizTalk server configuration was used, verify that utilities and applets in the following table are available.

Utilities and Applets Associated with BizTalk supporting QNXT Connect

Utility / Applet	Used to...
Tools\Custom 820 File Splitter	split large 820 files because of BizTalk file size limitations.
Tools\X12 File Merger	merge separate X12 documents into one file.
Outbound website	verify path/location of the Web service to receive the outbound transactions.

To view utilities and applets:

Select **Start** | **All Programs** | **TriZetto** | **QNXT Connect** | **Tools**.

Installing HPA Connect Business Services

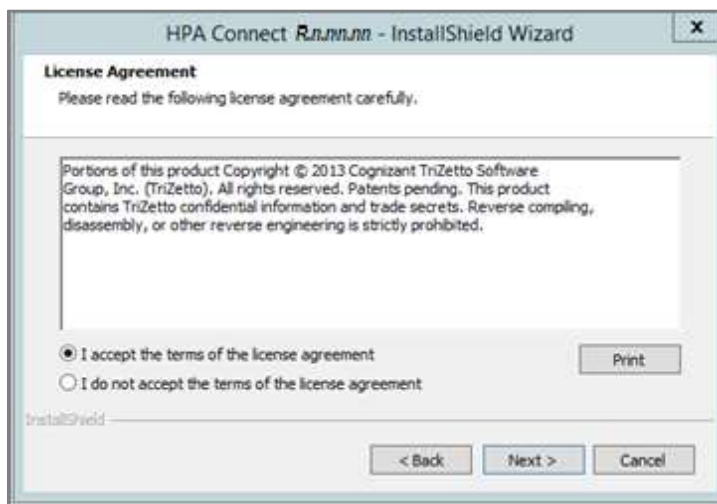
This section describes how to install HPA Connect Business Services on the Execution Unit or Distribution Hub servers. If you are using QFramework, HPA Connect Services must be installed on each Execution Unit server.

To install HPA Connect Business Services:

1. Browse to the QNXT Installation Package, then run the `HPA_Connect_<version>_x64.exe` file.

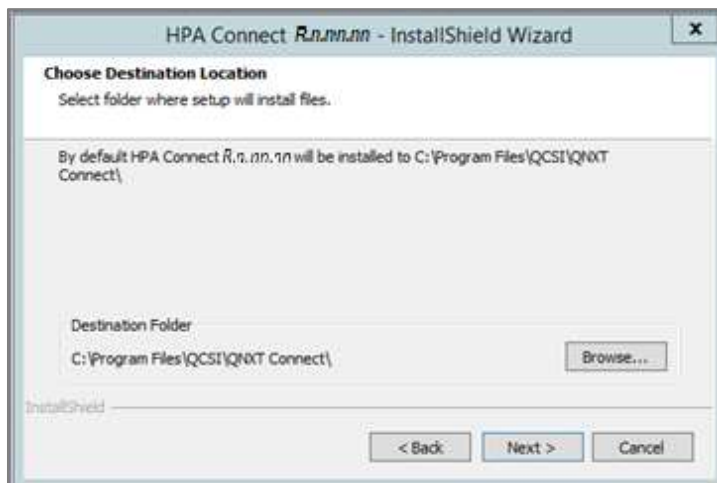
The InstallShield Wizard starts.

2. Click **Next** on the Welcome window.
3. Review the license agreement and select **I accept the terms in the license agreement**, then click **Next** on the License Agreement window.



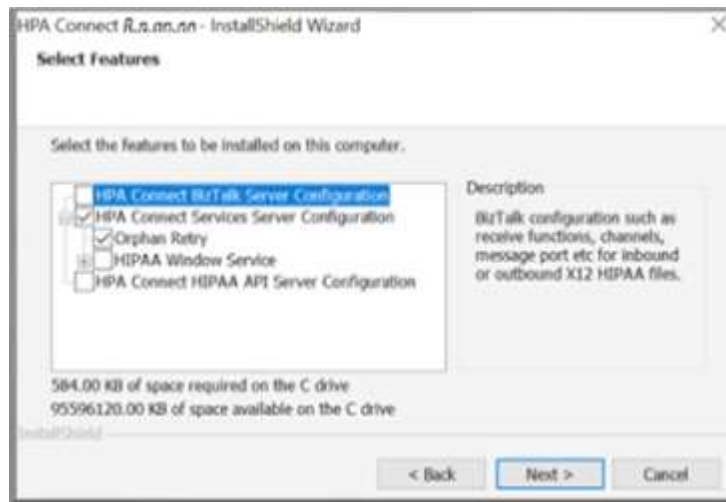
4. Accept the default destination folder or click **Browse** to select a different folder for the installation and click **Next** on the Choose Destination Location window.

NOTE: The destination folder must be the same for all HPA products installed.

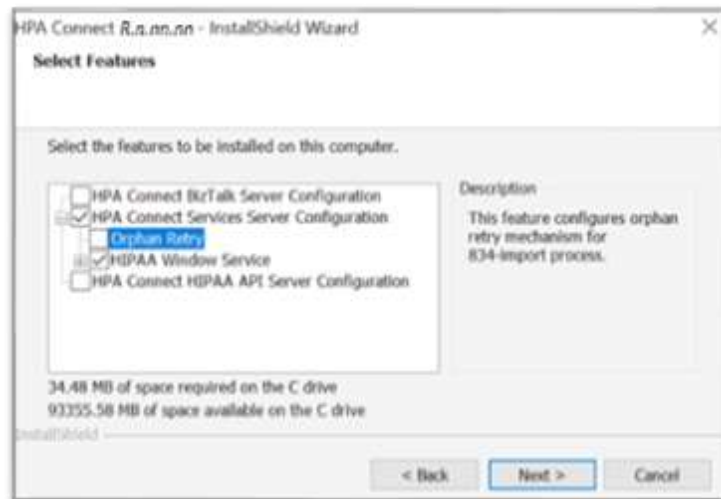


5. Use the following list and examples to make selections, then click **Next** on the Select Features window.

- If you are installing Business Services features on an Execution Unit, select **HPA Connect Services Server Configuration** and **HIPAA Window Service**.



- If you are installing Business Services features on a Distribution Hub Server, select **HPA Connect Services Server Configuration** and **Orphan Retry**. For configuration instructions, see [Configuring HPA Connect Orphan Retry](#).



6. Complete the following actions on the COM+ Identity and Security Dialog window.
 - 6.1 Type the domain user name (must be an administrator) in the User name box.
 - 6.2 Type the password in the Password box.
 - 6.3 Click **Next**.



7. Complete the following actions on the Transaction Processing Servers window.
 - 7.1 Type the name of the server where the HPA Connect BizTalk components **or** HPA Connect HIPAA API components are installed in the HIPAA Export Server box. For example, enter the BizTalk server name for the BizTalk option **or** HIPAA API server name for the non-BizTalk option.
 - 7.2 Type the name of the Distribution Hub server where the Orphan Retry Server is installed in the Orphan Retry Server box.
 - 7.3 Click **Next**.



8. Review installation details and click **Next** on the Start Copying Files window.

The progress indicator shows the installation status.

9. After the installation is complete, click **Finish** to exit the wizard.

Verifying Business Services Server Tools

If you are using Business Services server configuration, make sure that the following utilities and applets are available.

Utilities and Applets Associated with QNXT Connect

Utility / Applet	Description
QCSI Orphan Retry	Enables configuration of wait time intervals, etc., for the Orphan Retry (used for 834 processing).
Orphan Retry	Verifies path/location of the Web service that processes orphaned 834 transactions.
Tools \ 834 File Manager	Part of EDI Manager module functionality used to support HPA Connect 5.0 and allows you to manually submit enrollments imported using 834 Import process when marked Audit or Compare by QNXT during HIPAA 834 Import processing.

The HPA Connect 5.0 release includes Port/Channel configuration that supports transactions for the IG version 4010A1(Addenda), as indicated by presence of the A1 in the naming conventions used.

To verify the utilities and applets:

Select **Start** | **All Programs** | **TriZetto** | **QNXT Connect** | **Tools** to view the various tools that were installed.

Configuring HPA Connect Orphan Retry

Orphan Retry Service

The Orphan Retry feature in HPA Connect helps you manage situations in which dependent child member information may post to the HPA translation component before the parent member information is posted. When the Orphan Retry service is properly configured, the HPA 834 import process checks for conditions to ensure that when dependent child data arrives before parent member data, that data is flagged for processing after the parent member information has also arrived.

When an orphan child transaction is first detected, the 834 import process increments the retry count in the XML transaction and posts it to the orphan retry website. The ASP Orphan Retry application inspects the incoming XML document and checks to see if it is the first time the data has been received. If it is, the document is written to the \\Program Files\\QCSI\\QNXT Connect\\OrphanRetry\\Windows\\Wait folder.

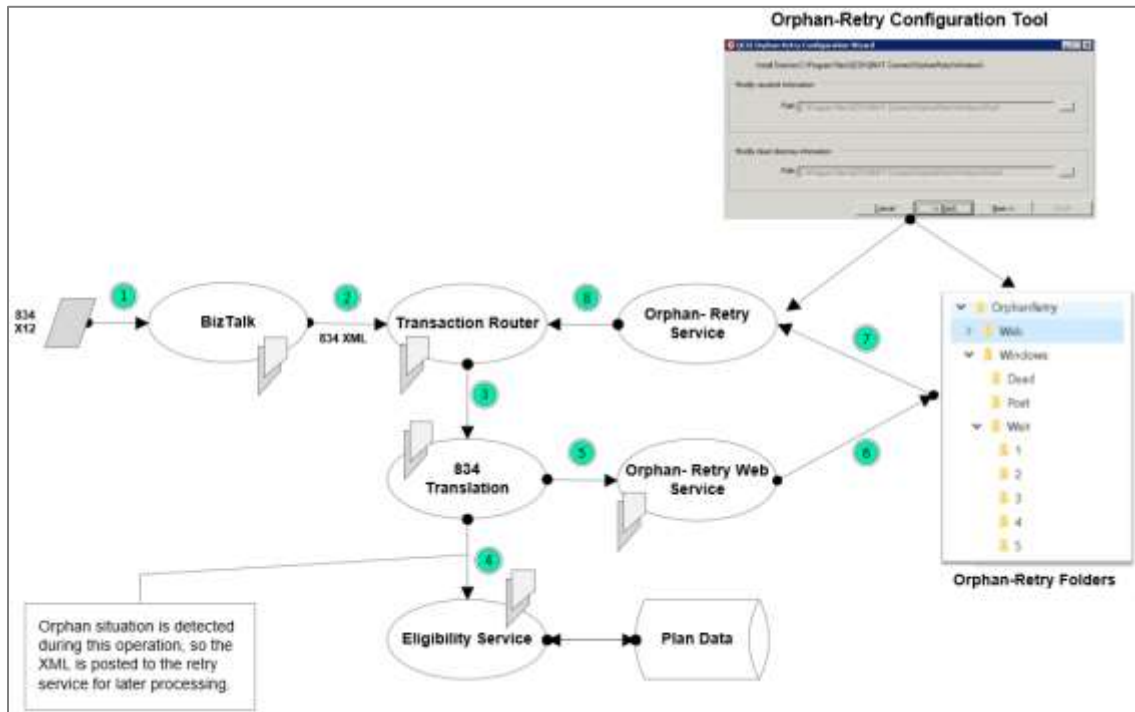
The orphan retry service monitors these folders and is responsible for moving XML documents after a user-configured elapsed time to another folder called Post in the same folder hierarchy. When files are available in the Post folder, the Orphan Retry service forwards them to the Transaction Router (QEnterprise Distribution Hub) for further processing. In this manner the orphan records are retried after a user-configurable elapsed time.

Even after the first delay, it is still possible the child data precedes parent data. The 834 import process detects this situation and increments a counter in the XML document to mark it as a second try and post it to the orphan retry site. The service monitoring the folders, detects the file, evaluates the counter, and places the file into a folder designated for second retries. The Orphan Retry service picks up the document after the user-specified elapsed time moves it to the Post folder as before and processing continues.

If the transaction is still an orphan after five retries, the 834-import process increments the counter to 6 and the transaction is relegated to the Dead folder. At this point, manual intervention is required.

Orphan-Retry Configuration Tool Flow

The following graphic shows the flow of the Orphan-Retry Configuration tool.



Configuring the Orphan Retry Service

After installing HPA Connect with the Orphan Retry service, you must configure Orphan Retry using the Orphan Retry configuration wizard. Currently, only the wait times between each retry can be altered.

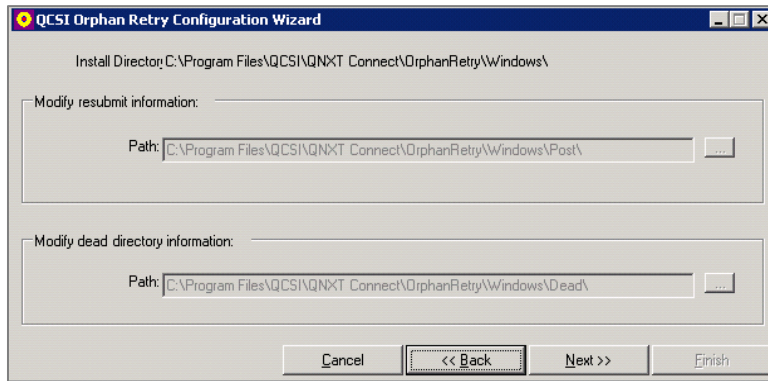
To configure the Orphan Retry service:

1. Select **Start | All Programs | TriZetto | QNXT Connect | Tools | QCSI Orphan Retry** to open the QCSI Orphan Retry Configuration Wizard.
2. Click **Next** to start the wizard.
3. Select each iteration to modify its value in the Wait Time box.



NOTE: You must click **Update** to save the change before modifying other values.

4. After making the applicable modifications, click **Next**. The options shown below are view only and cannot be changed.
5. Click **Next** on the following window with options that are view-only and cannot be changed.



6. Click **Finish** on the next window with consideration given to the following:
 - Use the default Trace setting, **Off (No logging)** or choose **Information (Minimal logging)** or **Verbose (Full logging)** to write trace information to the event log.

NOTE: Use of a non-default option is recommended to help troubleshoot issues.

- **Re-start orphan retry service** is selected by default.



7. Restart the Orphan Retry service to save the configuration changes.

Configuring HPA Connect

The following must be configured in HPA:

- **Configuration Options:** Allow use of HIPAA Transactions and Allow Non-NT Login into HPA configuration must be set to **Yes** for the working HPA environment.
- **Trading Partners:** Do the following:
 - Ensure the most recent copy of the `TradingPartnerCfg.xml` file is in `C:\Program Files\QCSI\QNXT Connect\Outbound\TradingPartnerCfg.xml` on the BizTalk server.
 - Identify trading partner-specific logins sent by trading partners in an inbound file (ISA06).
 - Set up corresponding accounts in HPA User Security. Refer to the *HPA Connect User Guide* for more information on configuring the ISA06 login and ISA04 password.
- **EDI Client:** Setup the corresponding EDI client through the HPA EDI Manager module.
- **EDI Formats:** Setup corresponding EDI formats for each trading partner. You must enable or disable EDI rules as applicable to the business application and set up translation values for transaction-specific translations.
- **Registry Keys:** Ensure the following registry keys on the application server are configured to the correct BizTalk server:
 - `HKEY_LOCAL_MACHINE\SOFTWARE\TriZetto\QNXT Connect\<version>\URLs\ExportHIPAA`
 - `HKEY_LOCAL_MACHINE\SOFTWARE\TriZetto\QNXT Connect\<version>\URLs\ExportServer`

In situations where HPA Connect has been reinstalled, you can import trading partner configurations that you exported (e.g., alias, receive port, receive location for the 837 and/or alias, receive port, receive location, send port for the 835) via BizTalk Administrator.

If you are a current HPA client, and you have existing trading partner configurations (e.g., alias, receive port, receive location for the 837 and/or alias, receive port, receive location, send port for the 835), you must export Configure the BizTalk server with the Application Server role. See [Configuring Security Permissions](#) for more information.

X Installing HPA Connect API Server

This section describes how to install HPA Connect Business Services on the Connect API server(s). The Connect API server replaces BizTalk, therefore a new server is recommended. In an existing installation, the BizTalk server can be repurposed by uninstalling the HPA and BizTalk products and completing the following steps.

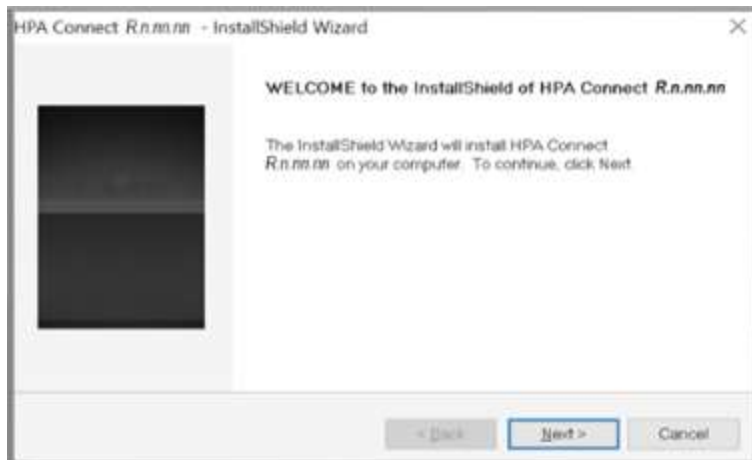
The Connect HIPAA API expects X12 XML format for import transactions; therefore, a standard X12 structured file is expected for conversion to X12 XML specifications before Connect HIPAA API is able to process the transaction. You can choose Biztalk or HIPAA API server configuration while installing HPA connect.

To install HPA Connect HIPAA API configuration:

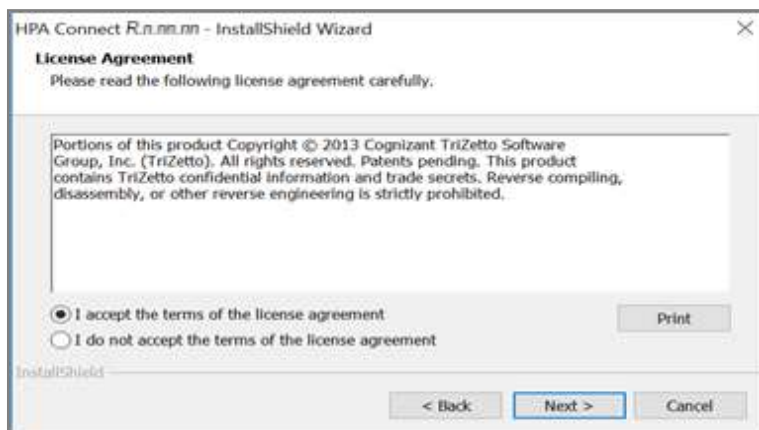
1. Browse to the QNXT Connect Installation Package, then run the following file to start the InstallShield Wizard.

```
HPA_Connect_<version>_x64.exe
```

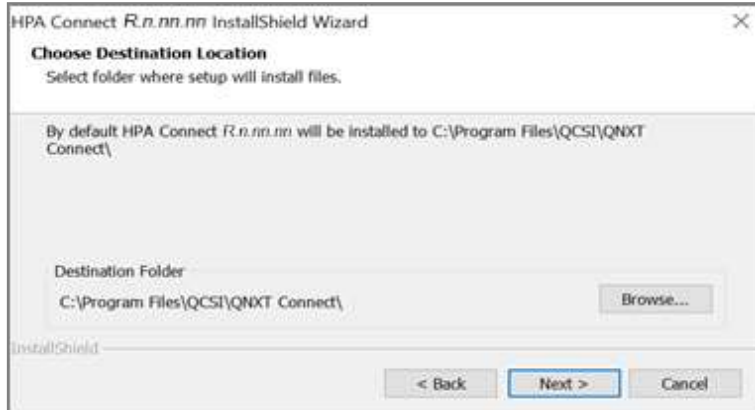
2. Click **Next** on the Welcome window.



3. Review the license agreement and select **I accept the terms of the license agreement**, then click **Next** on the License Agreement window.

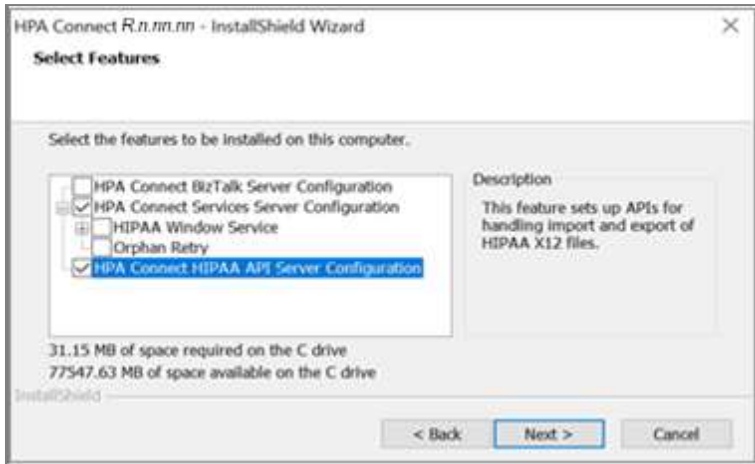


4. Accept the default destination folder or click **Browse** to select a different folder for the installation, then click **Next** on the Choose Destination Location window.

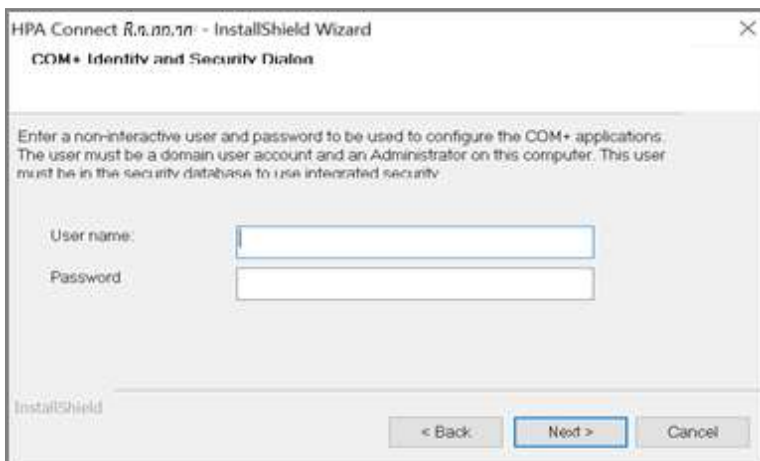


5. Select **HPA Connect HIPAA API Server Configuration** and click **Next** on the Select Features window.

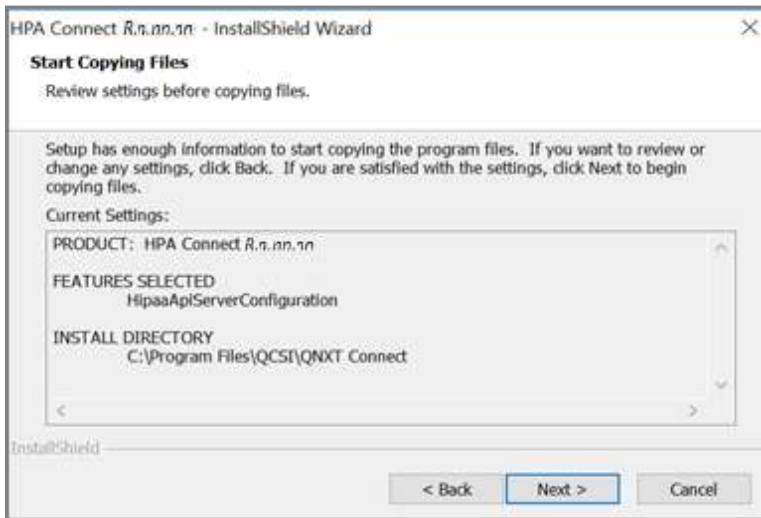
NOTE: You cannot select both BizTalk and API server configuration at the same time.



6. Complete the following actions on the COM+ Identity and Security Dialog window.
 - 6.1 Type the user name for the domain (must be an Claim Administrator service account) in the User name box.
 - 6.2 Type the password in the Password box, then click **Next**.

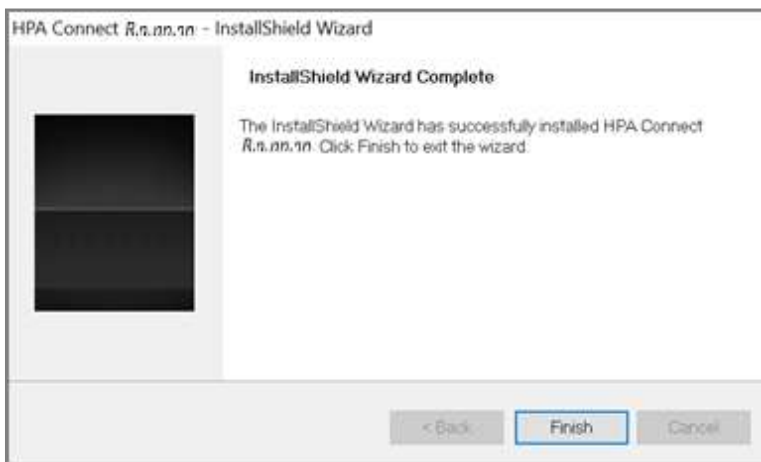


7. Click **Next** on the Start Copying Files window.



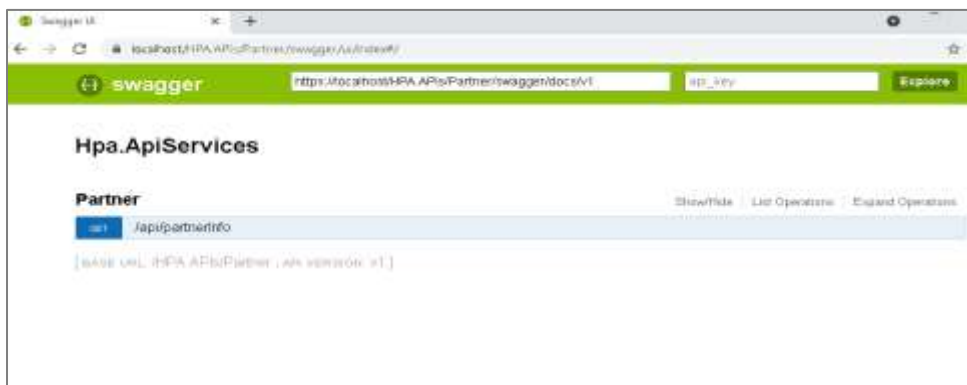
The progress indicator shows the installation status.

8. After the installation is complete, click **Finish** to exit the wizard.



To configure SSL for HPA connect API server, See [Configure SSL Certificate](#).

Test HPA Connect Partner API service to see if it responds on using SSL connection. In a browser URL bar paste the link below: <https://localhost/HPA.APIs/Partner/swagger/ui/index>. The browser should display a UI like the one below.



Once HPA Connect installation is done, verify EDI Fabric serial key is added in the following registry path:

`HKEY_LOCAL_MACHINE\SOFTWARE\TriZetto\QNXT Connect\<version>\EDIFabricKey`. This serial key will be validated during HPA Connect 6020 275/277 processing.

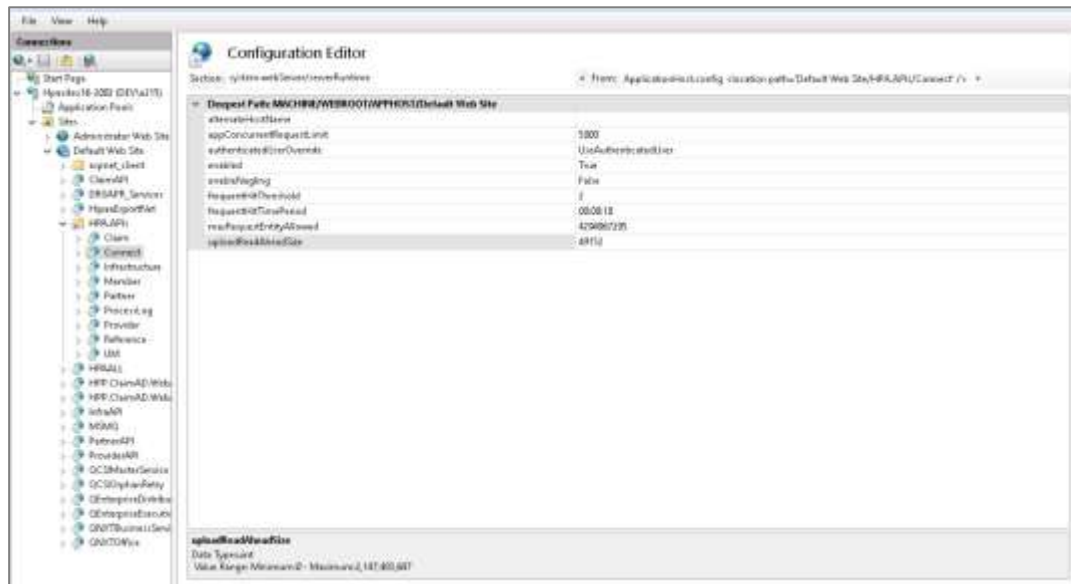
Configure UploadReadAheadSize for Connect API in IIS

The **UploadReadAheadSize** property establishes the number of bytes a Web server will read into a buffer. The default value is 49152 (49KB). The value must be between 0 and 2147483647.

To process large files in **Connect 6020_275** transaction, UploadReadAheadSize limit should be increased in IIS as per the request size. Recommended value is 1073741824(1GB).

To change the value of UploadReadAheadSize:

1. Open IIS Manager
2. Select the Connect application (Location Path: \HPA.APIs\Connect)
3. Double click “Configuration Editor.”
4. Select system.webServer and then serverRuntime.
5. Modify the uploadReadAheadSize value (specify in bytes).
6. Click “Apply”



XI HPA Reports Installation

This chapter includes the following topics:

- [HPA Reports Requirements](#)
- [Installing HPA Reports](#)
- [Configuring Model Reports](#)

HPA Reports Requirements

The following components must be installed and configured before installing HPA Reports:

- QFramework (see [QFramework Installation](#))
- HPA (see [HPA Installation](#))
- SQL Picklist database, created manually using the provided SQL script (`01 tlReportPickList Reference Data.sql`) against the ReportServer database (see [Installing the Picklist Table](#))
- SQL Server (refer to the Product documentation)
- SQL Reporting Services (refer to the Product documentation)

NOTE: HPA Reports only supports SQL Reporting Services 2022 on a single instance of SQL Server 2022.

Installing HPA Reports

The HPA Reports component provides reports and analytics to help monitor and plan system functions. Using roles and permissions, you can configure system reports to allow restricted access. See [Configuring Model Reports](#).

To install HPA reports:

1. Browse to the HPA Installation Package and run the `HPA_Reports_<version>_x64.exe` file.

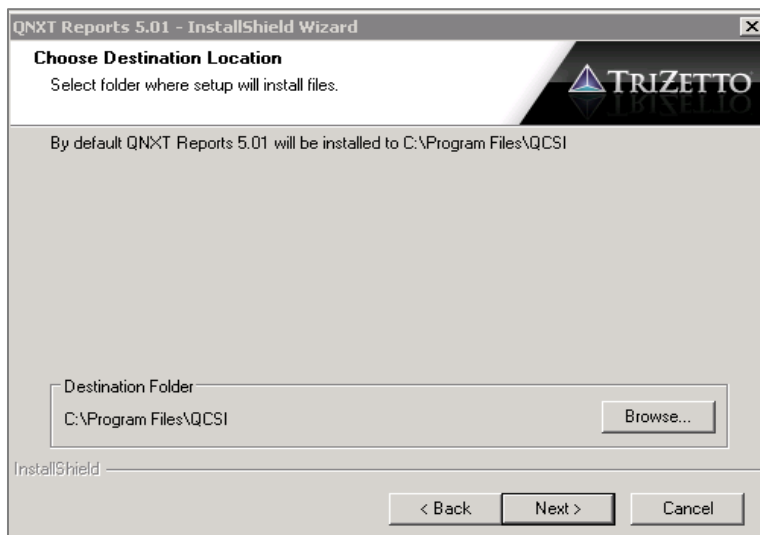
The InstallShield Wizard starts.

2. Click **Next** on the Welcome window.
3. Review the license agreement and select **I accept the terms of the license agreement**, then click **Next** on the License Agreement window.



4. Accept the default destination folder or click **Browse** to select a different folder for the installation, then click **Next** on the Choose Destination Location window.

NOTE: The destination folder must be the same for all HPA products installed.



5. Complete the following actions on the Plandata Location Information window.
 - 5.1 Type the name of the SQL server from which reports are processed in the SQL Server Name box.
 - 5.2 Type the name of the plandata database from which reports are processed in the Database Name box.
 - 5.3 Click **Next**.

6. Complete the following actions on the Picklist Location Information window.
 - 6.1 Type the name of the server from which reports are processed in the SQL Server Name box.

Tip! The Picklist table must be created manually using MS SQL Server before you can specify the database name as required in the next installation step. See [Installing the Picklist Table](#) for more information.

 - 6.2 Type the name of the picklist database from which reports are processed in the Database Name box.
 - 6.3 Click **Next**.

7. Complete the following actions on the Security Database Location Information window.
 - 7.1 Type the name of the SQL server from which reports are processed in the SQL Server Name box.

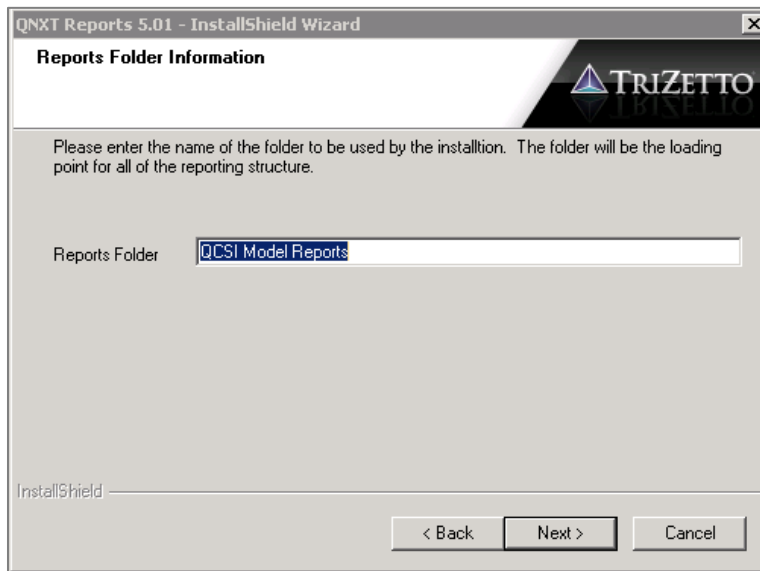
- 7.2 Type the name of the QCSIDB database from which reports are processed In the Database Name box.
- 7.3 Click **Next**.

The screenshot shows a window titled "QNXT Reports 5.01 - InstallShield Wizard". The main heading is "Security Database Location Information". Below the heading is a TriZetto logo. The text reads: "Please enter the name of the server and the database name of the QCSIDB database that the reports will be processed from." There are two input fields: "SQL Server Name:" and "Database Name:". At the bottom, there are three buttons: "< Back", "Next >", and "Cancel". The "InstallShield" logo is visible in the bottom left corner.

8. Complete the following actions on the Plan Integration Location Information window.
 - 8.1 Type the name of the SQL server from which reports are processed in the SQL Server Name box.
 - 8.2 Type the name of the Plan Integration database from which reports are processed in the Database Name box.
 - 8.3 Click **Next**.

The screenshot shows a window titled "QNXT Reports 5.01 - InstallShield Wizard". The main heading is "Plan Integration Location Information". Below the heading is a TriZetto logo. The text reads: "Please enter the name of the server and the database name of the Plan Integration database that the reports will be processed from." There are two input fields: "SQL Server Name:" and "Database Name:". At the bottom, there are three buttons: "< Back", "Next >", and "Cancel". The "InstallShield" logo is visible in the bottom left corner.

9. Accept the default or type the name of the folder where reports are stored in the Reports Folder box, then click **Next** on the Reports Folder Information window.



10. Review installation details and click **Next** on the Start Copy Files window.
The installation starts and a progress indicator shows the installation status.
11. After the installation is complete, click **Finish** to exit the wizard.

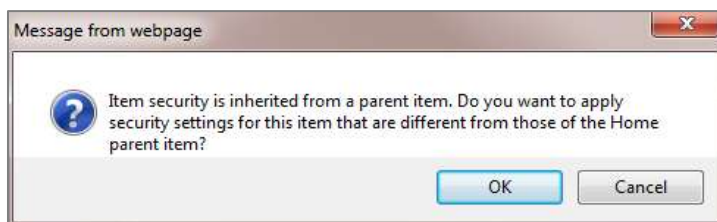
Configuring Model Reports

NOTE: When using SQL Server Reporting Services, you must have access to a domain user account to configure model reports.

To configure model report settings:

1. Start HPA, then navigate to the Reporting Services module.
2. Select **QCSI Model Reports**, then **Folder Settings**.
3. In the left pane, click **Security**.
4. Click **Edit Item Security**.

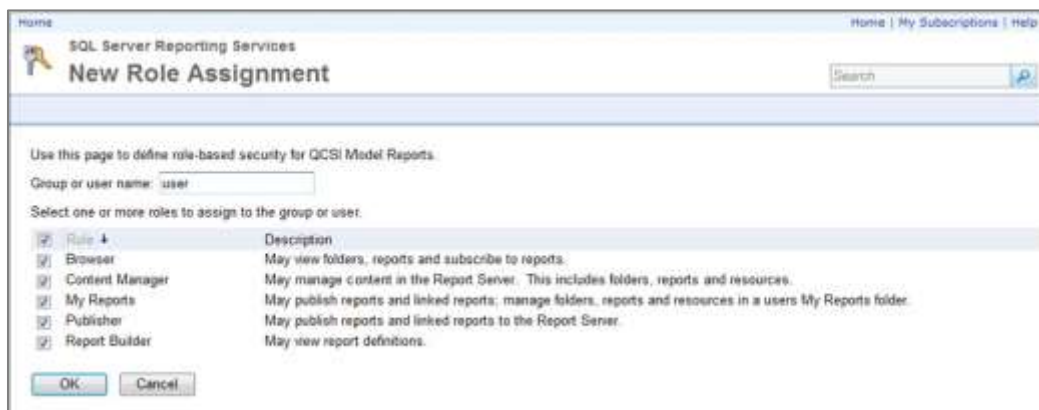
The following message may appear the first time you configure folder-level user security. If applicable, click **OK**.



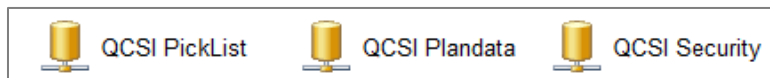
5. Click **New Role Assignment** and complete the following actions on the New Roll Assignment page.
 - 5.1 Type the name of the user or group to add as a Report Server user in the Group or user name box.

NOTE: The name entered is used later in this configuration process.

- 5.2 Select the applicable roles from the list of options.
- 5.3 Click **OK**.



6. From the folders area, select and configure **each** of the model reports (QCSI PickList, QCSI Plandata, and QCSI Security) as described in step 7.



7. Select the applicable report icon to view the configuration page and complete the following actions referring to the [QCSI Picklist configuration page example](#) as needed.
 - 7.1 Select **Credentials stored securely in the report server**.
 - 7.2 Type the domain user account from [step 5](#) in the User name box.
 - 7.3 Type the account password in the Password box.

NOTE: The accounts must also have SQL Administrator rights for each database. HPA recommends that a separate domain account be configured for reporting.

7.4 Select **Use as Windows credentials when connecting to the data source**.

7.5 Click **Apply** to save the changes.

Following is an example of the QCSI Picklist configuration page.

The screenshot shows the 'QCSI PickList' configuration page in SQL Server Reporting Services. The page has a left-hand navigation pane with 'Properties' selected. The main area contains the following fields and options:

- Name:** QCSI PickList
- Description:** (empty text box)
- ☐ Hide in tile view
- ☒ Enable this data source
- Data source type:** Microsoft SQL Server
- Connection string:** Data source=HPA; Initial Catalog=ReportServer
- Connect using:**
 - ☐ Credentials supplied by the user running the report
 - Display the following text to prompt user for a user name and password:
user
 - ☐ Use as Windows credentials when connecting to the data source
 - ☒ Credentials stored securely in the report server
 - User name: user
 - Password: ****
 - ☒ Use as Windows credentials when connecting to the data source
 - ☐ Impersonate the authenticated user after a connection has been made to the data source
 - ☐ Windows integrated security
 - ☐ Credentials are not required

Buttons at the bottom: Test Connection, Apply.

NOTE: For more information about report features and configuration options, refer to the application Help for HPA and SQL Server Reporting Services (click **Help** in the upper-right corner of the interface).

XII HPA Help Installation

This chapter includes the following topics:

- [HPA Help Requirements](#)
- [Installing HPA Help](#)

HPA Help Requirements

The following HPA components must be installed and configured before installing HPA Help:

- QFramework (see [QFramework Installation](#))
- HPA (see [HPA Installation](#))

If upgrading HPA, the HPA Help support component must be uninstalled, then reinstalled to receive the most current Help system. If you do not uninstall and reinstall HPA Help, you will not receive updates outlined in the Enhancement and Documentation Updates sections of the HPA 5.0 HPA Release Notes PDF.

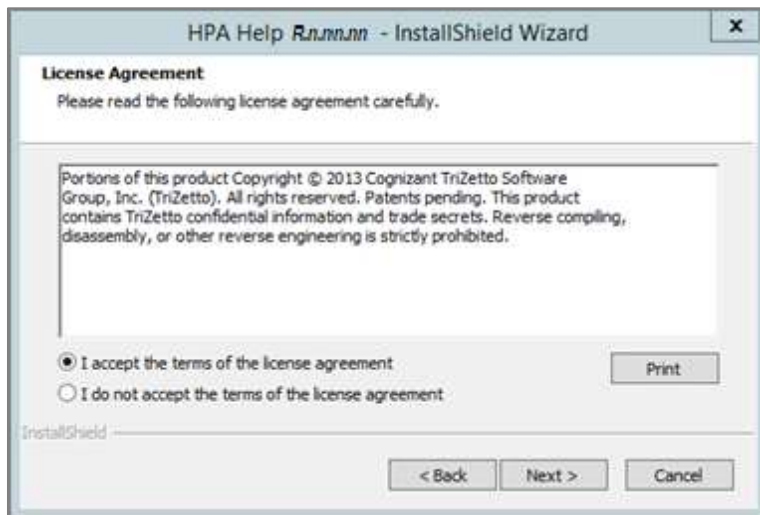
Installing HPA Help

To install the HPA Help application:

1. Browse to the HPA installation package, then run the `HPA_Help_<version>_x64.exe` installation file.

The InstallShield Wizard starts.

2. Click **Next** on the Welcome window.
3. Review the license agreement and select **I accept the terms in the license agreement**, then click **Next** on the License Agreement window.



4. Review installation details, then click **Next** on the Start Copying Files window.

The installation starts and a progress indicator shows the installation status.

5. After the installation is complete, click **Finish** to exit the wizard.

XIII HPA Rules Installation

This chapter includes the following topics:

- [HPA Rules Requirements](#)
- [Installing HPA Rules](#)

HPA Rules Requirements

The following HPA components must be installed and configured before installing HPA Rules:

- QFramework (see [QFramework Installation](#))
- HPA (see [HPA Installation](#))

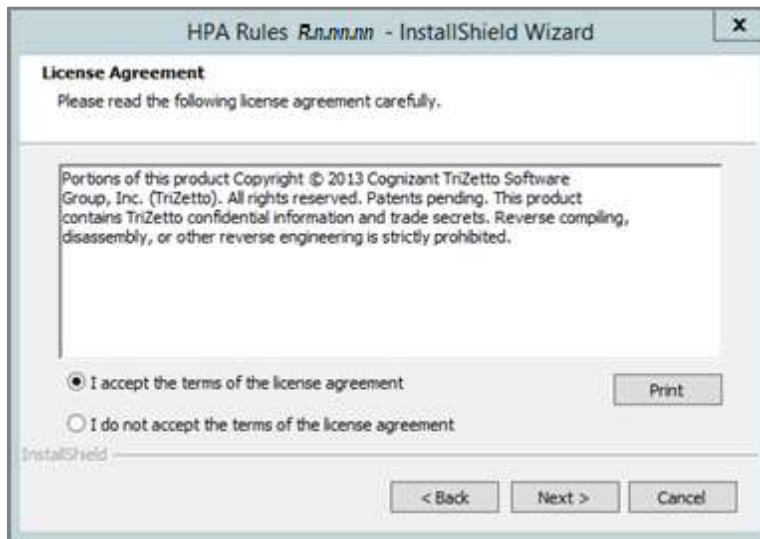
Installing HPA Rules

This section describes how to install the HPA Rules application. You must install HPA (HPA_<version>_x64.exe) before you can install the Rules component. See [HPA Installation](#).

1. Browse to the HPA Installation Package and run the HPA_Rules_<version>_x64.exe installation file.

The InstallShield Wizard starts.

2. Click **Next** on the Welcome window.
3. Review the license agreement and select **I accept the terms of the license agreement**, then click **Next** on the License Agreement window.



4. Review installation details and click **Next** on the Start Copying Files window.

The installation starts and a progress indicator shows the installation status.

5. After the installation is complete, click **Finish** to exit the wizard.

XIV Configure SSL Certificate

This section describes steps to secure the HPA components/ API web services using SSL configuration. Prior to configuration an SSL certificate must be acquired by creating a Certificate Request, followed by importing of the newly acquired certificate into the Windows Server 2019 and configuring IIS to use SSL certificate

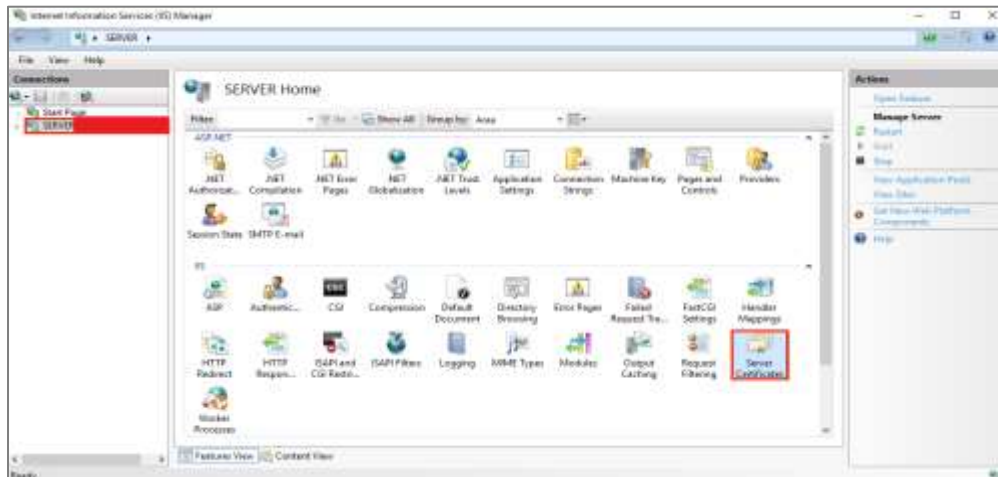
NOTE: When enabling HTTPS functionality, it is recommended to use a trusted certificate from a CA. A self-signed certificate will prevent certain features within Claims Administrator from functionality properly without changes to machine configuration to trust a self-signed certificate.

For More information, please see this article for Hostin secure applications in AWS:

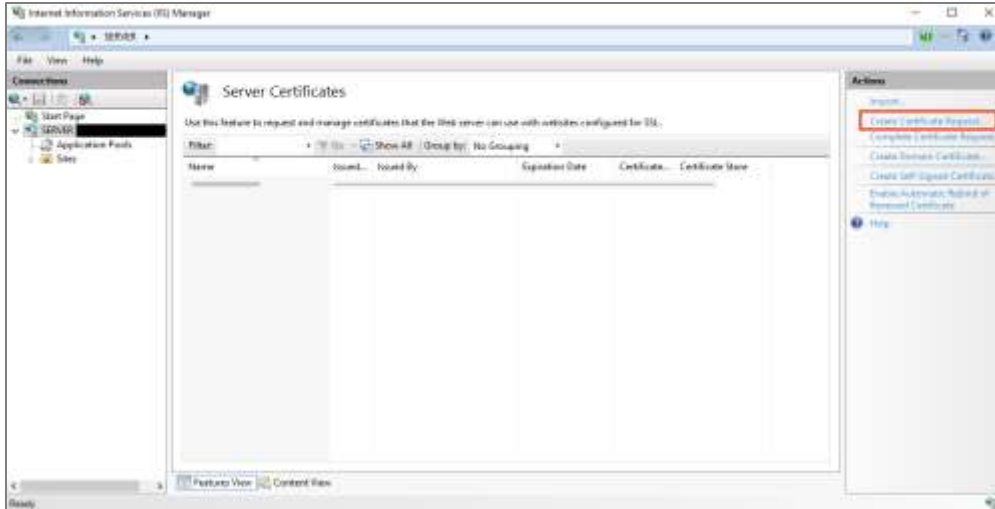
<https://docs.aws.amazon.com/privateca/latest/userguide/PcaWelcome.html>

Create SSL Certificate Request

1. Locate and click the server's name on Windows Server 2019 where HPA component is installed or will be installed, using IIS Manager in the Connections Menu tree (left pane).

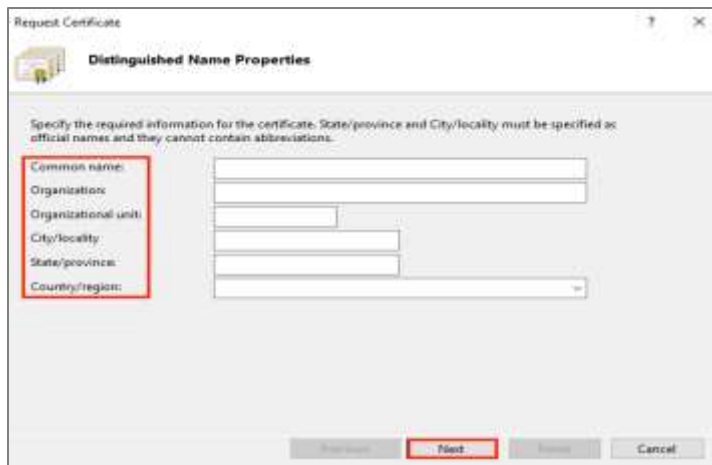


2. Click **Create Certificate Request** ("IIS 10: CSR Creation & SSL Certificate Installation") in the Actions menu (right pane) on the Server Certificates page.



- On the Distinguished Name Properties page of the Request Certificate wizard, provide the information specified below and then click **Next**

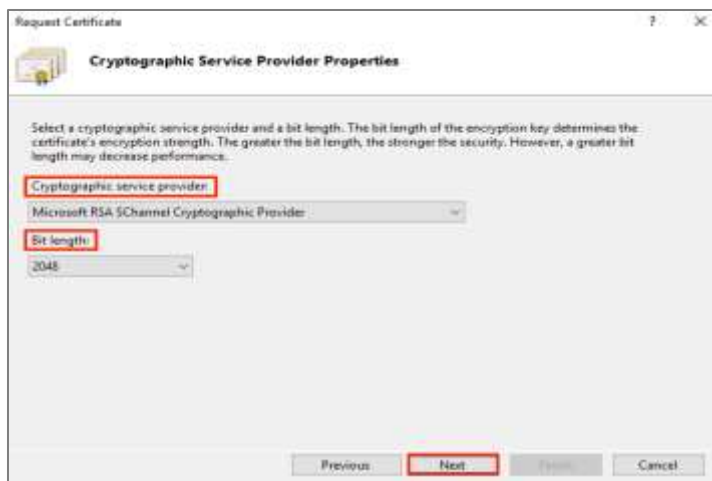
Field	Action/Description
Common name	Type the fully-qualified domain name (FQDN) (e.g., machinename.yourdomain.extension). The FQDN is the hosted data center domain prepended with the machine name. When installing the SSL certificate on multiple hosts, the common name should have a wildcard for the machine name. (e.g., *.yourdomain.extension).
Organization	Your company's legally registered name (e.g., <i>YourCompany, Inc.</i>)
Organizational unit	Name of your department within the organization. This is commonly listed as IT, Web Security , ("How to Generate Certificate Signing Request on Microsoft ...") or Leave blank.
City/locality	City where your company is legally located.
State/province	State/province where your company is legally located. ("Exchange 2016: CSR Creation & SSL Certificate Installation")
Country	Select the country where your company is legally located in the drop-down list. ("Microsoft Active Directory LDAP (2012) - Generate CSR")



The dialog box is titled "Request Certificate" and "Distinguished Name Properties". It contains a text box for "Common name:" and four text boxes for "Organization:", "Organizational unit:", "City/locality:", and "State/province:". A red box highlights the "Common name:" field. At the bottom, there are buttons for "Previous", "Next", "Finish", and "Cancel". The "Next" button is highlighted with a red box.

4. On the Cryptographic Service Provider Properties window, provide the following information and click **Next**.

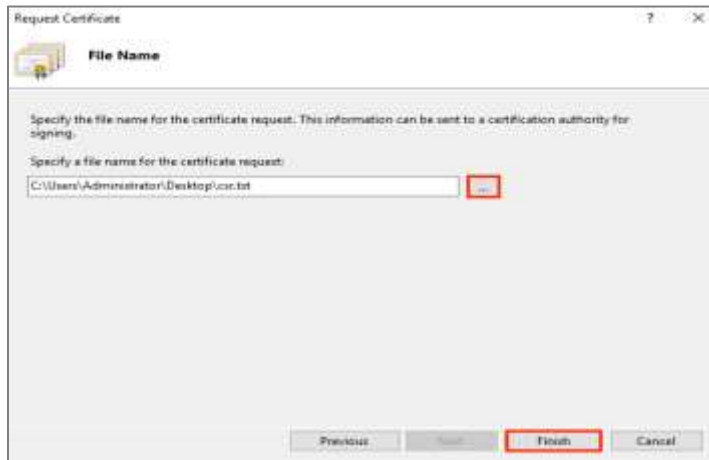
Field	Action/Description
Cryptographic service provider	Select a Microsoft RSA Channel Cryptographic Provider in the drop-down list, ("IIS 10 SSL Certificate Renewal Instructions -DigiCert.com") unless you have a specific cryptographic provider.
Bit length	Select 2048 in the drop-down list, unless you have a specific reason for opting for larger bit length. ("How to Generate a CSR for a Microsoft IIS 10 Web Server ...")



The dialog box is titled "Request Certificate" and "Cryptographic Service Provider Properties". It contains a text box for "Cryptographic service provider:" and a text box for "Bit length:". The "Cryptographic service provider:" field is highlighted with a red box. The "Bit length:" field is also highlighted with a red box. At the bottom, there are buttons for "Previous", "Next", "Finish", and "Cancel". The "Next" button is highlighted with a red box.

5. On the File Name window, click the ... (ellipsis) to browse to a location where you want to save your CSR under Specify a file name for the certificate request.

NOTE: Remember the filename you choose and location to which you save your csr.txt file. (“How to Generate a CSR for a Microsoft IIS 10 Web Server ...”)

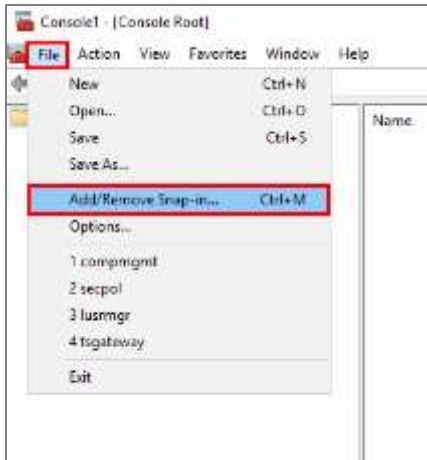


Use a text editor (such as Notepad) to open the file. (“CSR Creation :: DigiCert Utility Microsoft Server”) Then, copy the text, including the -----BEGIN NEW CERTIFICATE REQUEST----- and -----END NEW CERTIFICATE REQUEST----- tags when sending the Certificate Request to a certificate authority for SSL certificate fulfillment.

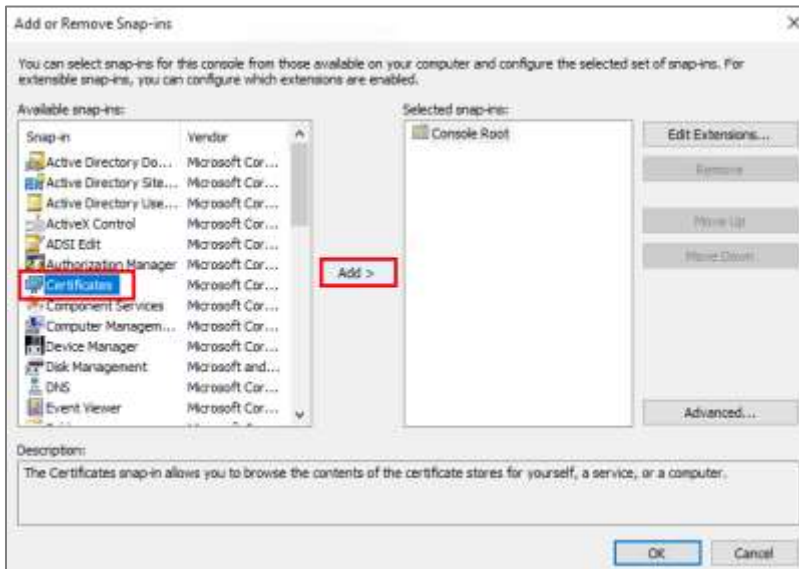
```
-----BEGIN NEW CERTIFICATE REQUEST-----
MIICvDCCAAQCAQAwZELMAkGA1UEBhMCVVMxEjAQBgNVBAgTCVlvdXJtZGF0ZTER
MA8GA1UEBxMIWH91ckNpdHkxGA1UEAaTAK1UMRowGAYDVQQKEwFZb3VyQ29t
cGueSwqSW5jLjEYMBYGA1UEAxMPd3d3LmV4YW1wbGUuY29tMIIBIjANBgkqhkiG
9w0BAQEFAAOCAQ8AMIIBCgKCAQEA379BFFxfACdKxUk2wrQka/nAlKbo+I9DAW32
+/SRxj/KtXVddscKW1obHSpMKPw4meJqOpQwJkIChYjSUQSpPKadGpccDMf/ef0
J7EaQ2szLv9AqdRQv2Aaek8SmocVmd3LxEOK4VvALBOMLHVrB5/vhYfGECLJbc31
RdEbdXyHdtHk1RAoIVQCfjTwHGNAD337vmHW7QOR6FYUoa4fcJh7Rv6jHSyvwqx
7pVfaDb2PuTgUhw7vkeHNFxcccG0xcTMr/+GrciHEuZ0chq86CBP9RIylpp2+RMSf
m6rMEYm9o65j7vEYaKEJUOJtA5MIz/ZjeXfS1LjXurLU0nCOQQIDAQABoAAwDQYJ
Ko2IhvcNAQEFBQADggEBAK159goyAYOpennrQ2EvCG11zrK1kS3D8JnAiP1NHrjB
/qdTYR+/8Dr/hMcwvU6ThQAVf68eMkk6tUNwAdp29C904Js2z+ENeb08GA0Fc4rw
1x7vb15vSKe3shGiJRGiazHVGRoR3z7xQtIuMaDar3x1V8jHbcvZTcpx0Kbq6H1G
NLA4CKsOI4KGuu4FXfSzJEGb3gEJD8HaMP8V8er5G0owv/g/9Z/1/b0g97kAcUwk
M2eDsvPhHx/pENGbnLPe4XMy7HP1EdsFnaYtUy2BDcXj3ZQEWxRWk1ERgg9/YcWI
obf5ziunMldf24NBt5tpCNzfGvIKI6/RYFWg3dMaKxc=
-----END NEW CERTIFICATE REQUEST-----
```

Import SSL Certificate into certificate store

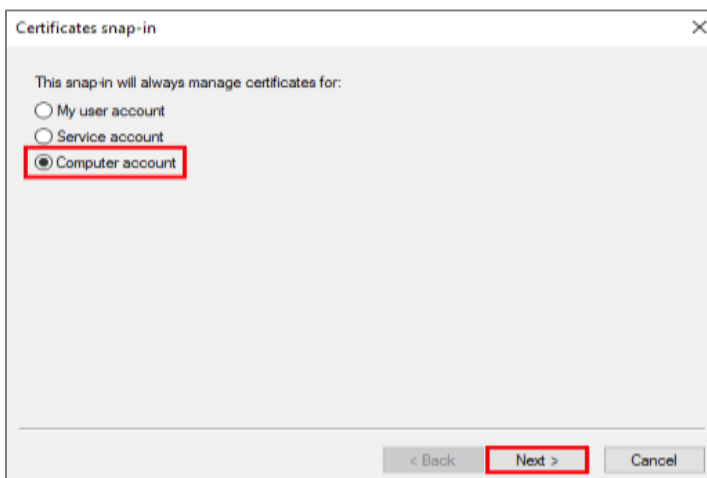
1. Open the Console (“Teadmistebaas - HOTServers”) on the Windows Server 2019 where you want to install the SSL certificate.
2. Type **mmc** and open on the Windows Start menu.
3. On the Console window, in the top menu, click **File > Add/Remove Snap-in**.



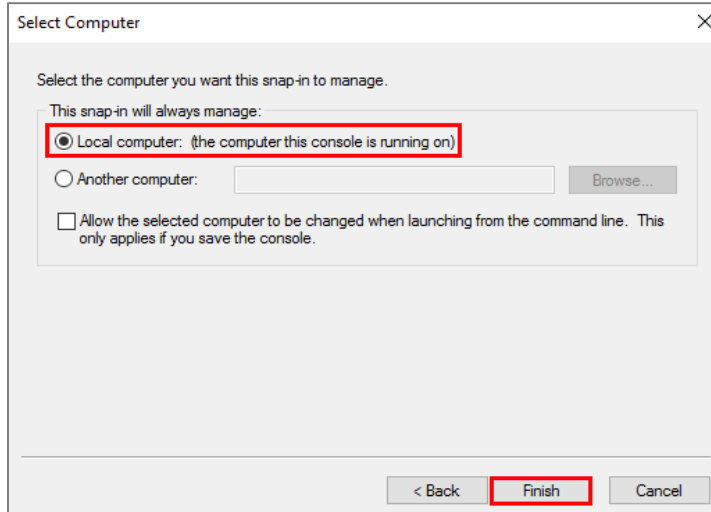
4. Select **Certificates** in the Available snap-ins pane (left pane) on the Add or Remove Snap-ins window, and then click **Add**.



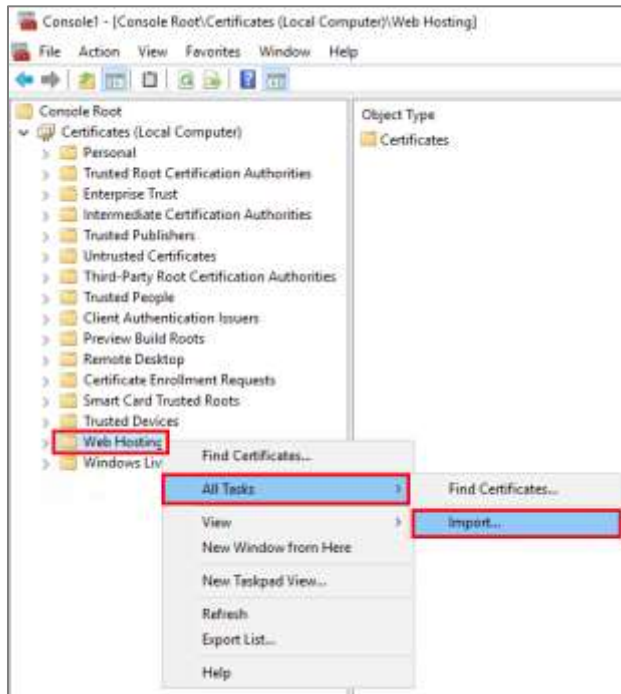
5. Select **Computer account** on the Certificate snap-in window and click **Next**.



6. Select **Local computer: (the computer this console is running on)** on the Select Computer window and click **Finish**.

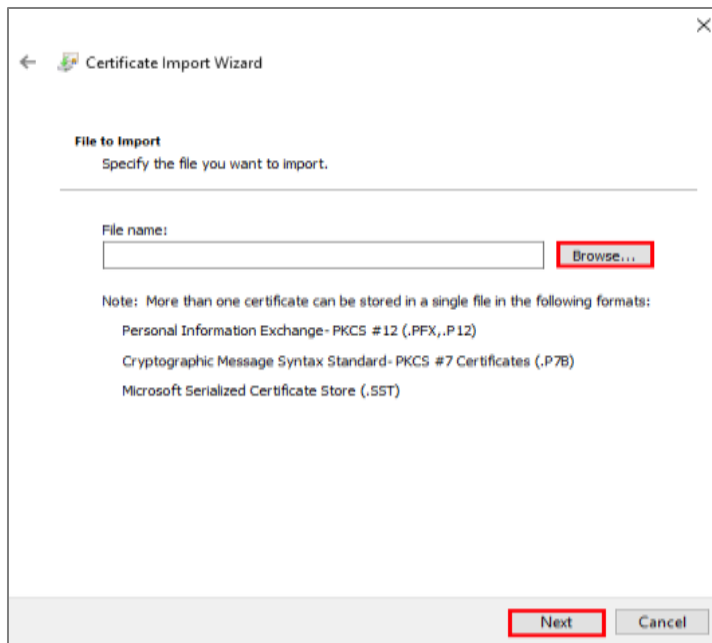


7. Click **OK** on the **Add or Remove Snap-ins** window.
8. Expand **Certificates (Local Computer)** in the Console Root pane (left side) on the Console window. Right-click the **Web Hosting** folder, and then click **All Tasks > Import**.

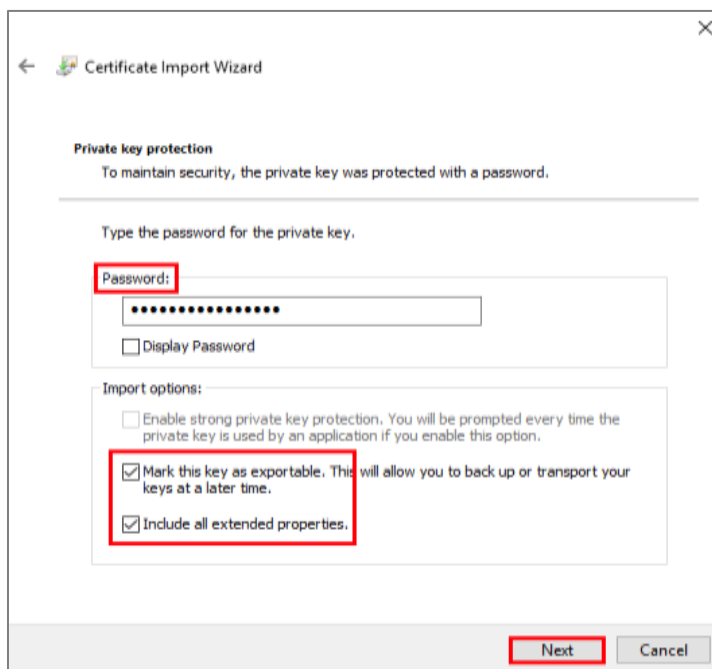


9. Click **Next** on the Welcome to the Certificate Import Wizard page in the Certificate Import Wizard.
10. Browse to and select the certificate file you want import on the File to Import page and click **Next**.

NOTE: Make sure to select **All Files (*.*)** in the file type drop-down on the **File Explorer** window. By default, it is set to search for **X.509 Certificate (*.cert;*.crt)** file types only.

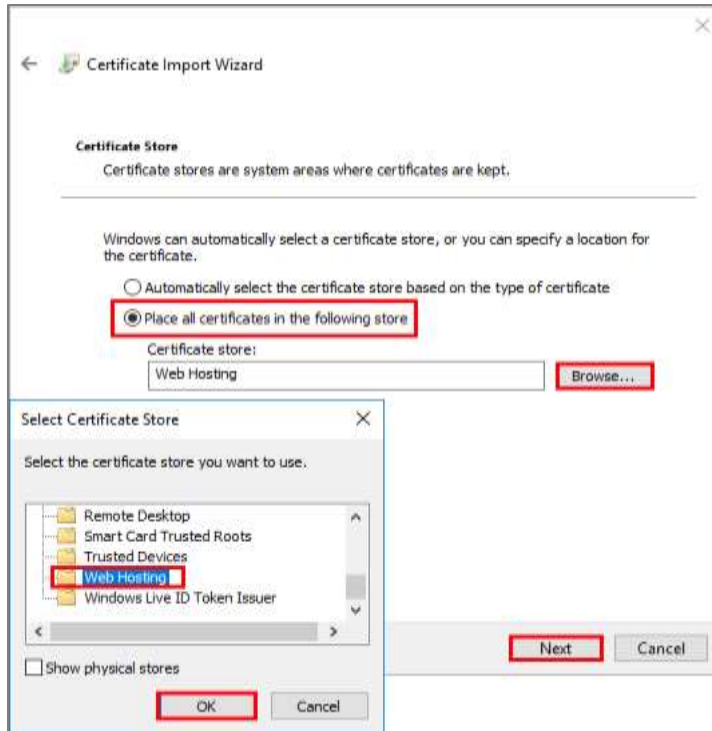


11. Type the password that was issued with the certificate on the Private key protection window.



12. Select **Place all certificates in the following store** and click **Browse**.

13. Select Web Hosting in the Select Certificate Store window and click **OK**.



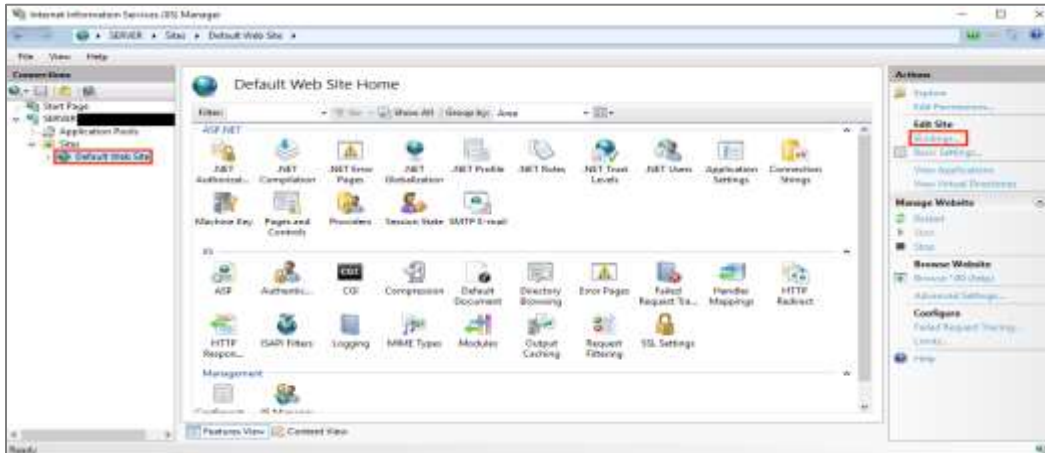
14. Click **OK**, **Next**, and **Finish** on the Certificate Import Wizard UI. You should see the message “The import was successful.”

Configure Windows Server 2019 to use SSL Certificate

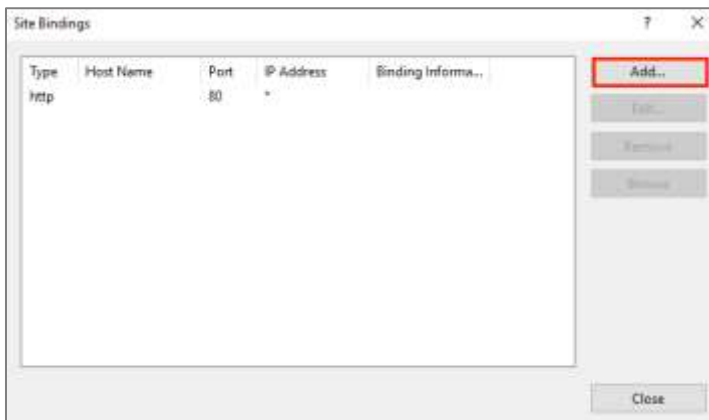
1. Open Internet Information Services (IIS) Manager on the Windows Server 2019 where SSL certificate installed.

Tip! Type Internet Information Services (IIS) Manager on the Windows Start menu and open it.

2. Expand the name of the server on which the certificate was installed, expand Sites, then click the site you want to use the SSL certificate to secure in the Connections menu tree (left pane) of Internet Information Services (IIS) Manager .



3. Click the **Bindings** link on the website Home page in the Actions menu (right pane), under Edit Site.
4. Click **Add** on the Site Bindings window.



5. Take actions listed in the following table in the Add Site Bindings window:

Field	Action/Description
Type	Select https in the drop-down list.
IP address	Select the IP address of the site or select All Unassigned in the drop-down list.
Port	Type port 443 . (Port over which traffic is secured by SSL is port 443.)
SSL certificate	Select your new SSL certificate in the drop-down list.

The screenshot shows the 'Add Site Binding' dialog box. It has a title bar with a question mark and a close button. The dialog contains the following fields and controls:

- Type:** A dropdown menu with 'https' selected.
- IP address:** A dropdown menu with 'All Unassigned' selected.
- Port:** A text box containing '443'.
- Host name:** An empty text box.
- Require Server Name Indication:** An unchecked checkbox.
- SSL certificate:** A dropdown menu that is currently empty.
- Select...:** A button next to the SSL certificate dropdown.
- View...:** A button next to the SSL certificate dropdown.
- OK:** A button at the bottom right, highlighted with a red box.
- Cancel:** A button at the bottom right, next to the OK button.

6. Click **OK** to install the SSL certificate and configure the website to accept secure connections.

XV Guidelines to Implement Single Sign On (SSO) Feature

Prerequisites and Requirements

This section provides the pre-requisites and other guidelines required before implementing the SSO feature.

- **HTTPS** protocol availability.
- See [QFramework Installation](#) & [HPA Installation](#), where the Protocol type will be set during installation.
- Name of the Identity Broker.

NOTE: HPAALL application Login (Authentication mode) Method is **WINDOWS** by default.

To enable the SSO feature:

1. Open Internet Information Services (IIS) Manager on the Windows Server 2019 where the SSL certificate is installed.
2. Select HPAALL option in the Connections window, and then select the Application UI.
3. Enter the following required information into the Identity Broker:
 - Client ID
 - Client Secret
 - End Point JSON Configuration
 - User Claims Json Configuration

Refer to the following **User Security SSO Configuration (UI Page Route)** template .

Field	Action/Description
Enable SSO Authentication checkbox	To Enable SSO Authentication active. Data will be saved to QCSIDB – SSOConfig table. If data is available in SSOconfig table with Active status, then the application will proceed with SSO Login method.
Authentication Protocol	Dropdown value: OIDC - Authorization Code Flow OIDC - Hybrid
Identity Broker	Provide the Name of the Identity Provider(IdP) . e.g ADFS , Azure B2C,OKTA etc.,.
Client ID & Client Secret	Client ID & Secret key provided by the IdP. Client Secret is mandatory only for OIDC - Authorization Code Flow Authentication Protocol.
End Point JSON Configuration	metadata_url [REQUIRED] : Once application registered in IdentityProviders like OKTA,ADFS,GW configuration details will be available under metadata and this would be hared by IdP Team after app configuration. Note : If there is any custom policy and if metadata URL has the custom policy, then it must not be available as query params in url. Reference: Custom Policy : B2C_1A_SUSI_HPACLAIMSADMINISTRATOR Correct custom policy - Metadata URL : https://devgainwellauth.b2clogin.com/DevGainwellAuth.onmicrosoft.com/B2C_1A_SUSI_HPACLAIMSADMINISTRATOR/v2.0/.well-known/openid-configuration Incorrect Metadata URL : https://devgainwellauth.b2clogin.com/DevGainwellAuth.onmicrosoft.com/v2.0/.well-known/openid-configuration

Field	Action/Description
	configuration?p=B2C_1A_SUSI_DC_DDI_PRD_HPA_CLAIMS_AD_MINISTRATOR This fails the metadata url validation in SSO configuration tab. <u>scopes_supported</u> [REQUIRED]: Scopes supported by the Identity Provider(IdP) . <u>postlogoutredirecturi</u> [NOT MANDATORY]: Value should be a valid https:// URL . 1. If any URL configured here, When user clicks on Logoff it will redirect to that PORTAL/website after Log Off from Claims Administrator. 2. If not configured and Empty , When user clicks on LogOff , Claims Administrator logged off and will redirect to IdP configured Sign-out redirect URIs Screen .
User Claims Json Configuration	<u>claim_quser_loginid</u> : Enter claim that maps to quser.loginid <u>claim_quser_firstname</u> : Claim that maps to quser.firstname <u>claim_quser_middlename</u> : Claim that maps to quser.middlename <u>claim_quser_lastname</u> : Claim that maps to quser.lastname <u>claim_quser_initials</u> : Claim that maps to quser.initials <u>claim_quser_SSOUserID</u> [REQUIRED] : Claim that maps to quser.SSOUserID - Used to identify user <u>claim_qenvironment_role_description</u> [REQUIRED] : Claim that maps to qenvironment.description [separator] qrole.description - used to assign environment\role for the user. <u>qenvironment_role_separator</u> [REQUIRED] : Enter [separator] used in claim_qenvironment_role_description
4. Click Validate – This perform validation against the user provided sso configuration details . Validation includes each field level and also on IDToken generation which consists of 5 steps verification. In this, if all the steps are in COMPLETED status then only users are allowed to SAVE the SSOConfiguration details . If any steps with ERROR , user has to re-configure and validate again . User can verify step status by clicking on EXPAND button at any time . 5. Click Save .	
NOTE: Do not use special character's which is already available in qenvironment.description / qrole.description and "\$#" as [separator] .	

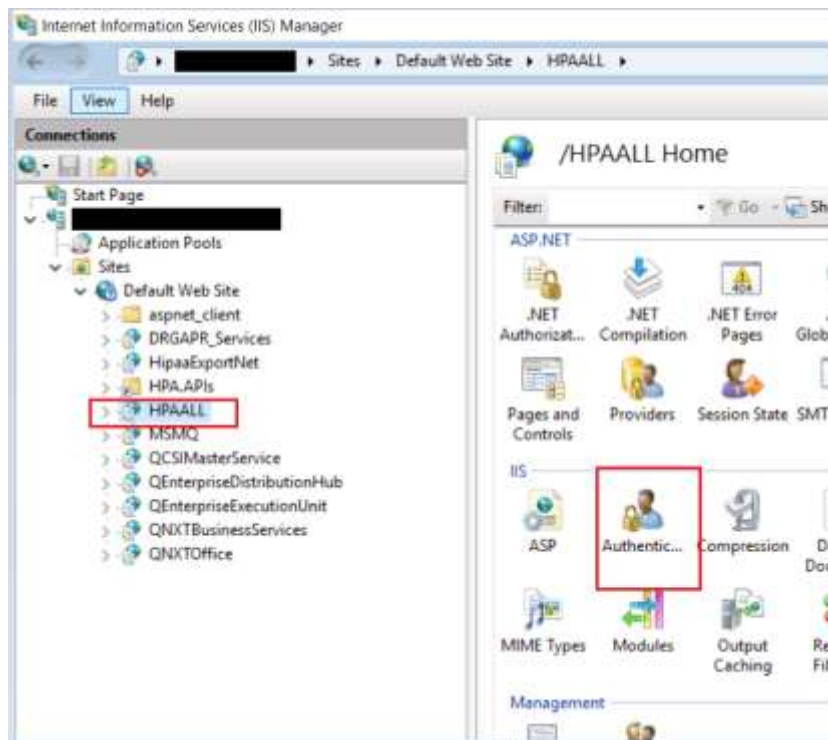
1. Open Internet Information Services (IIS) Manager on the Windows Server 2019 where the SSL certificate is installed.
2. Select HPAALL option in the Connections window, and then select the Application UI.

To check HPAALL - IIS Authentication Settings in Internet Information Services (IIS), set the HPAALL Application Authentication to Anonymous Authentication in IIS for SSO. This can be achieved through the **Playbook**, which will **Enable** the **Anonymous Authentication** in IIS

Playbook Details:

Field	Action/Description
Playbook Name	admin_dsc_prereq_config.yml
Parameters	config_sso : true / false (Value of true is SSO ,Value of false is Windows Authentication)
	hpaall_auth_config_type : 1 or 4 (Value of 1 is Anonymous Authentication (required when SSO is in play), Value of 4 is Windows Authentication)
	For Example, To enable SSO , param for playbook should be as like the below . config_sso: true hpaall_auth_config_type: 1

To check **HPAALL - Authentication Mode** in **Internet Information Services (IIS)**, use the following instructions.



1. Open **Internet Information Services (IIS)**.
2. Click **HPAALL** in the Connections window.
3. Select **Authentication** under IIS.

The enabled/disabled respective Authentication Mode will be available in the Authentication Screen.



Name	Status	Response Type
Anonymous Authentication	Enabled	
ASP.NET Impersonation	Disabled	
Forms Authentication	Disabled	HTTP 302 Login/Redirect
Windows Authentication	Disabled	HTTP 401 Challenge

XVI Configure Audit Logging-Certificate

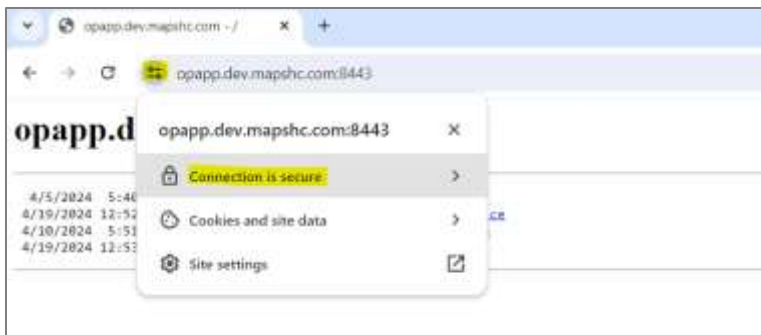
If configuration is enabled for “Enable Audit Logging - 986” (**Configuration** > **Configuration Options**) in the UI, do the following:

- Configure the API URL in the User Security tab (**User Security** > **Users**) UNC Path
- Install the certificates based on API URL
- This API certificate needs to be installed in both UI and EU servers for Audit Logging

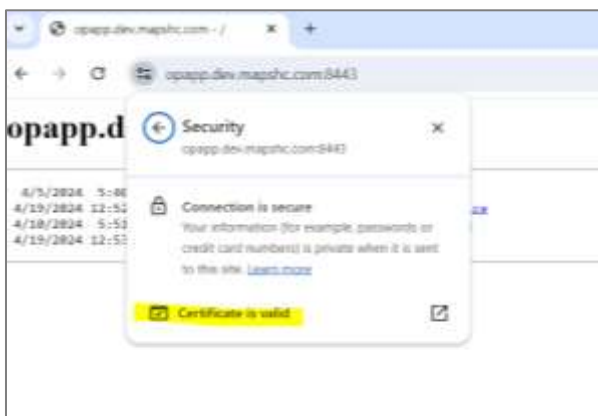
1. Access the API URL in the browser with host name (EU and UI server).



2. Click the highlighted **View site information**.



3. Click the **Connection is secure** option. **Certificate is valid** message displays.



NOTE: User must verify the root certificate has been installed if the certificate is not valid.

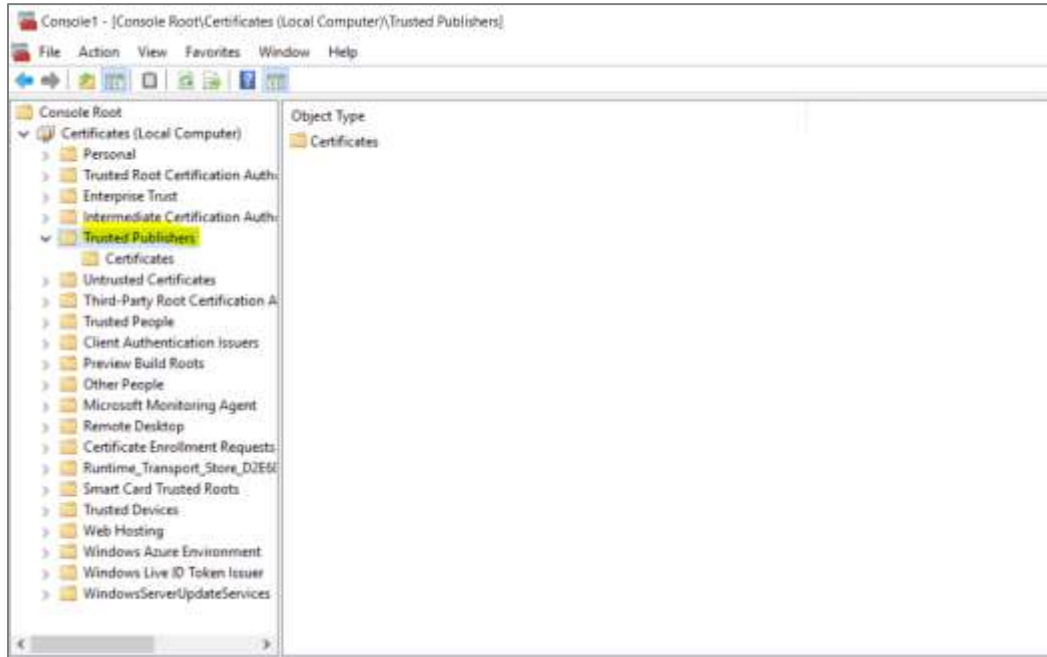
4. Verify the root certificate is present in the **Trusted Root Certification Authorities certificate store**.

5. Select **Trusted Root Certification Authorities**, and then click **Certificates**.

The root certificate displays in the Certificates folder.

6. Verify the root certificate is present in the Trusted Publishers certificate store.

7. Select Trusted Publishers, and then click **Certificates**.



8. Select the **Certificates** folder.

The root certificate is displayed in the Certificates folder.

XV Environment setup for Pharmacy Claims API Integration

The following configuration is required to successfully access the *Compound* and *ClaimKeys* endpoints in the Claim API.

In the Claims Administrator, **User Security > Environments** tab

1. Create a DB alias for HRX DB to the respective environment. It is recommended to set the DB alias name exactly as **hrx**.

User Security > Environment

Environments	BRT_RTYV
Users/DB Alias	DB Alias
Users	
Env UNC Access/	
Roles/User Activities	

Add Edit

DB Alias

Select	Description
☐	plandocument, HpasDBSql16-1/usl2022
☐	Planreport, HpasDBSql16-1/usl2022
☐	CX_Extensions, HpasDBSql16-1/usl2022
☐	healthgas, HpasDBSql16-1/usl2022
☒	hrs, HpasDBSql16-1/usl2022
☐	plandata, HpasDBSql16-1/usl2022
☐	planevent, HpasDBSql16-1/usl2022
☐	planintegration, HpasDBSql16-1/usl2022
☐	pentaprise, HpasDBSql16-1/usl2022

Environment ID	U1
Name	hrs
Server	HpasDBSql16-1/usl2022
Database	DEV_HRS
Description	hrs database
Schema Version Major	3
Schema Version Minor	40
Sql Server Version	SQL Server 2008

Configuration Options

Configuration > Configuration Options

Configuration Option Category: MISC

Configurations

Select	ID	Description	Configuration Value	Configuration ID
☐	100	Enable Impact Pro	No	986
☐	101	Interface to Millman CareWeb Guidelines	Yes	Configuration Name
☐	510	Enable HSS ECM Pro Integration	No	Enable Audit Logging
☐	511	Maintain Service Code History	No	Configuration Value
☐	518	Generate SCOF # on SF Export	No	Yes
☐	519	Send Private Room Rate	No	
☐	606	Allow setup of HIPAA transaction processing	Yes	
☐	621	Create payments based on provider's configured pay schedule	No	
☐	902	Use Windows Task Scheduler to schedule processes	Yes	
☐	904	Suppress Private Information	Yes	
☐	905	Integrate to ICD Translation Manager	No	
☐	909	Enable Rebill Carrier Processing	Yes	
☐	979	MHS State		
☐	980	Enable Claim Archive Announcement		
☐	985	Enable US Application Tracking	No	
☒	986	Enable Audit Logging	Yes	

Users Security – UNC Access and UNC Path

User Security > Users

Users: User_AuditLogging

QEnvironments: DEV_Connect

DB Alias/Env UNC Access/Roles/User Activities: Env UNC Access

Add Edit Delete

Env UNC Access

Select	Description
☒	Envuncaccess, https://spapp-dev.maphc.com:8443/GWT/Core/AuditListenerService/Audit/...

User ID: 352

Login ID: AuditLogging

Last Name: User

First Name: AuditLogging

Environment ID: 31

Description: DEV_Connect

UNC Path: https://spapp-dev.maphc.com:8443/GWT/Core/AuditListenerService/Audit/...

UNC Default:

Effective Date: 9/15/2024

Termination Date:

Appendix A HPA Silent Installation

Topics for HPA Silent Installation include:

- [Silent Installation Overview](#)
- [Silent Installation/Uninstallation](#)
- [Performing Silent Install/Uninstall Using Response File](#)
- [Connect Installation on BizTalk Server](#)

Silent Installation Overview

A silent installation allows you to install HPA components using command-line interface (CLI) commands. This helps to automate future installations.

Before you can perform a silent installation, the HPA component must have been previously installed on the server. This allows the system to retrieve configuration information (for example, user and server credentials) necessary for the silent installation.

The silent installation then uses the stored configuration information to automatically complete the installation process.

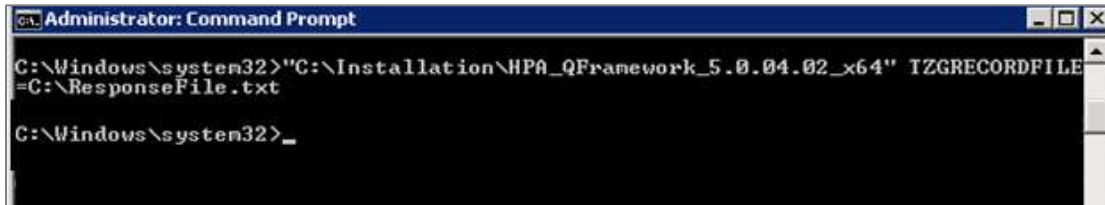
NOTE: When performing a silent installation, you must install one component at a time. You cannot install multiple components simultaneously.

Silent Installation/Uninstallation

To perform a silent installation, you must first create a “response file” to record and save the configuration information entered when the feature was originally installed. You then use a separate CLI command to install the feature from the response file.

To generate a silent installation response file:

1. Open Command Prompt as Administrator.
2. Enter the following command as shown in the following and defined below:



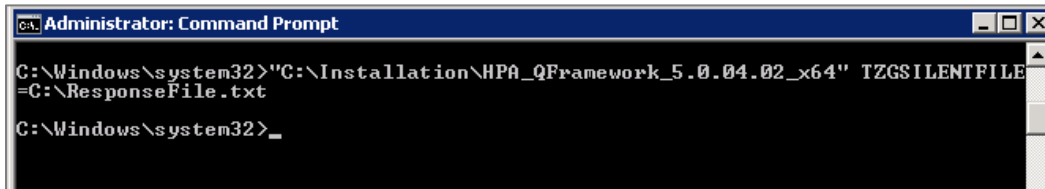
```
Administrator: Command Prompt
C:\Windows\system32>"C:\Installation\HPA_QFramework_5.0.04.02_x64" TZGRECORDFILE
=C:\ResponseFile.txt
C:\Windows\system32>_
```

- "C:\Installation\HPA_QFramework_5.0.04.02_x64" is the full path of the HPA component to be installed.
- TZGRECORDFILE is the command for generating the response file.
- C:\ResponseFile.txt or C:\ResponseFile.ini is the response file to be generated.

Once successfully executed the response file in .txt or .ini format is generated and can be used for future installations.

Performing Silent Install/Uninstall Using Response File

1. Open Command Prompt as Administrator.
2. Write the following command as shown below.



- `C:\Installation\HPA_QFramework_5.0.04.02_x64` is the full path of the HPA component to be installed.
 - `TZGSILENTFILE` is the command for silent installation/uninstallation.
 - `C:\ResponseFile.txt` or `C:\ResponseFile.ini` is the previously generated response file.
3. Verify the installation was successful by making sure the HPA component is added to the server Programs list.
 4. Repeat as needed for each HPA component.

QFramework Silent Installation

The following table lists QFramework responses file properties.

Properties	Description
License Agreement=TRUE	Accepting the term of the license agreement
TARGETDIR=C:\Program Files\QCSI\QFrame	Destination folder, by default HPA QFramework will be installed to C:\Program Files\QCSI\QFrame
QCSI_Technology_Framework=SELECTED	Installs all the QCSI Technology Framework
QEnterprise_Distribution_Hub=SELECTED	Installs the QEnterprise Distribution Hub, which includes QEnterprise tools NOTE: A Distribution hub cannot be installed on the same machine as an execution unit.
QEnterprise_Execution_Unit=NOT SELECTED	Installs the QEnterprise Execution Unit, which includes QEnterprise tools NOTE: An Execution Unit cannot be installed on the same machine as the distribution hub.
hidden=SELECTED	
g_svSecurityDBServer= <<Security DB Server>>	Name of the server on which the security database is located
g_svSecurityDBName= <<Security DB Name>>	Name of the security database
g_svAppUser= <<Service Account Name>>	Username of the user account that will logon to use this application
g_svAppUserDomain= <<Service Account Domain>>	Domain of the user account that will logon to use this application
g_svAppUserPassword= <<Service Account Password>>	Password for the user account that will logon to use this application

Properties	Description
g_svDHServer= <<Distribution Hub Server>>	Name of the server where the QEnterprise Distribution Hub is installed
g_svMessageStoreDBServer= <<Message Store DB Server>>	Name of the server to communicate with the Message Store Database
g_svMessageStoreDBName= <<Message Store DB Name>>	Name of the Message Store Database

QFramework on Distribution Hub

QFramework silent installation configuration on the Distribution Hub Server shown in the following Response file example.

```
[Params]
License Agreement=TRUE
TARGETDIR=C:\Program Files\QCSI\QFrame
QCSI_Technology_Framework=SELECTED
QEnterprise_Distribution_Hub=SELECTED
QEnterprise_Execution_Unit=NOT SELECTED
hidden=SELECTED
g_svSecurityDBServer=<<Security DB Server>>
g_svSecurityDBName=<<Security DB Name>>
g_svAppUser=<<Service Account Name>>
g_svAppUserDomain=<<Service Account Domain>>
g_svAppUserPassword=<<Service Account Password>>
g_svDHServer=<<Distribution Hub Server>>
g_svMessageStoreDBServer=<<Message Store DB Server>>
g_svMessageStoreDBName=<<Message Store DB Name>>
```

QFramework on Execution Unit

QFramework silent installation configuration on the Execution Unit Server is shown in the following Response file example.

```
[Params]
License Agreement=TRUE
TARGETDIR=C:\Program Files\QCSI\QFrame
QCSI_Technology_Framework=SELECTED
QEnterprise_Distribution_Hub=NOT SELECTED
QEnterprise_Execution_Unit=SELECTED
hidden=SELECTED
g_svSecurityDBServer=<<Security DB Server>>
g_svSecurityDBName=<<Security DB Name>>
g_svAppUser=<<Service Account Name>>
g_svAppUserDomain=<<Service Account Domain>>
g_svAppUserPassword=<<Service Account Password>>
g_svDHServer=<<Distribution Hub Server>>
g_svMessageStoreDBServer=<<Message Store DB Server>>
g_svMessageStoreDBName=<<Message Store DB Name>>
```

HPA Silent Installation

HPA UI on Application Server

HPA UI silent installation configuration on the Application Server is shown in the following Response file example.

```
BS\QNXT_BS\Partner\Microdyn=SELECTED
BS\QNXT_BS\Partner\ICES=NOT SELECTED
BS\QNXT_BS\Partner\Impact_Pro=NOT SELECTED
BS\QNXT_BS\Partner=SELECTED
BS\QNXT_BS\TMS=NOT SELECTED
BS\QNXT_BS=SELECTED
BS\QNXT_AR_BS=SELECTED
BS\QNXT_CM_BS=SELECTED
BS\Developers_Toolkit=NOT SELECTED
BS\BS_GAC=SELECTED
BS=SELECTED
UI\QNXTALL=SELECTED
UI\QNXT_CM\QNXT_CM_UI=SELECTED
UI\QNXT_CM\QNXT_CM_Services=SELECTED
UI\QNXT_CM=SELECTED
UI\QNXT_AR\QNXT_AR_UI=SELECTED
UI\QNXT_AR\QNXT_AR_Services=SELECTED
UI\QNXT_AR=SELECTED
UI\QNXT\QNXT_UI=SELECTED
UI\QNXT\Services=SELECTED
UI\QNXT=SELECTED
UI=SELECTED
hidden=SELECTED
g_svNetServer=<<HPA PlanEvent DB Server>>
g_svNetDatabase=<<HPA PlanEvent DB Name>>
g_svAppUser=<<Service Account Name>>
g_svAppUserDomain=<<Service Account Domain>>
g_svAppUserPassword=<<Service Account Password>>
g_svReportingServer=<<HPA Reporting Server>>
```

The following table lists HPA UI Responses file properties.

Properties	Description
License Agreement=TRUE	Accepting the term of the license agreement
TARGETDIR=C:\Program Files\QCSI	Destination folder, by default HPA QFramework will be installed to C:\Program Files\QCSI\QFrame
BS\QNXT_BS\Core\Core_GAC=SELECTED	QNXT Core Services
BS\QNXT_BS\Core\MS_Core_GAC=SELECTED	QNXT Core Services
BS\QNXT_BS\Core=SELECTED	QNXT Core Services
BS\QNXT_BS\Partner\ClaimsXten=SELECTED	ClaimsXten Web Services
BS\QNXT_BS\Partner\MyHB\MyHB_GAC=NOT SELECTED	Configuration and files required to communication with MyHealthBank Software
BS\QNXT_BS\Partner\MyHB=NOT SELECTED	Configuration and files required to communication with MyHealthBank Software
BS\QNXT_BS\Partner\Microdyn=SELECTED	Microdyn Services
BS\QNXT_BS\Partner\ICES=NOT SELECTED	iCES Services

Properties	Description
BS\QNXT_BS\Partner\Impact_Pro=NOT SELECTED	Impact Pro Services
BS\QNXT_BS\Partner=SELECTED	Microdyn, iCES, Impact_Pro are under Partner Services
BS\QNXT_BS\TMS=NOT SELECTED	Integration for TMS
BS\QNXT_BS=SELECTED	QNXT Business Service
BS\QNXT_AR_BS=SELECTED	QNXT A/R
BS\QNXT_CM_BS=SELECTED	QNXT Case Management
BS\Developers_Toolkit=NOT SELECTED	Developers Toolkit
BS\BS_GAC=SELECTED	QNXT Business Services
BS=SELECTED	All Business Services
UI\QNXTALL=SELECTED	QNXT ALL
UI\QNXT_CM\QNXT_CM_UI=SELECTED	QNXT Case Management addon product User Interface
UI\QNXT_CM\QNXT_CM_Services=SELECTED	QNXT Case Management related services
UI\QNXT_CM=SELECTED	QNXT Case Management
UI\QNXT_AR\QNXT_AR_UI=SELECTED	QNXT A/R addon product User Interface
UI\QNXT_AR\QNXT_AR_Services=SELECTED	QNXT A/R related services
UI\QNXT_AR=SELECTED	QNXT A/R
UI\QNXT\QNXT_UI=SELECTED	QNXT User Interface
UI\QNXT\Services=SELECTED	QNXT Services
UI\QNXT=SELECTED	Core QNXT Product
UI=SELECTED	Aspx and related web pages and files
hidden=SELECTED	
g_svNetServer=<<HPA PlanEvent DB Server>>	Name of the server where the PlanEvent Database
g_svNetDatabase=<<HPA PlanEvent DB Name>>	Name of the PlanEvent Database
g_svAppUser=<<Service Account Name>>	Username of the user account that will logon to use this application
g_svAppUserDomain= <<Service Account Domain>>	Domain of the user account that will logon to use this application
g_svAppUserPassword= <<Service Account Password>>	Password of the user account that will logon to use this application
g_svReportingServer= <<HPA Reporting Server>>	Name of the server where the QNXT Reporting Software has been Installed

HPA UI on Execution Unit/Distribution Hub

HPA UI silent installation configuration on the Execution Unit/Distribution Hub is shown in the following Response file example.

```
[Params]
License Agreement=TRUE
TARGETDIR=C:\Program Files\QCSI
BS\QNXT_BS\Core\Core_GAC=SELECTED
BS\QNXT_BS\Core\MS_Core_GAC=SELECTED
BS\QNXT_BS\Core=SELECTED

BS\QNXT_BS\Partner\MyHB\MyHB_GAC=NOT SELECTED
BS\QNXT_BS\Partner\MyHB=NOT SELECTED
BS\QNXT_BS\Partner\Microdyn=NOT SELECTED
BS\QNXT_BS\Partner\ICES=NOT SELECTED
BS\QNXT_BS\Partner\Impact_Pro=NOT SELECTED
BS\QNXT_BS\Partner=NOT SELECTED
BS\QNXT_BS\TMS=NOT SELECTED
BS\QNXT_BS=SELECTED
BS\QNXT_AR_BS=SELECTED
BS\QNXT_CM_BS=SELECTED
BS\Developers_Toolkit=NOT SELECTED
BS\BS_GAC=SELECTED
BS=SELECTED
UI\QNXTALL=NOT SELECTED
UI\QNXT_CM\QNXT_CM_UI=NOT SELECTED
UI\QNXT_CM\QNXT_CM_Services=NOT SELECTED
UI\QNXT_CM=NOT SELECTED
UI\QNXT_AR\QNXT_AR_UI=NOT SELECTED
UI\QNXT_AR\QNXT_AR_Services=NOT SELECTED
UI\QNXT_AR=NOT SELECTED
UI\QNXT\QNXT_UI=NOT SELECTED
UI\QNXT\Services=NOT SELECTED
UI\QNXT=NOT SELECTED
UI=NOT SELECTED
hidden=SELECTED
g_svNetServer=<<HPA PlanEvent DB Server>>
g_svNetDatabase=<<HPA PlanEvent DB Name>>
g_svAppUser=<<Service Account Name>>
g_svAppUserDomain=<<Service Account Domain>>
g_svAppUserPassword=<<Service Account Password>>
```

HPA Rules Silent Installation

HPA rules for silent installation configuration are shown in the following Response file example.

```
[Params]
Init=1
License Agreement=TRUE
```

The following table lists Rules Response file properties.

Properties	Description
License Agreement=TRUE	Accepting the term of the license agreement

HPA Connect Silent Installation

The following table lists HPA Connect Response file properties.

Properties	Description
License Agreement=1	Accepting the term of the license agreement
g_sUser_Name=<<Service Account Name>>	Username of the user account that will logon to use this application.
g_sUser_Company=<<Service Account Domain>>	Company
g_sComPlusUID=<<Service Account Name>>	Enter a non-interactive domain/user name to be used to configure the COM+ applications. The user must be a domain user account and an administrator on this computer. This user must be in the security database to use integrated security
g_sComPlusPWD=<<Service Account Password>>	Enter a non-interactive password to be used to configure the COM+ applications. The user must be a domain user account and an administrator on this computer. This user must be in the security database to use integrated security
TARGETDIR=C:\Program Files\QCSI\QNXT Connect	Destination Folder, By Default HPA Connect will be installed to C:\Program Files\QCSI\ QNXT Connect
g_sBizTalkTranServer=<<BizTalk Transaction Server>>	BizTalk Transaction server
g_sRetryTranServer=<<Retry Transaction Server>>	Retry Transaction server
g_bImportConfig=FALSE	Import Configuration
g_sBTSSenderId=<<Trading Partner ID>>	Sender ID
BizTalkServerConfiguration=NOT SELECTED	Biz Talk Server Configuration
ServicesServerConfiguration\Common=SELECTED	Service Server Configuration
ServicesServerConfiguration\HIPAAWindow\HIPAAImport=SELECTED	HIPPA Import Service Server Configuration
ServicesServerConfiguration\HIPAAWindow\HIPAAExport=SELECTED	HIPPA Export Service Server Configuration
ServicesServerConfiguration\HIPAAWindow\HIPAACommon=SELECTED	HIPPA Common Service Server Configuration
ServicesServerConfiguration\HIPAAWindow\HIPAA_ALWAYS=SELECTED	HIPPA Service Server Configuration
ServicesServerConfiguration\HIPAAWindow=SELECTED	HIPPA Window Service Server Configuration
ServicesServerConfiguration\OrphanRetry=NOT SELECTED	Orphan Retry Service Server Configuration
ServicesServerConfiguration=SELECTED	All Services Server Configuration
g_bInstall_BizTalkConfig=FALSE	Biz Talk Configuration
g_bInstall_HIPAA=TRUE	HIPPA Transaction
g_bInstall_HIPAA_Import=TRUE	Name of the server that will manage the HIPPA and QNXT Connect related transactions.
g_bInstall_HIPAA_Export=TRUE	Name of the server that will manage the HIPPA and QNXT Connect related transactions.
g_bInstall_OrphanRetry=FALSE	Name of the server that will manage the HIPPA and QNXT Connect related transactions.
Icon=SELECTED	

Connect installation on Non BizTalk Server

Connection installation configuration on a Non BizTalk server is shown in the following Response file example.

```
[Params]
LicenseAgreement=1
g_sUser_Name=<<Service Account Name>>
g_sUser_Company=<<Service Account Domain>>
g_sComPlusUID=<<Service Account Name>>
g_sComPlusPWD=<<Service Account Password>>
TARGETDIR=C:\Program Files\QCSI\QFrame\QNXT Connect
g_sBizTalkTranServer=<<Biztalk Transaction Server>>
g_sRetryTranServer=<<Retry Transaction Server>>
g_bImportConfig=FALSE
g_sBTSSenderId=<<Trading Partner ID>>
BizTalkServerConfiguration=NOT_SELECTED
ServicesServerConfiguration\Common=SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAAImport=SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAAExport=SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAACommon=SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAA_ALWAYS=SELECTED
ServicesServerConfiguration\HIPAAWindow=SELECTED
ServicesServerConfiguration\OrphanRetry=NOT_SELECTED
ServicesServerConfiguration=SELECTED
g_bInstall_BizTalkConfig=FALSE
g_bInstall_HIPAA=TRUE
g_bInstall_HIPAA_Import=TRUE
g_bInstall_HIPAA_Export=TRUE
g_bInstall_OrphanRetry=FALSE
Icon=SELECTED
```

Connect Installation on BizTalk Server

Connection installation configuration on a BizTalk server is shown in the following Response file example.

```
[Params]
License Agreement=1
g_sUser_Name=<<Service Account Name>>
g_sUser_Company=<<Service Account Domain>>
g_sComPlusUID=<<Service Account Name>>
g_sComPlusPWD=<<Service Account Password>>
TARGETDIR=C:\Program Files\QCSI\QNXT Connect
g_sBizTalkTranServer=<<Biztalk Transaction Server>>
g_sRetryTranServer=<<Retry Transaction Server>>
g_sBTSSenderId=<<Trading Partner ID>>
g_bImportConfig=FALSE
BizTalkServerConfiguration=SELECTED
ServicesServerConfiguration\Common=NOT SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAAImport=NOT SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAAExport=NOT SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAACommon=NOT SELECTED
ServicesServerConfiguration\HIPAAWindow\HIPAA_ALWAYS=NOT SELECTED
ServicesServerConfiguration\HIPAAWindow=NOT SELECTED
ServicesServerConfiguration\OrphanRetry=NOT SELECTED
ServicesServerConfiguration=NOT SELECTED
g_bInstall_BizTalkConfig=TRUE
g_bInstall_HIPAA=FALSE
g_bInstall_HIPAA_Import=FALSE
g_bInstall_HIPAA_Export=FALSE
g_bInstall_OrphanRetry=FALSE
Icon=SELECTED
```

Help Silent Installation

Help silent installation configuration are shown in the following Response file example.

```
[Params]
Init=1
TARGETDIR=C:\Program Files\QCSI\QFrame
License Agreement=1
```

Help Response File Properties

The following table lists HPA Connect Response file properties.

Properties	Description
TARGETDIR=C:\Program Files\QCSI\QFrame	Destination folder, Help will be installed by default to C:\Program Files\QCSI\QFrame
License Agreement=1	Accepting the term of the license agreement

Appendix B Micro-Dyn Preparation and Setup

Topics for Micro-Dyn Preparation and Setup include:

- [Installing Micro-Dyn Products](#)
- [Updating the Micro-Dyn Web.config File Versions](#)

Installing Micro-Dyn Products

The following Micro-Dyn components can be used with QNXT:

- MCEActive
- OCEActive
- APCActive Enterprise Pricer DLL
- APCActive Enterprise Code Editor DLL
- DRGActive Enterprise
- PricerActive™ Enterprise
- DRG AP Enterprise
- IRF PricerActive Enterprise

Configuring Micro-Dyn Components

Follow the Micro-Dyn installation instructions, with the following exceptions:

- Install Micro-Dyn applications on both the QNXT web server and Execution Units (EU). This process registers the necessary .dll files so that Administrator and Micro-Dyn products work in concert.
- The Micro-Dyn database must match the collation setting of the QCSI SQL server. Otherwise, the database restoration will fail.

Administrator SQL Server databases are configured with the **Dictionary order, case- insensitive, with uppercase preference for use with the 1252 character set** collation setting selected.

Micro-Dyn provides a database for sort order 53. Sort order 53 is the Micro-Dyn equivalent of the SQL Server 2022 collation setting mentioned above.

NOTE: Micro-Dyn provides specific sort order 53 backups. Always restore the database with the applicable collation setting.

- If you want to install and use the DRGActive or PricerActive 2-year component, you must uninstall the DRGActive or PricerActive 5-year component. Delete or rename the registry entry under:

```
o HKEY_LOCAL_MACHINE\SOFTWARE\MicroDyn\DRGActiveNet Enterprise 5-YR DLL
```

or

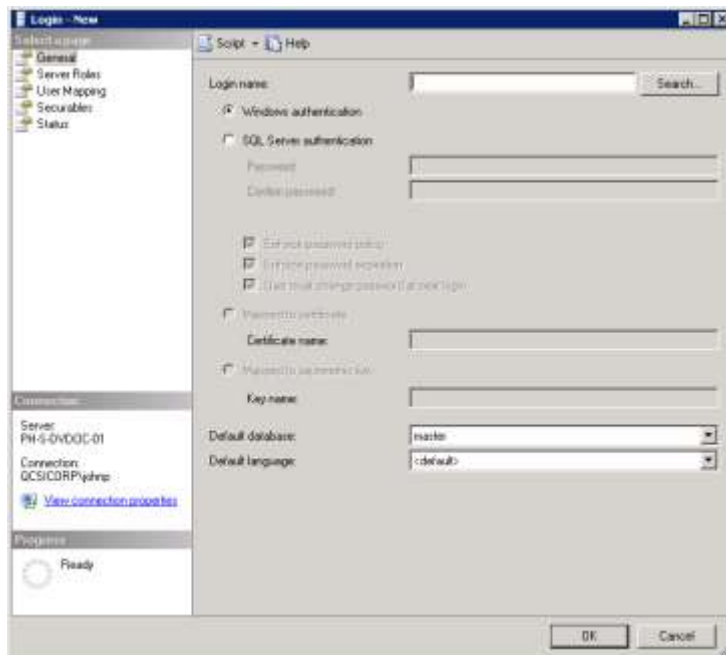
```
o HKEY_LOCAL_MACHINE\SOFTWARE\MicroDyn\PRICERActiveNet Enterprise 5-YR  
DLL
```

Defining a Micro-Dyn SQL Database User Name

After restoring the Micro-Dyn SQL database with sort order 53, create a unique user name as the owner of the Micro-Dyn database.

To create a unique user name as the owner of the Micro-Dyn database:

1. Open the SQL Server Management Studio.
2. Expand the Security folder in the Object Explorer section.
3. Right-click **Logins**, then select **New Login**. The Login - New section appears with General selected as the default Select a page option.



4. Type the unique Login name and select **SQL Server authentication**.
5. Select **User Mapping** in the Select a page area.
6. Select all Micro-Dyn databases that can be accessed by this login.
7. Select **public** and **db_owner** for each database selected in the Database role membership for section.
8. Click **OK** after all roles have been selected for all Micro-Dyn databases. A dialog box displays to confirm the password.
9. Re-enter the password and click **OK** to confirm.

NOTE: The user name created for the Micro-Dyn database is retained even after subsequent restoration. However, if the database is restored to a different server, repeat this process to create the user name and assign database roles on that server.

Installing Micro-Dyn Software

After restoring the Micro-Dyn database and configuring applicable database roles, log on to the HPA web application and Execution Unit servers to install the Micro-Dyn software.

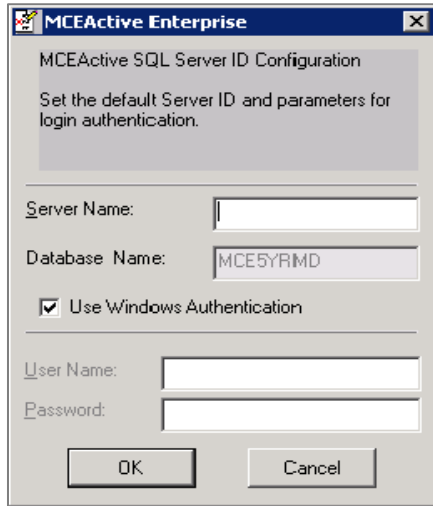
NOTE: Micro-Dyn installation instructions are not included in this guide. For installation instructions, refer to the product documentation.

Testing the Micro-Dyn Connectivity

Run the SQL Server ID Utility according to the Micro-Dyn installation instructions to test the connectivity between Micro-Dyn .dll files and the Micro-Dyn SQL database.

To run the SQL Server ID Utility:

1. Select Start | All Programs | {Micro-Dyn Product} | SQL Server ID Utility.



2. Type the Server Name (SQL server name where the Micro-Dyn database was restored.)
3. Make sure that **Use NT Authentication (recommended)** is unchecked.
4. Type the User Name and Password created in [Defining a Micro-Dyn SQL Database User Name](#).
5. Click **OK**.
 - If the connection is successful, the Test SQL Connection dialog box appears.
 - If the connection is **not** successful, the following message is displayed:



Check for the following possible issues:

- Database role has not been properly set up. Verify that the user name and password are set up with both the **Public** and **db_Owner** database roles.
- Installed database does not match the Micro-Dyn software installed on the web and EU servers.
- License for the Micro-Dyn software or database is expired.

After you have installed all of the Micro-Dyn products, continue with the installation of HPA, beginning with QFramework. See [QFramework Installation](#) for more information.

Updating the Micro-Dyn Web.config File Versions

This section describe how to change the component versions of Micro-Dyn products in the Web.config files. You must complete these steps after installing HPA because required folders are not in place until after the HPA installation.

For information on the installation and configuration of Micro-Dyn products see [Micro-Dyn Preparation and Setup](#).

Updating the Micro-Dyn Configuration Files

To change the Micro-Dyn Component version in the Web.config file:

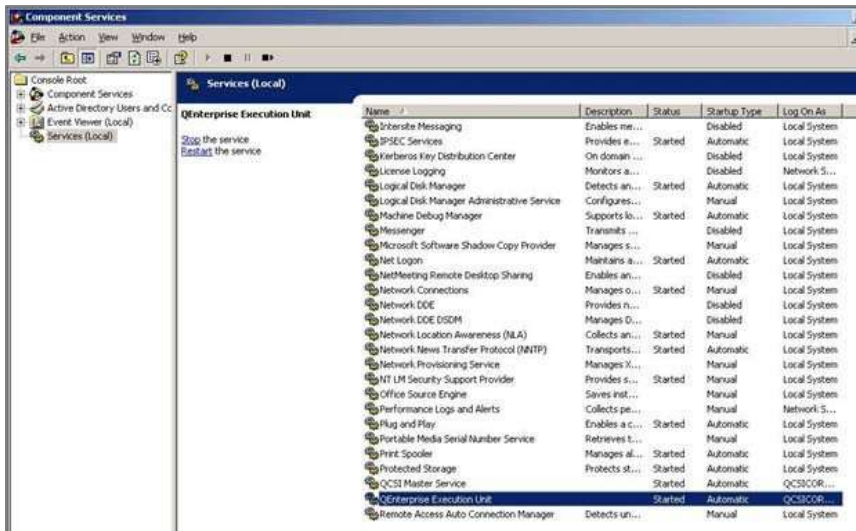
1. Run IISRESET on the Web Server where HPA resides.

NOTE: Complete the following steps only if your machine is also serving as the QNXT Execution Unit.

2. Open the `Execution Unit.exe.config` file in the Execution Unit; for example—
`\Program Files\QCSI\QFrame\Services\Execution Unit\Windows\`.
3. Locate the following text.

```
<!-- TraceObjects value= false -->
<!-- Microdyn Components-->
<add key="DRG2YRTypeName" value="NetDRG2YR.NetDLLM2, NetDRG2YR, Version=2.0.67.0, Culture=neutral, PublicKeyToken=
<add key="DRG5YRTypeName" value="NetDRG5YR.NetDLLM5, NetDRG5YR, Version=2.0.137.0, Culture=neutral, PublicKeyToken=
<add key="IRFPrc2YRTypeName" value="NetIRFPrc2YR.NetIRFPrcM2, NetIRFPrc2YR, Version=2.0.67.1, Culture=neutral, Pub
<add key="MCE5YRTypeName" value="NetMCE5YR.NetMCEM5, NetMCE5YR, Version=2.0.137.0, Culture=neutral, PublicKeyToken=
<add key="OCE5YRTypeName" value="NetOCE5YR.NetOCEM5, NetOCE5YR, Version=2.0.137.0, Culture=neutral, PublicKeyToken=
<add key="APCCeditTypeName" value="NetAPCCedit.NetAPCDLL, NetAPCCedit, Version=2.0.7.300, Culture=neutral, PublicKeyT
<add key="APCCalcTypeName" value="NetAPCCalc.NetAPCPay, NetAPCCalc, Version=2.0.7.301, Culture=neutral, PublicKeyT
<add key="PricerActive5YRTypeName" value="NetPrc2YR.NetPRCM2, NetPrc2Yr, Version=2.0.67.1, Culture=neutral, Public
<!-- ClaimCheck Component-->
<add key="ClaimCheckIwTypeName" value="ClmChk.ClmAudit, ClmChk, Version=1.0.2559.20353, Culture=neutral, Publick
</appSettings>
```

4. Update the version number to the Micro-Dyn version numbers you just recorded.
5. Restart the Execution Unit from the Console Root on the machine where the Execution Unit resides.



Appendix C Configuring GPCS Certificate for APR-DRG and EAPG

Complete the following steps to install the GPCS certificate:

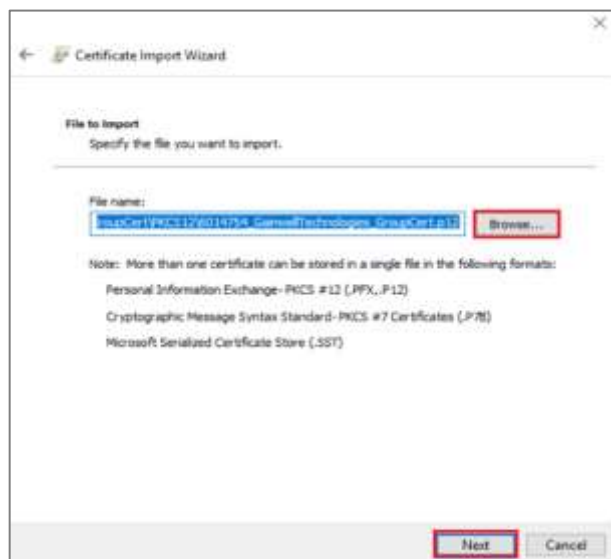
1. Download the client and root GPCS certificate (either Individual or Group) and import the certificate on all App (Core) and EU servers.

NOTE: Both certificates—Individual and Group, should **not** be installed in the same server.

2. Select **Local Machine** and click **Next** on the Certificate Import Wizard window.



3. Verify the file path of the certificate or click **Browse** for the path where the certificate is located and click **Next**.

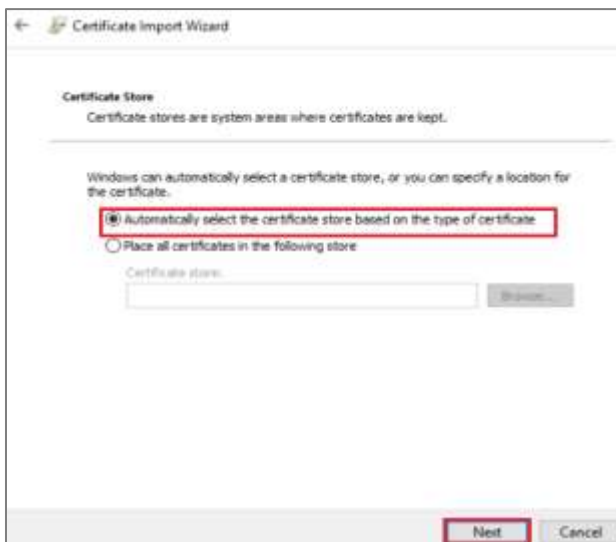


4. Type the Password issued with the certificate and click **Next**.



The screenshot shows the 'Certificate Import Wizard' window. The title bar says 'Certificate Import Wizard'. The main heading is 'Private key protection'. Below it, a message states: 'To maintain security, the private key was protected with a password.' The instruction says: 'Type the password for the private key.' There is a 'Password:' label followed by a text box containing eight asterisks. Below the text box is a checkbox labeled 'Display Password'. Underneath is the 'Import options:' section with three checkboxes: 'Enable strong private key protection. You will be prompted every time the private key is used by an application if you enable this option.' (unchecked), 'Mark this key as exportable. This will allow you to back up or transport your keys at a later time.' (unchecked), and 'Include all extended properties.' (checked). At the bottom right are 'Next' and 'Cancel' buttons. The 'Next' button is highlighted with a red rectangle.

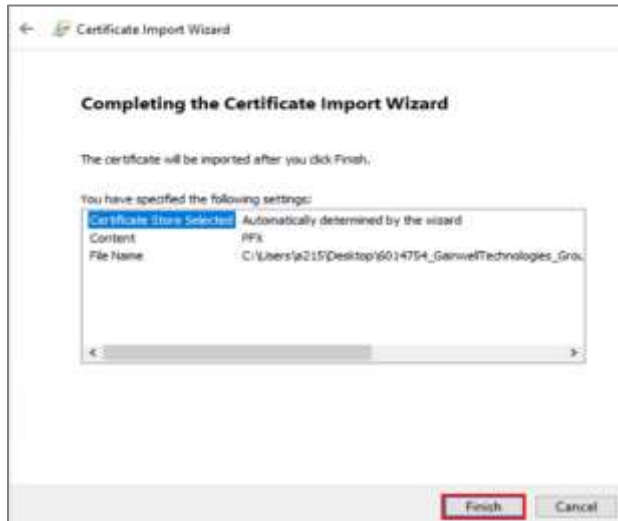
5. Select the **Automatically select the certificate store based on the type of certificate** option and click **Next**.



The screenshot shows the 'Certificate Import Wizard' window. The title bar says 'Certificate Import Wizard'. The main heading is 'Certificate Store'. Below it, a message states: 'Certificate stores are system areas where certificates are kept.' The instruction says: 'Windows can automatically select a certificate store, or you can specify a location for the certificate.' There are two radio button options: 'Automatically select the certificate store based on the type of certificate.' (selected and highlighted with a red rectangle) and 'Place all certificates in the following store'. Below the second option is a text box labeled 'Certificate store:' and a 'Browse...' button. At the bottom right are 'Next' and 'Cancel' buttons. The 'Next' button is highlighted with a red rectangle.

- Click **Finish** on the Completing the Certificate Import Wizard window.

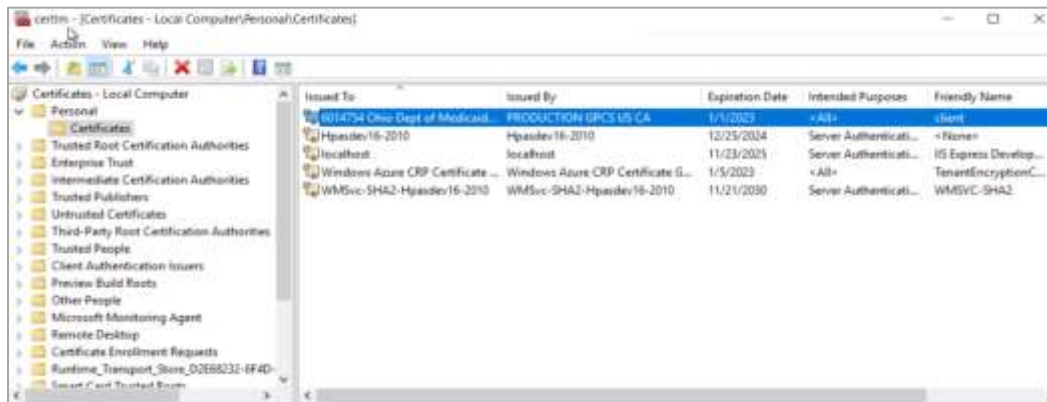
Tip! You should see the message “The import was successful.”



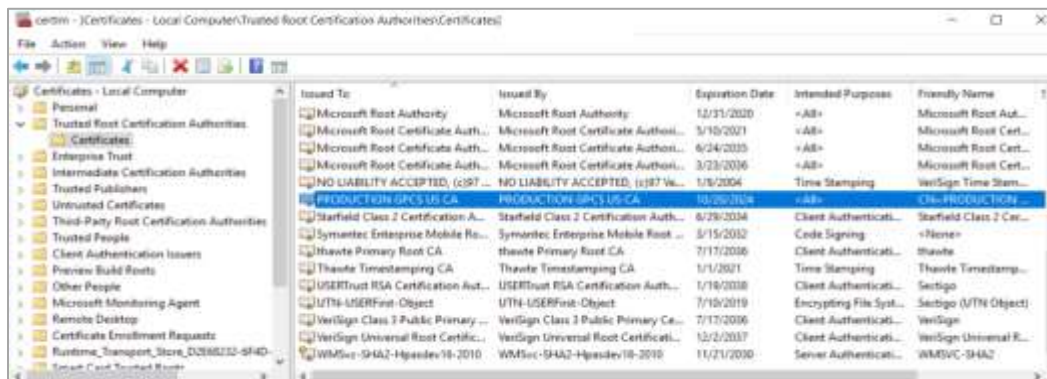
Two certificates will be added to the server in the Certificates – Local Computer folder.

- Verify that the client and root certificates are successfully imported.

NOTE: In the Certificates – Local Computer folder, select **Personal**, then **Certificates**. The Issued To column should include a file with your plan's name.



- Verify the root certificate is successfully imported under Trusted Root Certification Authorities – Certificates with a name like **CN=PRODUCTION GPCS US CA, O=3M Health Information Systems, C=US**.



9. After the new certificates are installed, restart the following services:

- IIS Admin Service
- QCSI Master Service
- QEnterprise Distribution Hub
- QEnterprise Execution Unit

NOTE: You must also provide a GPCS Service URL, Authorized Site Name, and related configuration in the HPA UI (Ancillary Claim Editing on the GPCS Configuration tab). See Help documentation for more information.
